SABR: EDA and Community Diversity Report

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About this report

This Quarto report executes two existing analysis scripts:

- `003_eda_analysis.R` — Exploratory Data Analysis of Microbial Communities
- `004_alpha_diversity.R` — Community Diversity Analyses

Both scripts assume project-root working directory (for sourcing `R/utils/000_setup.R`). Place this `.qmd` at the repository root so relative paths resolve as expected.

If your environment needs it, ensure all packages used by the scripts are installed before rendering.

> **Note**
>
> Render from the repository root:
>
> - CLI: `quarto render sabr_exploratory_and_diversity.qmd --execute-dir .`
> - R: `quarto::quarto_render("sabr_exploratory_and_diversity.qmd")`

Section 1 — Exploratory Data Analysis

This section runs the full workflow in 003_eda_analysis.R, including:

- Dataset exploration (metadata, taxonomic distribution)
- Read count summaries and visualizations
- Good's coverage estimates
- ASV prevalence/core taxa summaries
- Feature summaries, QQ plots, ACF, and correlations

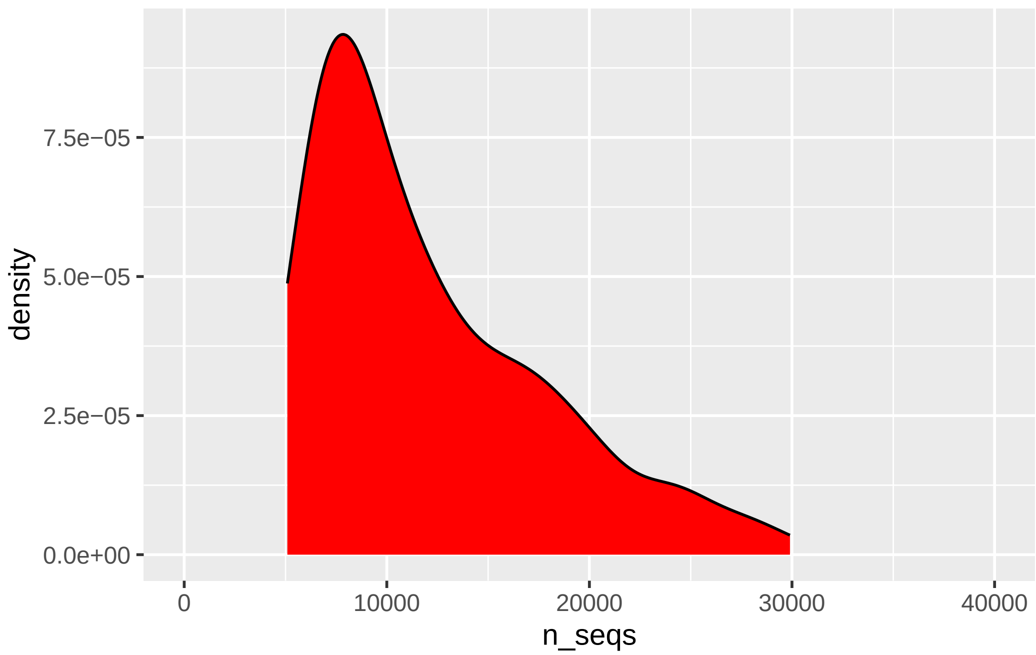
```r
source(
  "R/003_eda_analysis.R",
  local = knitr::knit_global(),
  echo = TRUE,
  print.eval = TRUE,
  skip.echo = 31
)
```

```
> source("R/utils/000_setup.R")
```

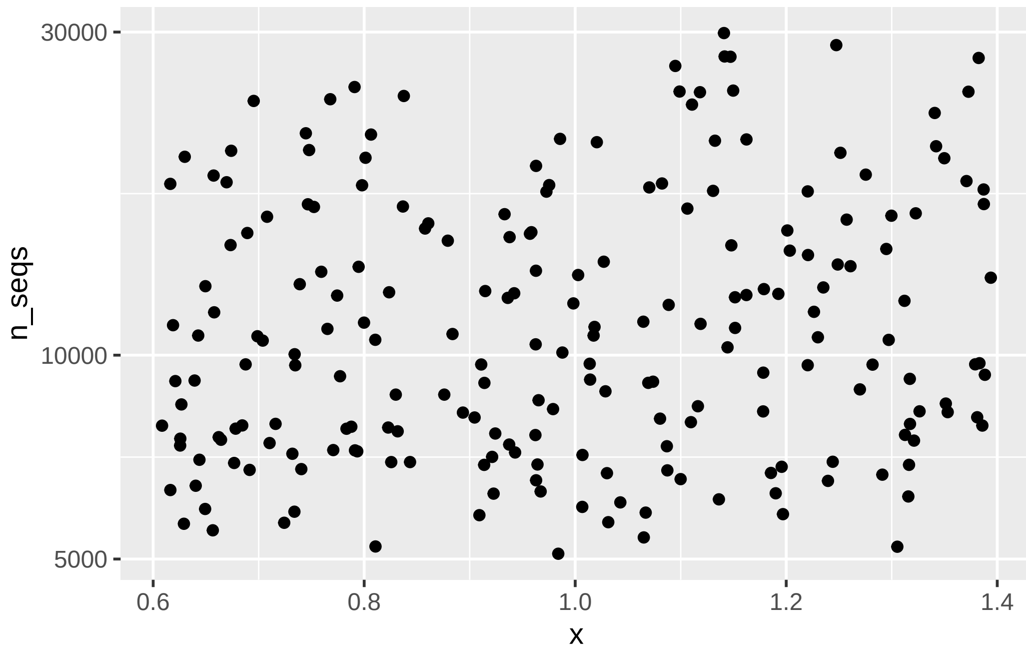
```
> base::colnames(tax_table(sabr_2023_physeq))
[1] "sequence" "kingdom"  "phylum"   "class"    "order"    "family"   "genus"
[8] "species"

> base::colnames(sample_data(sabr_2023_physeq))
 [1] "samples"          "nucleotide"       "year"
 [4] "plot"             "plant"            "sampling_location"
 [7] "replicate"        "sampling_date"    "fertilization"
[10] "nitrogen_conc"    "dna_conc_ng_ul"   "x16s_copies_g"
[13] "nmol_1m_2s_1"     "gnha"             "amo_a_337"
[16] "cnor_b"           "nitrate_ppm"      "ammonia_ppm"
[19] "n_available"      "gwc_g_g"

> names_list <- base::rownames(sample_data(sabr_2023_physeq))

> metagMisc::phyloseq_summary(sabr_2023_physeq, more_stats = F,
+     long = F)
                           Parameter        Phys1
1    Average number of reads per OTU     152.4735
2 Average number of reads per sample   12082.6338
3                     Number of OTUs   16879.0000
4                  Number of samples     213.0000
5              Total number of reads 2573601.0000

> microbiome::summarize_phyloseq(sabr_2023_physeq)


[[1]]
[1] "1] Min. number of reads = 5091"
```

```
[[2]]
[1] "2] Max. number of reads = 29899"

[[3]]
[1] "3] Total number of reads = 2573601"

[[4]]
[1] "4] Average number of reads = 12082.6338028169"

[[5]]
[1] "5] Median number of reads = 10375"

[[6]]
[1] "7] Sparsity = 0.975927806505681"

[[7]]
[1] "6] Any OTU sum to 1 or less? YES"

[[8]]
[1] "8] Number of singletons = 35"

[[9]]
[1] "9] Percent of OTUs that are singletons \n        (i.e. exactly one read detected across

[[10]]
[1] "10] Number of sample variables are: 20"

[[11]]
 [1] "samples"           "nucleotide"        "year"
 [4] "plot"              "plant"             "sampling_location"
 [7] "replicate"         "sampling_date"     "fertilization"
[10] "nitrogen_conc"     "dna_conc_ng_ul"    "x16s_copies_g"
[13] "nmol_1m_2s_1"      "gnha"              "amo_a_337"
[16] "cnor_b"            "nitrate_ppm"       "ammonia_ppm"
[19] "n_available"       "gwc_g_g"


> percent_phyla_clean <- summarise(group_by(mutate(as.data.frame(phyloseq_ntaxa_by_tax(sabr_
+     TaxRank = "phylum", relative = F, add_ .... [TRUNCATED]

> percent_phyla_clean
# A tibble: 54 x 2
   phylum                     occurance_in_samples
```

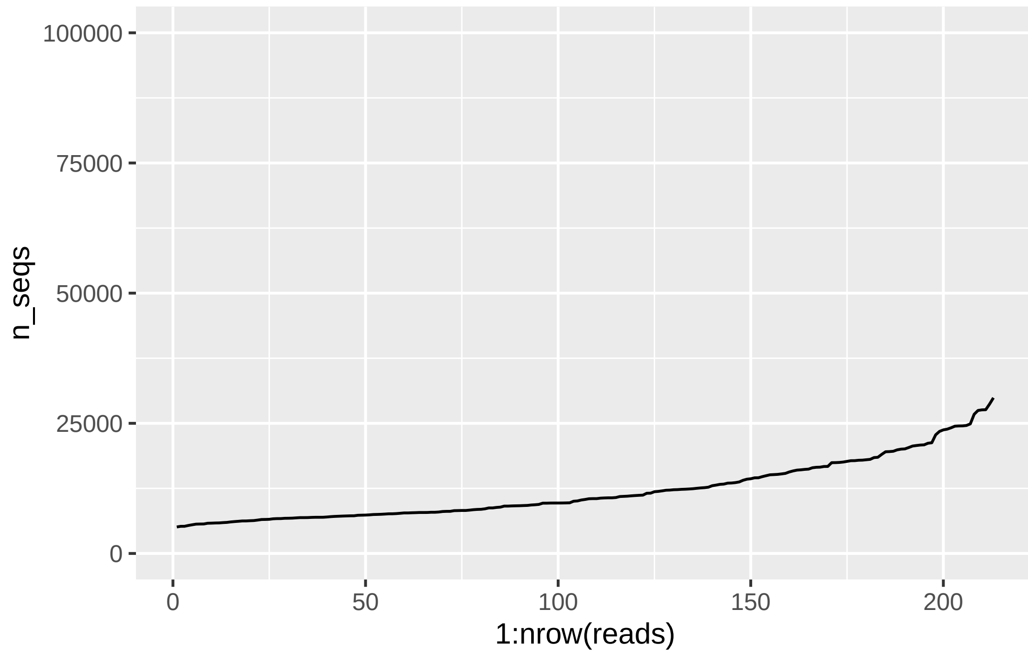
```
   <chr>                                      <int>
 1 Abditibacteriota                              38
 2 Acidobacteriota                              213
 3 Actinomycetota                               213
 4 Armatimonadota                               199
 5 Bacillota                                    208
 6 Bacteroidota                                 213
 7 Bdellovibrionota                             204
 8 Candidatus Eremiobacterota                    13
 9 Candidatus Kapabacteria                       68
10 Candidatus Kryptonia                          17
# i 44 more rows

> reads <- dplyr::left_join(rename(rownames_to_column(as.data.frame(readcount(sabr_2023_phys
+     var = "sample_id"), n_seqs = "readcount(sabr_ ..." ... [TRUNCATED]

> reads_sum <- arrange(summarise(filter(group_by(reads,
+     sample_id), sample_id %in% names_list), n_seqs = sum(n_seqs)),
+     n_seqs)

> ggplot(reads, aes(x = n_seqs)) + geom_density(position = "identity",
+     stat = "density", fill = "red") + coord_cartesian(xlim = c(0,
+     400 .... [TRUNCATED]
```



```
> ggplot(reads, aes(x = 1, y = n_seqs)) + geom_jitter() +
+     scale_y_log10()
```




```
> ggplot(filter(arrange(reads, n_seqs), sample_id %in%
+     names_list), aes(x = 1:nrow(reads), y = n_seqs)) + geom_line() +
+     coord_cartesian( .... [TRUNCATED]
```




```
> cover_chao <- phyloseq_coverage(sabr_2023_physeq,
+     correct_singletons = T, add_attr = T)

> cover_chao
          SampleID SampleCoverage
1   SABR_1-P01-B-a      0.9994320
2   SABR_1-P01-B-b      1.0000000
3   SABR_1-P01-R-a      1.0000000
4   SABR_1-P01-R-b      0.9997014
5   SABR_1-P02-B-a      0.9994946
6   SABR_1-P02-B-b      0.9985520
7   SABR_1-P02-R-a      0.9979631
8   SABR_1-P02-R-b      1.0000000
9   SABR_1-P03-B-a      0.9974886
10  SABR_1-P03-B-b      0.9998982
11  SABR_1-P03-R-a      0.9998119
12  SABR_1-P04-B-a      0.9998507
13  SABR_1-P04-B-b      0.9972731
14  SABR_1-P04-R-a      0.9995256
15  SABR_1-P04-R-b      0.9993472
16  SABR_1-P05-B-a      0.9997818
17  SABR_1-P05-B-b      1.0000000
```

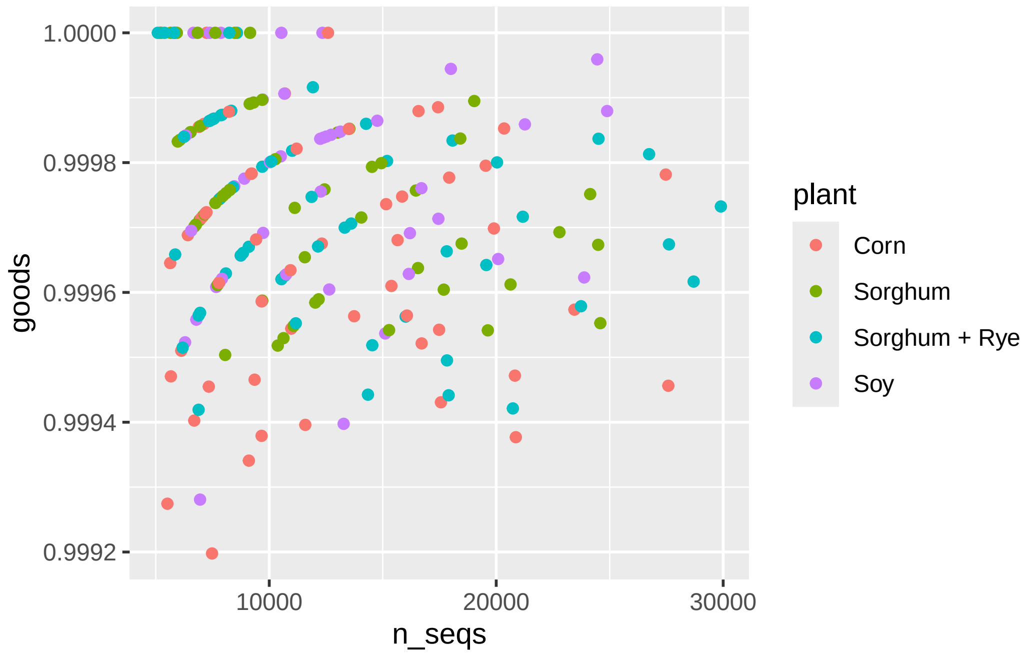
18	SABR_1-P05-R-a	0.9988468
19	SABR_1-P05-R-b	0.9998673
20	SABR_1-P06-B-a	0.9947230
21	SABR_1-P06-R-a	0.9997136
22	SABR_1-P06-R-b	0.9998502
23	SABR_1-P07-B-a	1.0000000
24	SABR_1-P07-B-b	0.9998088
25	SABR_1-P07-R-a	0.9994719
26	SABR_1-P07-R-b	0.9983674
27	SABR_1-P08-B-a	0.9991587
28	SABR_1-P08-B-b	0.9995861
29	SABR_1-P08-R-a	0.9996885
30	SABR_1-P08-R-b	0.9951908
31	SABR_1-P09-B-a	1.0000000
32	SABR_1-P09-B-b	1.0000000
33	SABR_1-P09-R-b	0.9997009
34	SABR_1-P10-R-b	1.0000000
35	SABR_1-P11-B-a	0.9996383
36	SABR_1-P11-B-b	0.9997478
37	SABR_1-P11-R-a	0.9998240
38	SABR_1-P11-R-b	1.0000000
39	SABR_1-P12-B-b	1.0000000
40	SABR_1-P12-R-a	0.9998902
41	SABR_1-P12-R-b	0.9996864
42	SABR_1-P13-B-b	0.9995023
43	SABR_1-P14-B-a	0.9997712
44	SABR_1-P14-B-b	1.0000000
45	SABR_1-P14-R-b	1.0000000
46	SABR_1-P15-B-a	0.9989254
47	SABR_1-P15-B-b	0.9993062
48	SABR_1-P15-R-a	0.9974384
49	SABR_1-P15-R-b	0.9994532
50	SABR_2-P01-B-a	0.9992385
51	SABR_2-P01-B-b	0.9971082
52	SABR_2-P01-R-a	0.9995634
53	SABR_2-P01-R-b	0.9992838
54	SABR_2-P02-B-a	0.9986724
55	SABR_2-P02-B-b	0.9977347
56	SABR_2-P02-R-a	0.9997080
57	SABR_2-P02-R-b	0.9990369
58	SABR_2-P03-B-a	0.9994112
59	SABR_2-P03-B-b	0.9991878
60	SABR_2-P03-R-a	0.9989267

61	SABR_2-P03-R-b	0.9959177
62	SABR_2-P04-B-a	0.9997462
63	SABR_2-P04-B-b	0.9993224
64	SABR_2-P04-R-a	0.9993575
65	SABR_2-P04-R-b	0.9971461
66	SABR_2-P05-B-a	1.0000000
67	SABR_2-P05-B-b	0.9998372
68	SABR_2-P05-R-a	1.0000000
69	SABR_2-P05-R-b	0.9990972
70	SABR_2-P06-B-a	0.9998974
71	SABR_2-P06-B-b	0.9991542
72	SABR_2-P06-R-a	0.9993480
73	SABR_2-P06-R-b	0.9994707
74	SABR_2-P07-B-a	0.9995153
75	SABR_2-P07-B-b	0.9989346
76	SABR_2-P07-R-a	0.9997939
77	SABR_2-P07-R-b	0.9986195
78	SABR_2-P08-B-a	0.9997230
79	SABR_2-P08-R-a	0.9992236
80	SABR_2-P08-R-b	0.9991063
81	SABR_2-P09-B-a	1.0000000
82	SABR_2-P09-B-b	0.9990840
83	SABR_2-P09-R-a	0.9998883
84	SABR_2-P09-R-b	1.0000000
85	SABR_2-P10-B-a	0.9990743
86	SABR_2-P10-B-b	0.9989797
87	SABR_2-P10-R-a	0.9995142
88	SABR_2-P10-R-b	0.9990517
89	SABR_2-P11-B-a	0.9988091
90	SABR_2-P11-B-b	0.9986640
91	SABR_2-P11-R-b	0.9996795
92	SABR_2-P12-B-a	0.9988798
93	SABR_2-P12-B-b	0.9991426
94	SABR_2-P12-R-a	0.9991465
95	SABR_2-P12-R-b	0.9994569
96	SABR_2-P13-B-a	0.9994000
97	SABR_2-P13-B-b	0.9996328
98	SABR_2-P13-R-a	0.9998776
99	SABR_2-P13-R-b	0.9992353
100	SABR_2-P14-B-a	1.0000000
101	SABR_2-P14-B-b	0.9996825
102	SABR_2-P14-R-a	1.0000000
103	SABR_2-P14-R-b	1.0000000

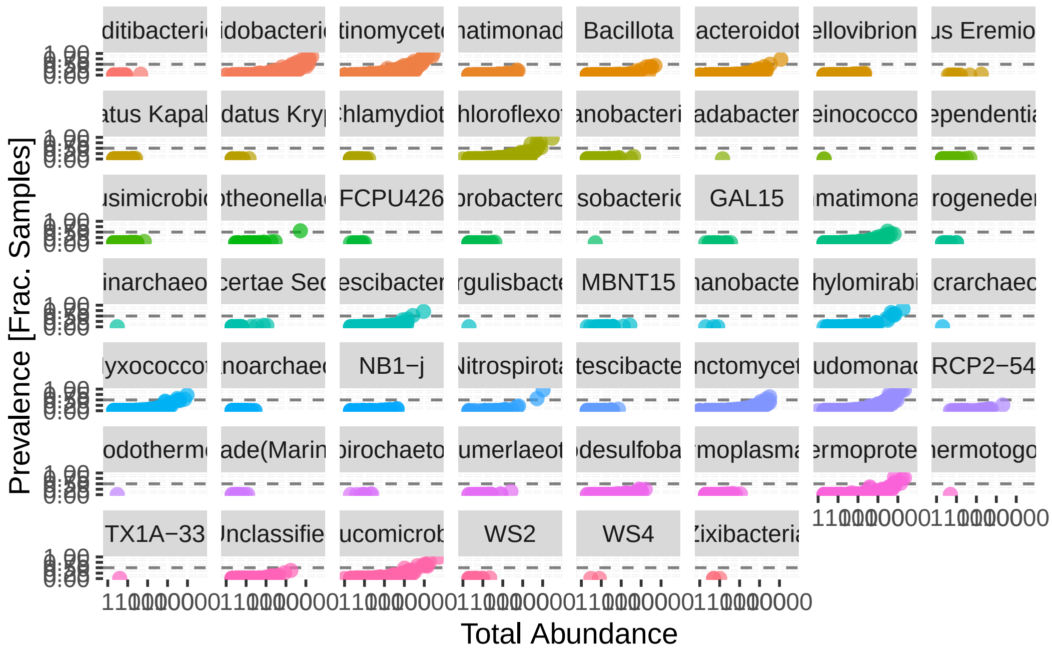
104	SABR_2-P15-B-a	0.9994413
105	SABR_2-P15-B-b	1.0000000
106	SABR_2-P15-R-a	0.9987643
107	SABR_2-P15-R-b	1.0000000
108	SABR_3-P01-B-b	0.9991474
109	SABR_3-P02-B-a	0.9998088
110	SABR_3-P02-B-b	0.9988772
111	SABR_3-P02-R-a	0.9989689
112	SABR_3-P02-R-b	0.9981673
113	SABR_3-P03-B-b	1.0000000
114	SABR_3-P03-R-a	1.0000000
115	SABR_3-P04-B-a	0.9927358
116	SABR_3-P04-B-b	0.9984085
117	SABR_3-P04-R-a	0.9991880
118	SABR_3-P04-R-b	0.9994094
119	SABR_3-P05-B-a	0.9983282
120	SABR_3-P05-B-b	1.0000000
121	SABR_3-P05-R-a	0.9895542
122	SABR_3-P06-B-a	0.9995960
123	SABR_3-P06-B-b	1.0000000
124	SABR_3-P06-R-a	0.9992046
125	SABR_3-P07-B-a	1.0000000
126	SABR_3-P07-B-b	1.0000000
127	SABR_3-P07-R-a	0.9964776
128	SABR_3-P07-R-b	0.9994250
129	SABR_3-P08-B-b	0.9994436
130	SABR_3-P08-R-a	1.0000000
131	SABR_3-P09-B-a	1.0000000
132	SABR_3-P09-B-b	0.9998551
133	SABR_3-P09-R-b	1.0000000
134	SABR_3-P10-B-a	0.9971992
135	SABR_3-P10-B-b	1.0000000
136	SABR_3-P10-R-a	0.9993153
137	SABR_3-P10-R-b	0.9989627
138	SABR_3-P11-B-b	0.9994561
139	SABR_3-P11-R-a	0.9992911
140	SABR_3-P11-R-b	0.9994821
141	SABR_3-P12-B-a	0.9979624
142	SABR_3-P12-B-b	0.9964570
143	SABR_3-P12-R-a	1.0000000
144	SABR_3-P12-R-b	0.9997486
145	SABR_3-P13-B-a	0.9990132
146	SABR_3-P13-B-b	0.9998739

147	SABR_3-P13-R-a	1.0000000
148	SABR_3-P13-R-b	0.9992337
149	SABR_3-P14-B-b	0.9983468
150	SABR_3-P14-R-b	0.9994722
151	SABR_3-P15-B-b	0.9993310
152	SABR_3-P15-R-a	0.9989956
153	SABR_3-P15-R-b	0.9989113
154	SABR_4-P01-B-a	1.0000000
155	SABR_4-P01-B-b	0.9997124
156	SABR_4-P01-R-a	0.9997276
157	SABR_4-P01-R-b	0.9995198
158	SABR_4-P02-B-a	0.9981507
159	SABR_4-P02-B-b	0.9992357
160	SABR_4-P02-R-a	0.9989476
161	SABR_4-P02-R-b	0.9978333
162	SABR_4-P03-B-a	1.0000000
163	SABR_4-P03-B-b	0.9993189
164	SABR_4-P03-R-a	1.0000000
165	SABR_4-P03-R-b	0.9997411
166	SABR_4-P04-B-a	0.9997524
167	SABR_4-P04-B-b	0.9996772
168	SABR_4-P04-R-a	0.9998646
169	SABR_4-P04-R-b	0.9991055
170	SABR_4-P05-B-a	0.9995871
171	SABR_4-P05-B-b	1.0000000
172	SABR_4-P05-R-a	0.9983396
173	SABR_4-P05-R-b	0.9995420
174	SABR_4-P06-B-a	1.0000000
175	SABR_4-P06-B-b	1.0000000
176	SABR_4-P06-R-a	0.9995020
177	SABR_4-P06-R-b	0.9993213
178	SABR_4-P07-B-a	1.0000000
179	SABR_4-P07-B-b	0.9988703
180	SABR_4-P07-R-a	1.0000000
181	SABR_4-P07-R-b	0.9995819
182	SABR_4-P08-B-a	1.0000000
183	SABR_4-P08-B-b	0.9994617
184	SABR_4-P08-R-a	0.9998939
185	SABR_4-P08-R-b	0.9970406
186	SABR_4-P09-B-a	0.9996038
187	SABR_4-P09-B-b	1.0000000
188	SABR_4-P09-R-a	0.9998315
189	SABR_4-P09-R-b	1.0000000

```
190 SABR_4-P10-B-a      0.9986986
191 SABR_4-P10-B-b      0.9992426
192 SABR_4-P10-R-a      1.0000000
193 SABR_4-P10-R-b      0.9998803
194 SABR_4-P11-B-a      0.9998756
195 SABR_4-P11-B-b      0.9995579
196 SABR_4-P11-R-a      0.9996097
197 SABR_4-P11-R-b      0.9998720
198 SABR_4-P12-B-a      0.9992420
199 SABR_4-P12-B-b      0.9986424
200 SABR_4-P12-R-a      0.9995150
201 SABR_4-P12-R-b      1.0000000
202 SABR_4-P13-B-a      1.0000000
203 SABR_4-P13-B-b      0.9998625
204 SABR_4-P13-R-a      0.9993176
205 SABR_4-P13-R-b      0.9988808
206 SABR_4-P14-B-a      0.9988124
207 SABR_4-P14-B-b      0.9982645
208 SABR_4-P14-R-a      1.0000000
209 SABR_4-P14-R-b      0.9989688
210 SABR_4-P15-B-a      0.9995154
211 SABR_4-P15-B-b      1.0000000
212 SABR_4-P15-R-a      0.9996371
213 SABR_4-P15-R-b      0.9998377

> ps_melt <- select(rename(janitor::clean_names(psmelt(sabr_2023_physeq)),
+     sample_id = sample, asv = otu), !nitrogen_conc)

> cover_goods <- summarise(group_by(select(ps_melt,
+     sample_id, plant, asv, abundance), sample_id, plant), n_seqs = sum(abundance),
+     n_sin .... [TRUNCATED]


> ggplot(cover_goods, aes(x = n_seqs, y = goods, color = plant)) +
+     geom_point()
```

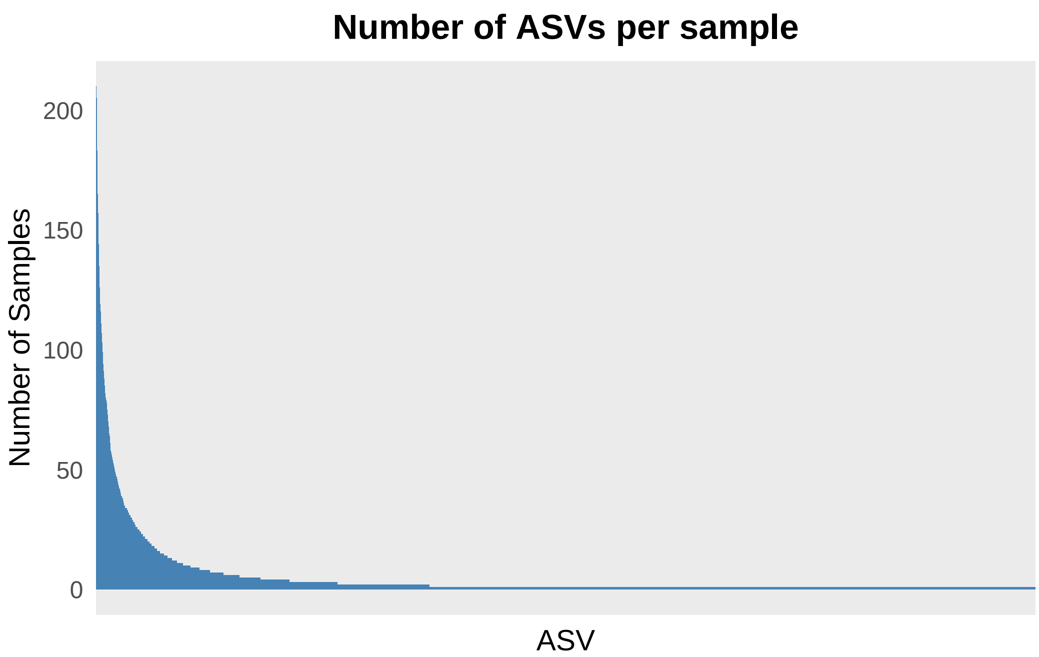



```
> arrange(filter(cover_goods, n_seqs > 750), goods)
# A tibble: 213 x 5
# Groups:   sample_id [213]
   sample_id      plant          n_seqs n_singletons goods
   <chr>          <chr>           <int>        <int> <dbl>
 1 SABR_2-P11-B-b Corn             7479            6 0.999
 2 SABR_3-P11-B-b Corn             5513            4 0.999
 3 SABR_1-P04-B-b Soy              6952            5 0.999
 4 SABR_4-P08-R-b Corn             9102            6 0.999
 5 SABR_4-P01-B-b Corn            20864           13 0.999
 6 SABR_1-P08-B-b Corn             9662            6 0.999
 7 SABR_2-P08-R-a Corn            11588            7 0.999
 8 SABR_2-P04-B-b Soy             13278            8 0.999
 9 SABR_1-P01-R-b Corn             6695            4 0.999
10 SABR_3-P09-B-b Sorghum + Rye    6887            4 0.999
# i 203 more rows

> metagMisc::phyloseq_prevalence_plot(sabr_2023_physeq,
+     prev.trh = 0.5, taxcolor = "phylum", facet = TRUE, point_alpha = 0.7,
+     showplot = .... [TRUNCATED]
```




```
> ps_average <- metagMisc::phyloseq_average(sabr_2023_physeq,
+     avg_type = "arithmetic", acomp_zero_impute = NULL, aldex_samples = 213,
+     al .... [TRUNCATED]

  |
  |                                                                      |   0%

  |
  |==================                                                    |  25%

  |
  |===================================                                   |  50%

  |
  |====================================================                  |  75%

  |
```

```
  |======================================================================| 100%

> ps_average
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16879 taxa and 4 samples ]
tax_table()   Taxonomy Table:    [ 16879 taxa by 8 taxonomic ranks ]

> ggplot(sabr_2023_physeq_relab %>% phyloseq::otu_table(.) %>%
+     t(.) %>% as.data.frame(.) %>% {
+     function(df) colSums(df > 0)
+ }() %>% {
+ .... [TRUNCATED]
```

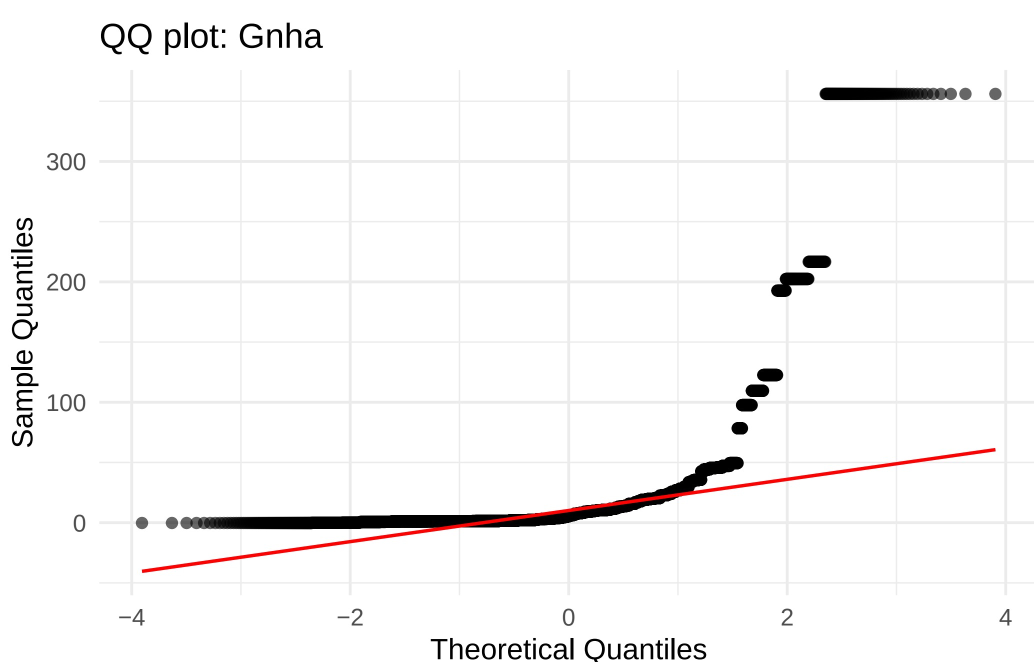



```
> core_abun <- microbiome::core_abundance(sabr_2023_physeq@otu_table,
+     detection = 0, prevalence = 50/100, include.lowest = FALSE)

> rare_abun <- microbiome::rare_abundance(sabr_2023_physeq@otu_table,
+     detection = 0, prevalence = 50/100, include.lowest = FALSE)

> core_tax <- microbiome::core_members(sabr_2023_physeq,
+     detection = 0, prevalence = 90/100, include.lowest = FALSE)
```

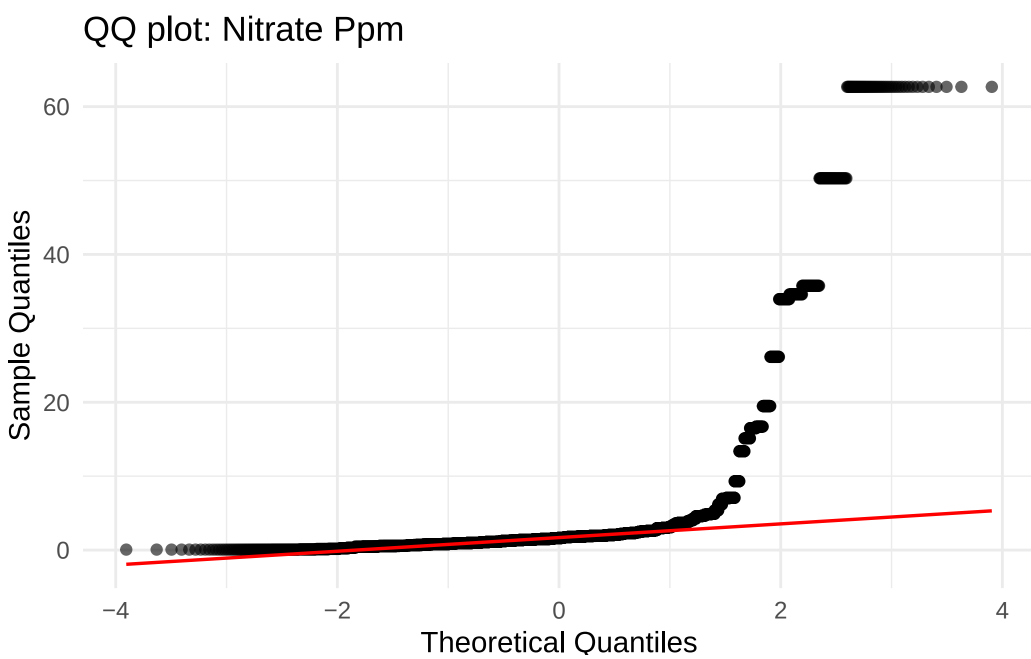
```
> core_tax
 [1] "ASV_1"  "ASV_2"  "ASV_3"  "ASV_6"  "ASV_11" "ASV_12" "ASV_19" "ASV_26"
 [9] "ASV_29" "ASV_36"

> rare_tax <- microbiome::rare_members(sabr_2023_physeq,
+     detection = 0, prevalence = 50/100, include.lowest = FALSE)

> ps_list_raw <- get_prevalent_rare(sabr_2023_physeq,
+     thresholds = c(90, 80, 70, 60, 30), detection = 0/100, include.lowest = FALSE)

> ps_list_relab <- get_prevalent_rare(sabr_2023_physeq_relab,
+     thresholds = c(90, 80, 70, 60, 30), detection = 0/100, include.lowest = FALSE)

> ps_list_rarefied <- get_prevalent_rare(main_rarefied_physeq,
+     thresholds = c(90, 80, 70, 60, 30), detection = 0/100, include.lowest = FALSE)

> main_physeq_list <- list(raw_phyloseq_lists = ps_list_raw,
+     relab_phyloseq_lists = ps_list_relab, rarefied_phyloseq_lists = ps_list_rarefied)

> ps_list_raw
$prevalent_90
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 10 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 10 taxa by 8 taxonomic ranks ]

$rare_90
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16869 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 16869 taxa by 8 taxonomic ranks ]

$prevalent_80
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 19 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 19 taxa by 8 taxonomic ranks ]

$rare_80
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16860 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 16860 taxa by 8 taxonomic ranks ]
```

```
$prevalent_70
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 38 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 38 taxa by 8 taxonomic ranks ]

$rare_70
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16841 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 16841 taxa by 8 taxonomic ranks ]

$prevalent_60
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 56 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 56 taxa by 8 taxonomic ranks ]

$rare_60
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16823 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 16823 taxa by 8 taxonomic ranks ]

$prevalent_30
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 237 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 237 taxa by 8 taxonomic ranks ]

$rare_30
phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 16642 taxa and 213 samples ]
sample_data() Sample Data:       [ 213 samples by 20 sample variables ]
tax_table()   Taxonomy Table:    [ 16642 taxa by 8 taxonomic ranks ]


> features <- c("gnha", "nitrate_ppm", "ammonia_ppm",
+     "amo_a_337", "cnor_b", "gwc_g_g")

> parameters <- c("plot", "plant", "sampling_location")
```

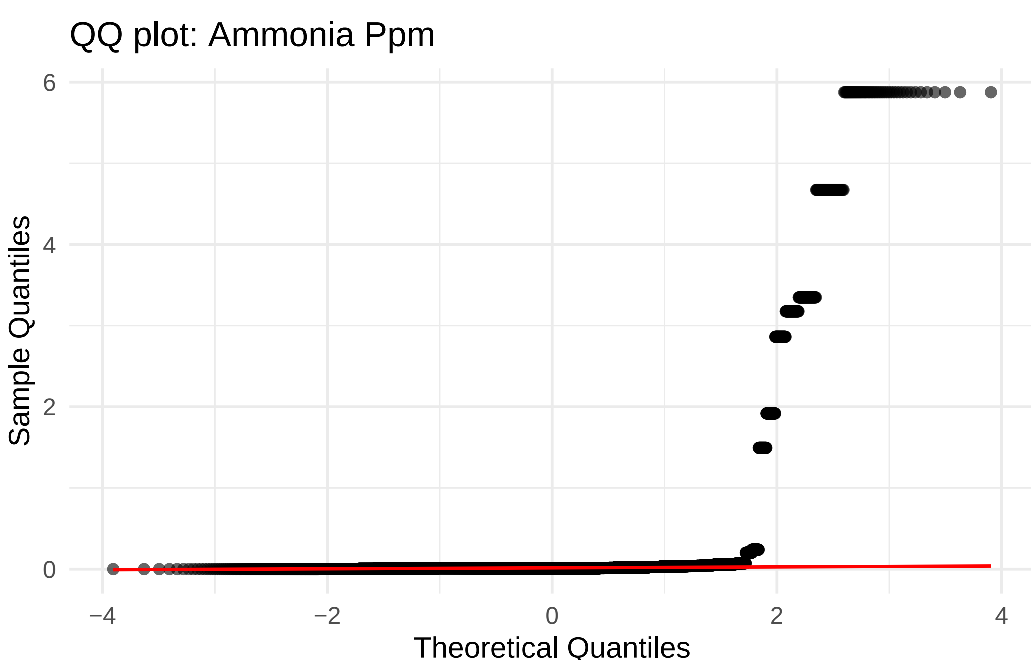
```
> time_col <- "sampling_date"

> use_log_y <- TRUE

> feature_summary <- map_dfr(features, function(f) {
+     set.seed(123)
+     x <- suppressWarnings(as.numeric(main_rarefied_df[[f]]))
+     x <- x[i .... [TRUNCATED]

> feature_summary
# A tibble: 6 x 11
  feature          n    mean      sd      min     max shapiro_W shapiro_p shapiro_n
  <chr>        <int>   <dbl>   <dbl>    <dbl>   <dbl>     <dbl>     <dbl>     <int>
1 gnha         10650 2.02e+1 4.66e+1 -2.83e-1 3.56e2      0.415  1.40e-83      5000
2 nitrate_p~ 10600 3.29e+0 7.16e+0  6.09e-2 6.27e1      0.334  1.78e-86      5000
3 ammonia_p~ 10600 1.28e-1 6.46e-1  1.03e-4 5.88e0      0.165  1.28e-91      5000
4 amo_a_337  10650 6.03e+7 3.58e+7  6.41e+6 3.09e8      0.796  5.81e-62      5000
5 cnor_b     10650 3.34e+9 1.64e+9  5.20e+8 9.97e9      0.920  1.85e-45      5000
6 gwc_g_g      9250 1.78e-1 2.69e-1  5.97e-2 3.79e0      0.118  6.56e-93      5000
# i 2 more variables: shapiro_note <chr>, qqplot <list>

> feature_summary$qqplot
[[1]]
```



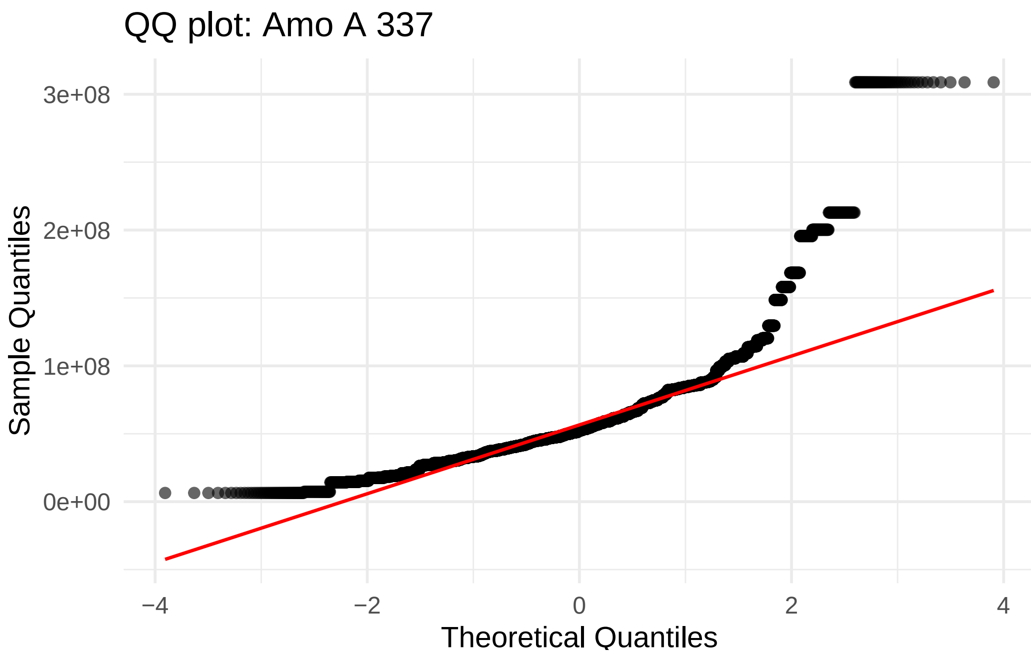
[[2]]


QQ plot: Nitrate Ppm
60
40
20
0
Sample Quantiles
−4
−2
0
2
4
Theoretical Quantiles

[[3]]

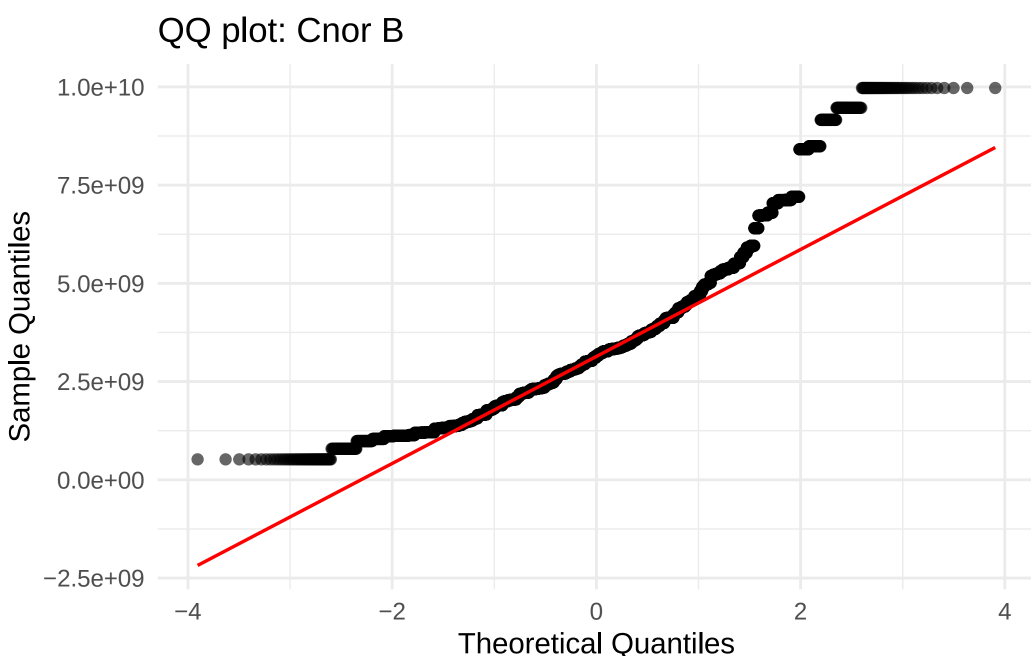

QQ plot: Ammonia Ppm
6
4
2
0
Sample Quantiles
−4
−2
0
2
4
Theoretical Quantiles

[[4]]



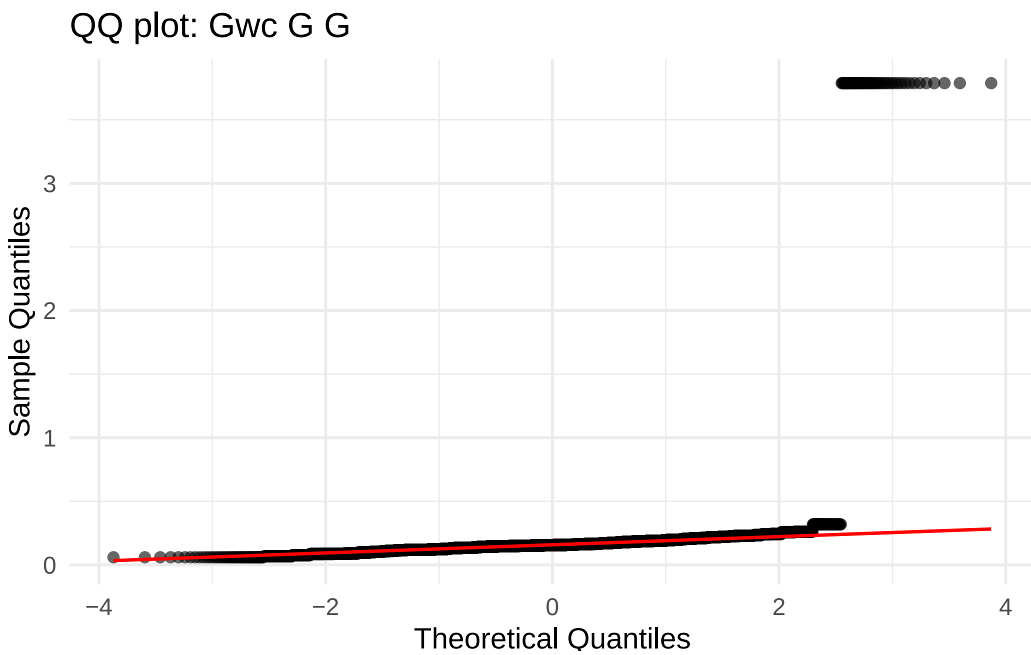
QQ plot: Amo A 337
3e+08
2e+08
1e+08
0e+00
Sample Quantiles
−4
−2
0
2
4
Theoretical Quantiles

[[5]]

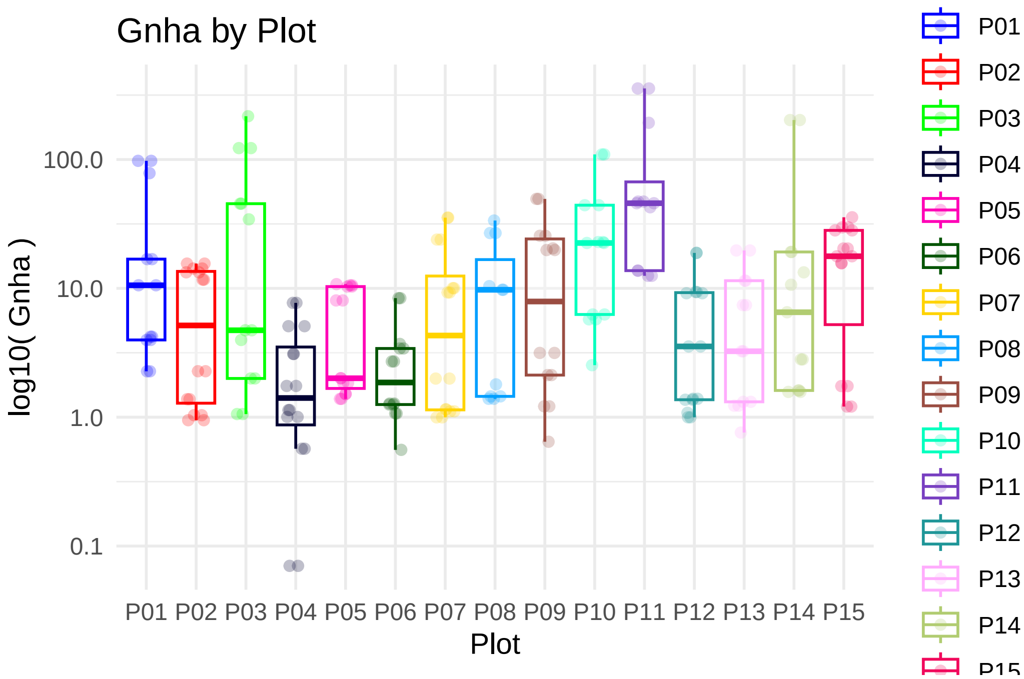


QQ plot: Cnor B
1.0e+10
7.5e+09
5.0e+09
2.5e+09
0.0e+00
−2.5e+09
Sample Quantiles
−4
−2
0
2
4
Theoretical Quantiles

```
[[6]]
```

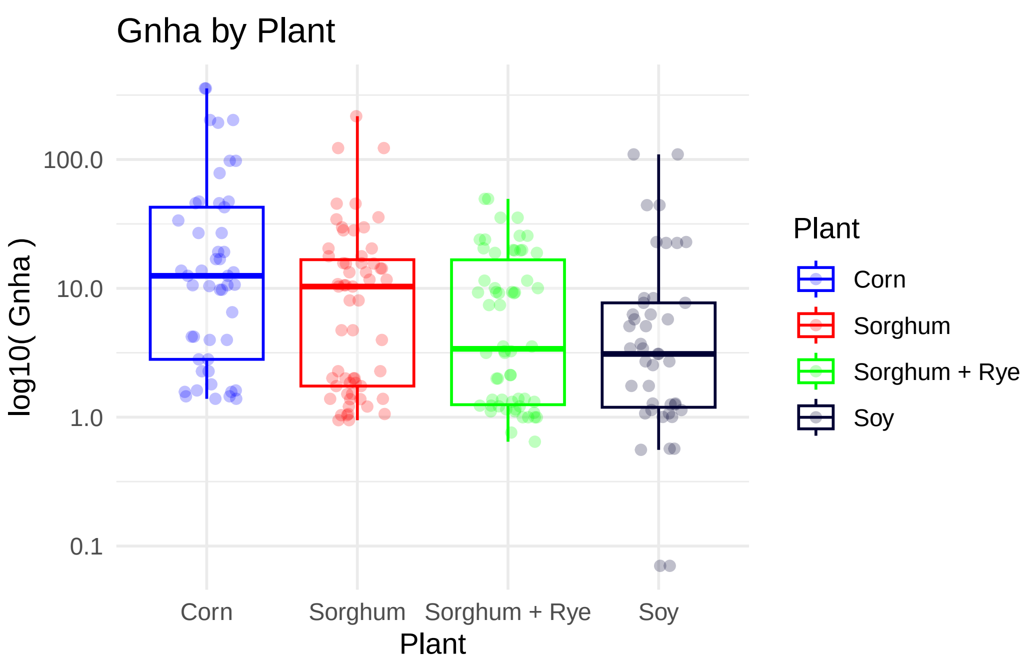



```
> feature_boxplots <- features %>% set_names() %>% map(function(feat) {
+     parameters %>% set_names() %>% map(function(par) {
+         df <- sabr_ .... [TRUNCATED]

> purrr::iwalk(feature_boxplots, ~{
+     feature_name <- .y
+     cat(rep("-", 40), "\n")
+     cat("FEATURE:", toupper(feature_name), "\n")
+     ca .... [TRUNCATED]
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
FEATURE: GNHA
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$plot
```



Gnha by Plot
log10(Gnha)
100.0
10.0
1.0
0.1
P01 P02 P03 P04 P05 P06 P07 P08 P09 P10 P11 P12 P13 P14 P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14

$plant

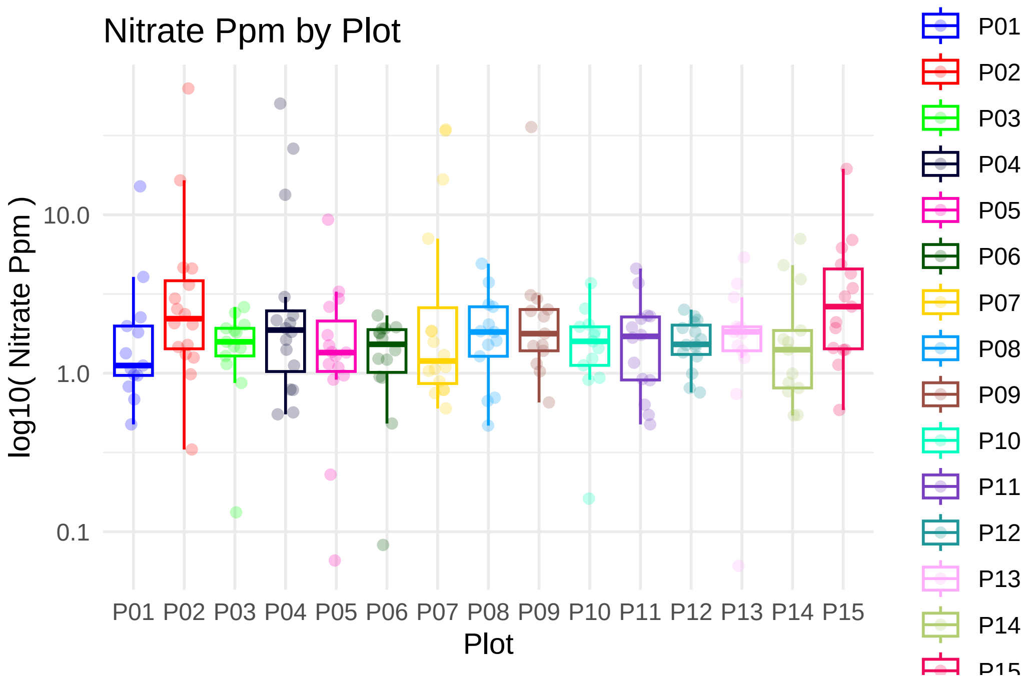


Gnha by Plant
log10(Gnha)
100.0
10.0
1.0
0.1
Corn
Sorghum
Sorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

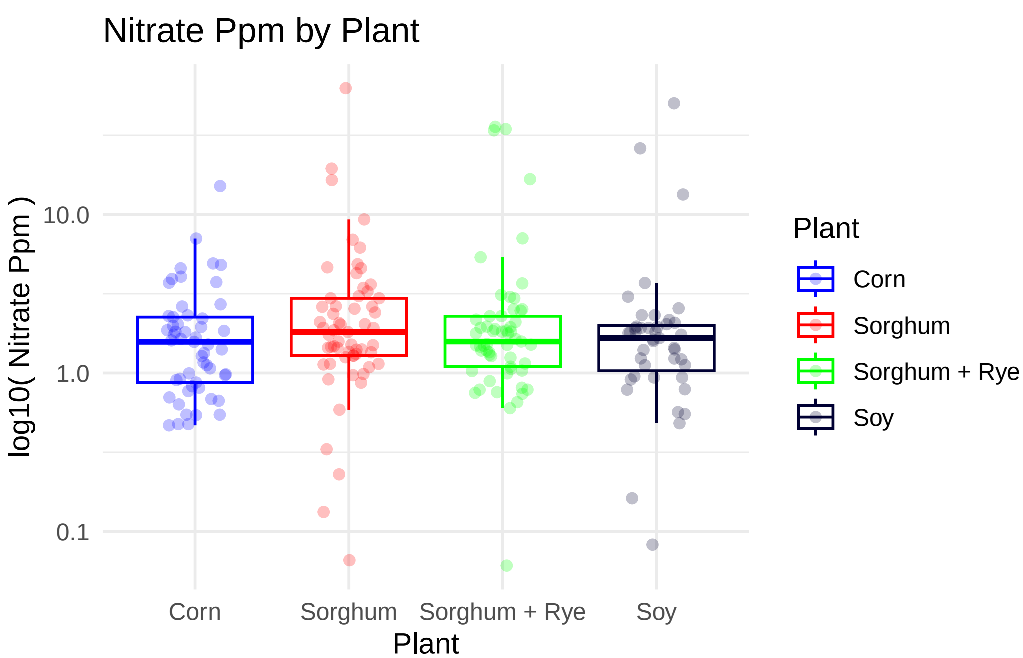
$sampling_location


```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
FEATURE: NITRATE_PPM
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

$plot

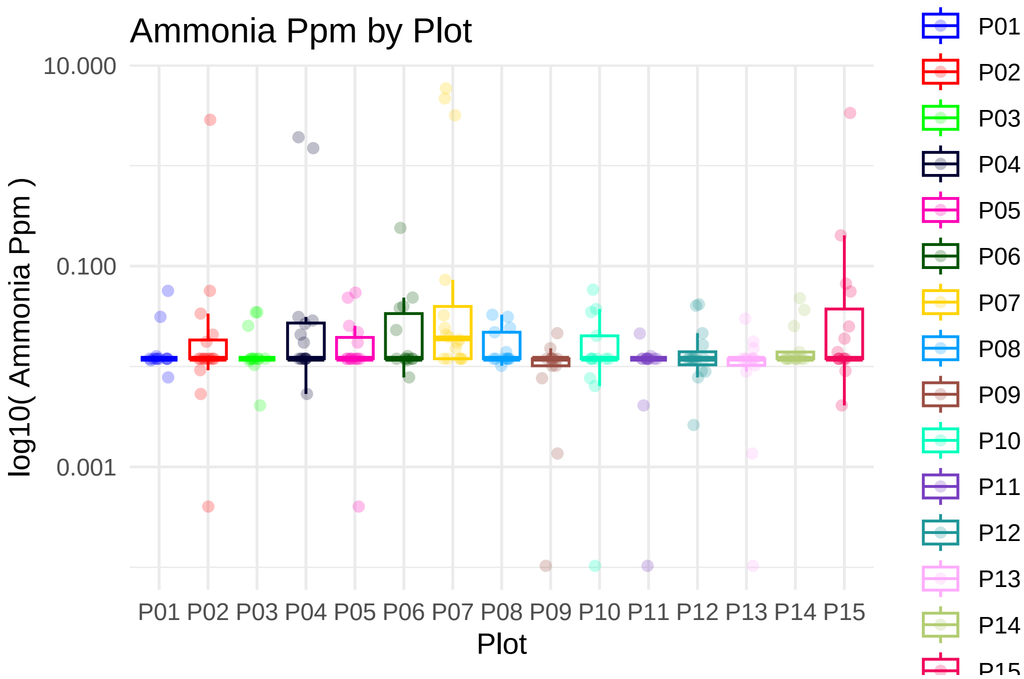

Nitrate Ppm by Plot
log10(Nitrate Ppm)
10.0
1.0
0.1
P01 P02 P03 P04 P05 P06 P07 P08 P09 P10 P11 P12 P13 P14 P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14

$plant


Nitrate Ppm by Plant
log10(Nitrate Ppm)
10.0
1.0
0.1
Corn
Sorghum
Sorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

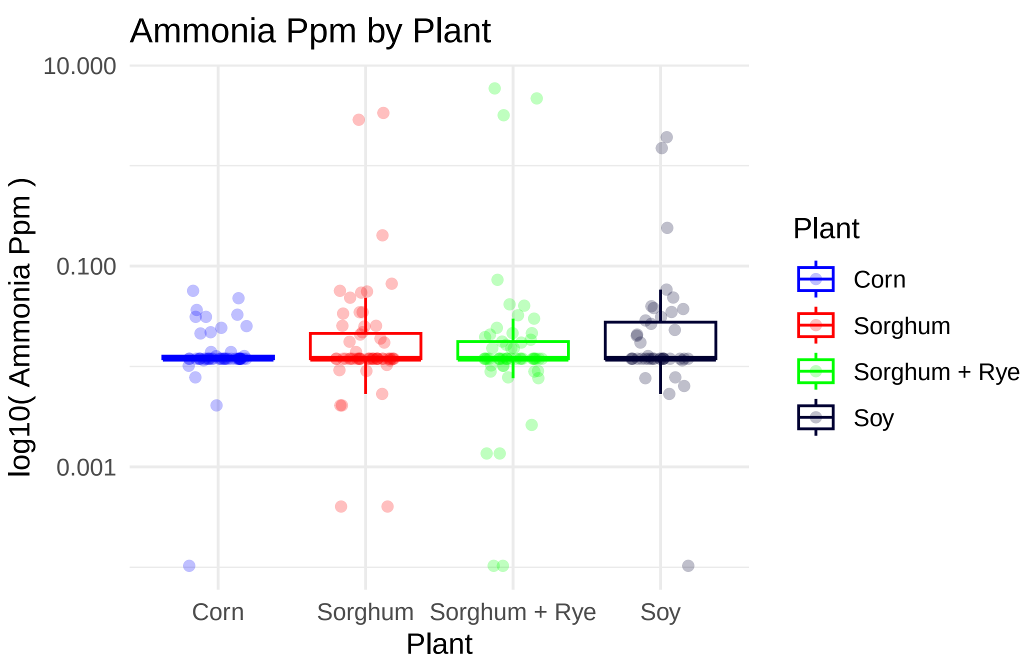
```
$sampling_location
```


```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
FEATURE: AMMONIA_PPM
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$plot
```



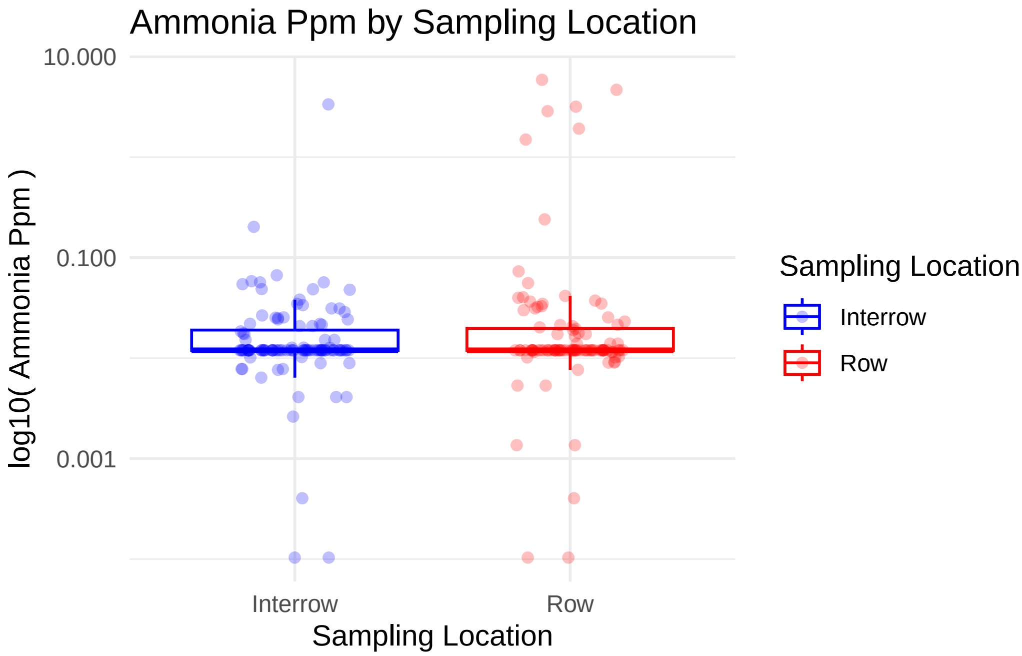
Ammonia Ppm by Plot
10.000
0.100
0.001
log10(Ammonia Ppm)
P01 P02 P03 P04 P05 P06 P07 P08 P09 P10 P11 P12 P13 P14 P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14
P15

$plant

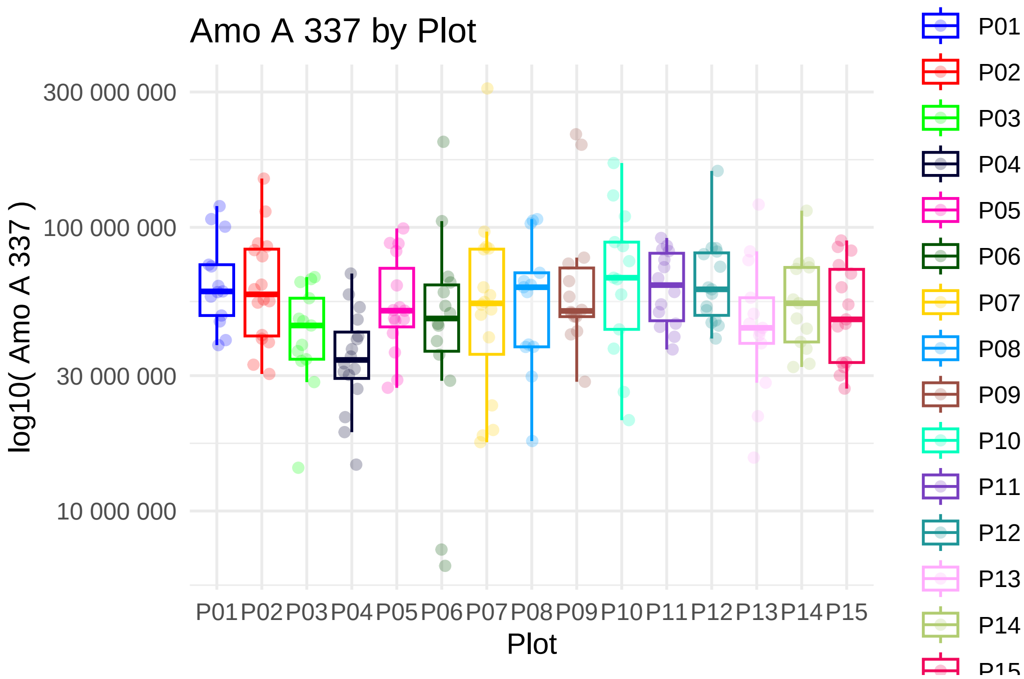


Ammonia Ppm by Plant
10.000
0.100
0.001
log10(Ammonia Ppm)
Corn
Sorghum
Sorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

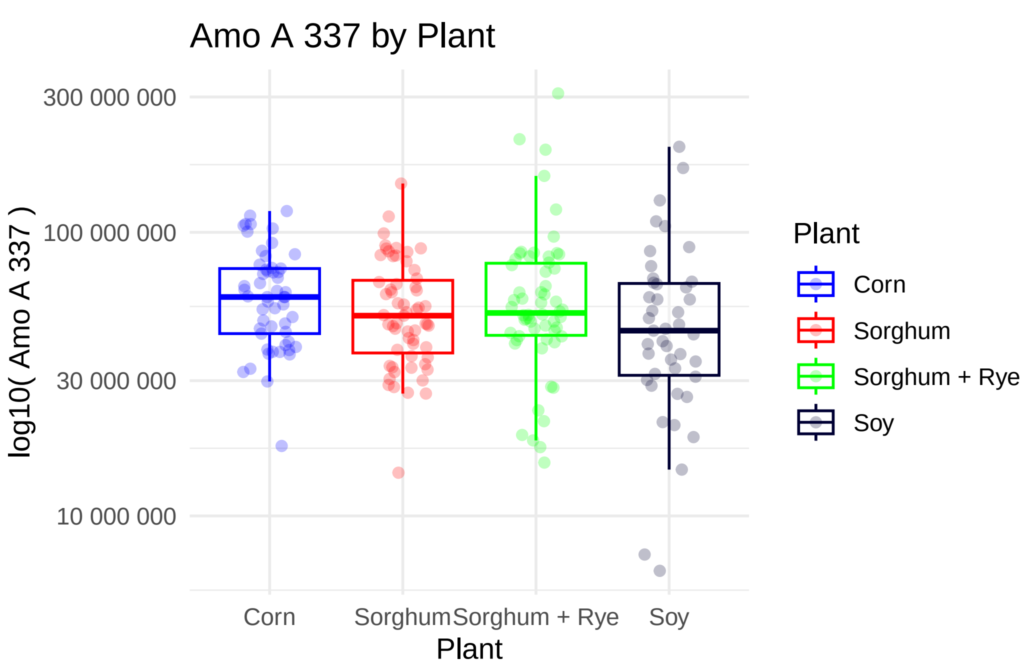
$sampling_location



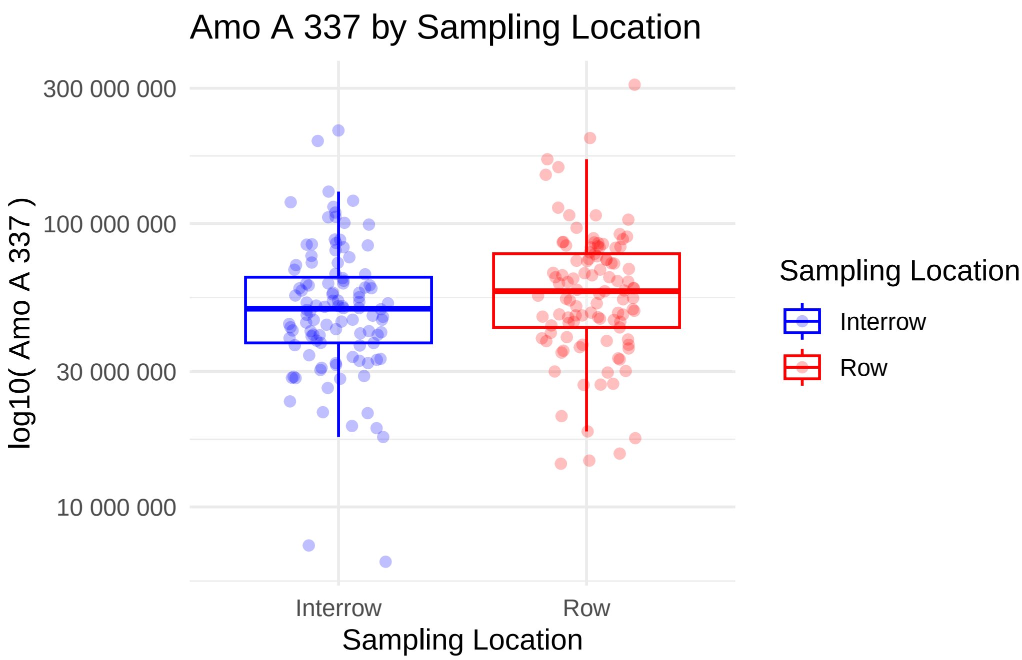
- -
FEATURE: AMO_A_337
- -
$plot


Amo A 337 by Plot
300 000 000
100 000 000
30 000 000
10 000 000
log10(Amo A 337)
P01 P02 P03 P04 P05 P06 P07 P08 P09 P10 P11 P12 P13 P14 P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14
P15

$plant


Amo A 337 by Plant
300 000 000
100 000 000
30 000 000
10 000 000
log10(Amo A 337)
Corn
Sorghum
Sorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

$sampling_location

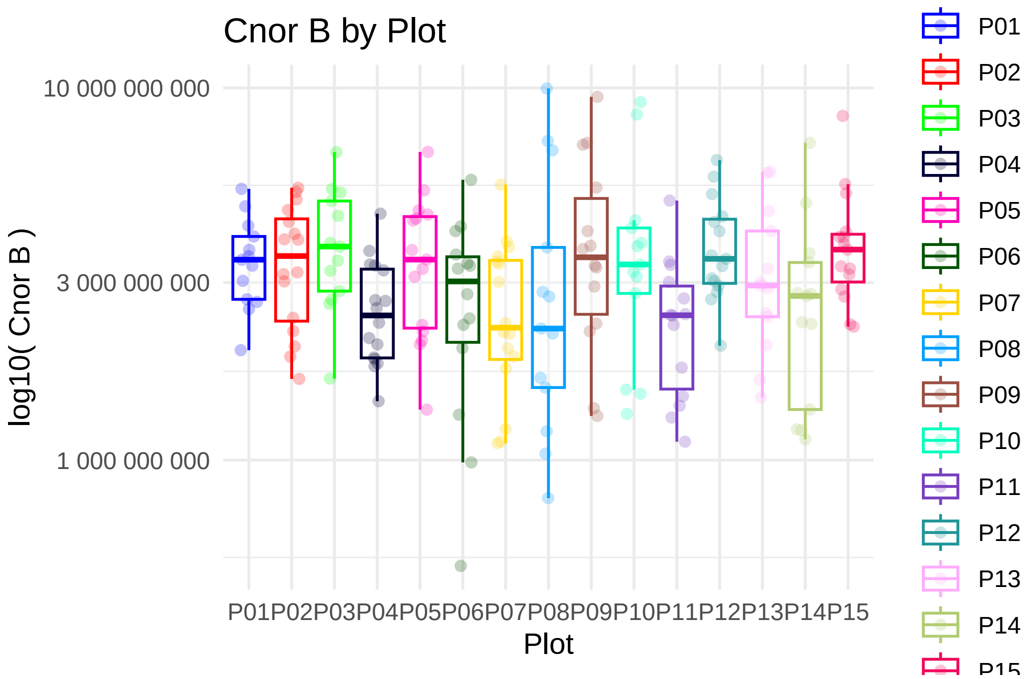


- -

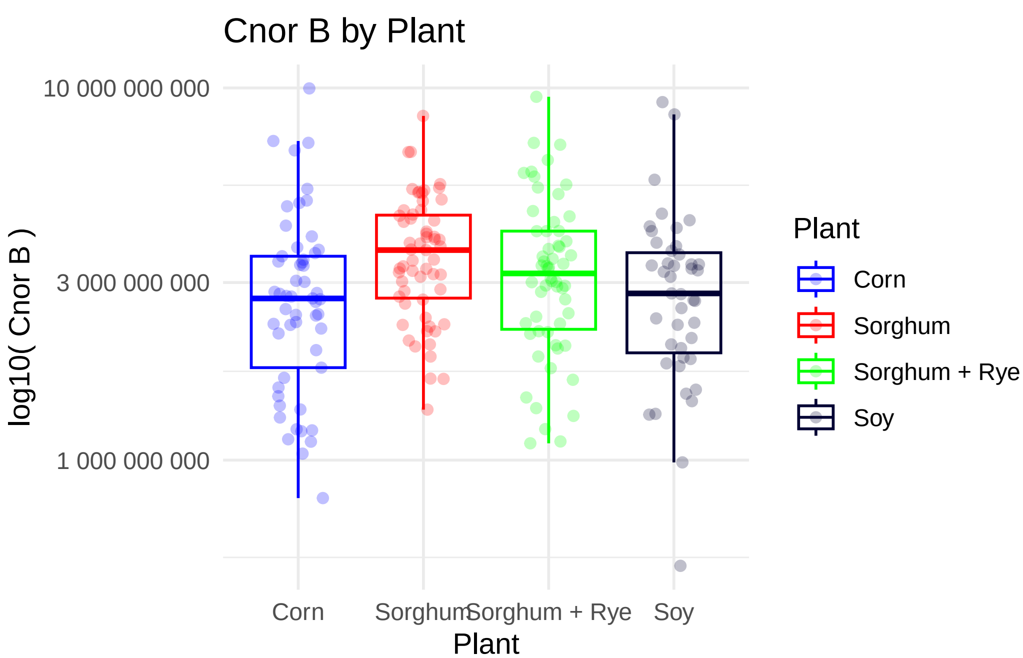
FEATURE: CNOR_B

- -

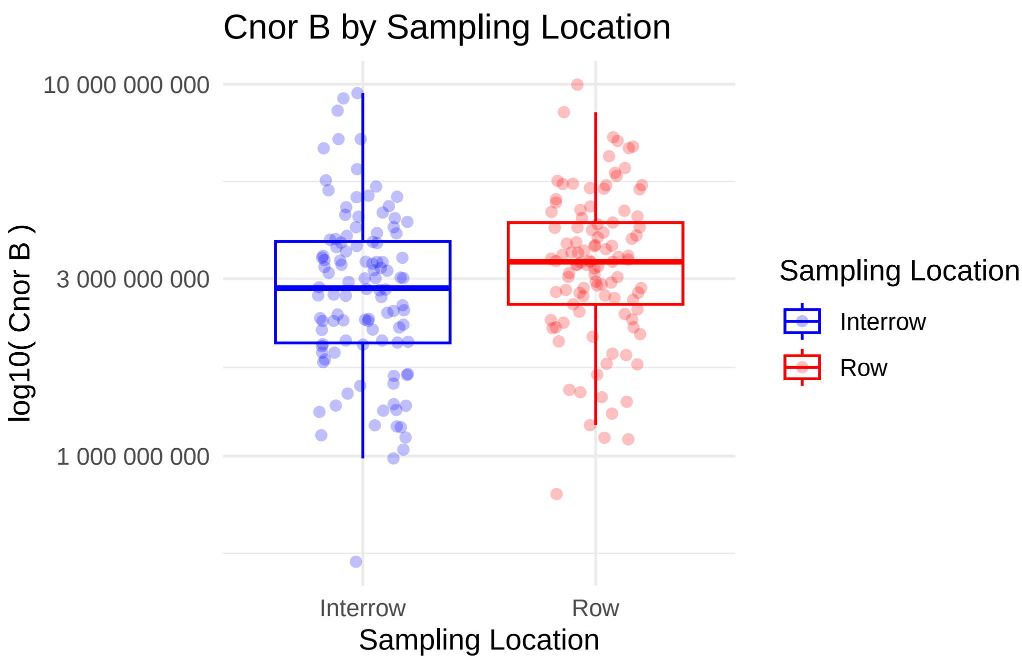
$plot


Cnor B by Plot
10 000 000 000
3 000 000 000
1 000 000 000
log10(Cnor B)
P01P02P03P04P05P06P07P08P09P10P11P12P13P14P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14
P15

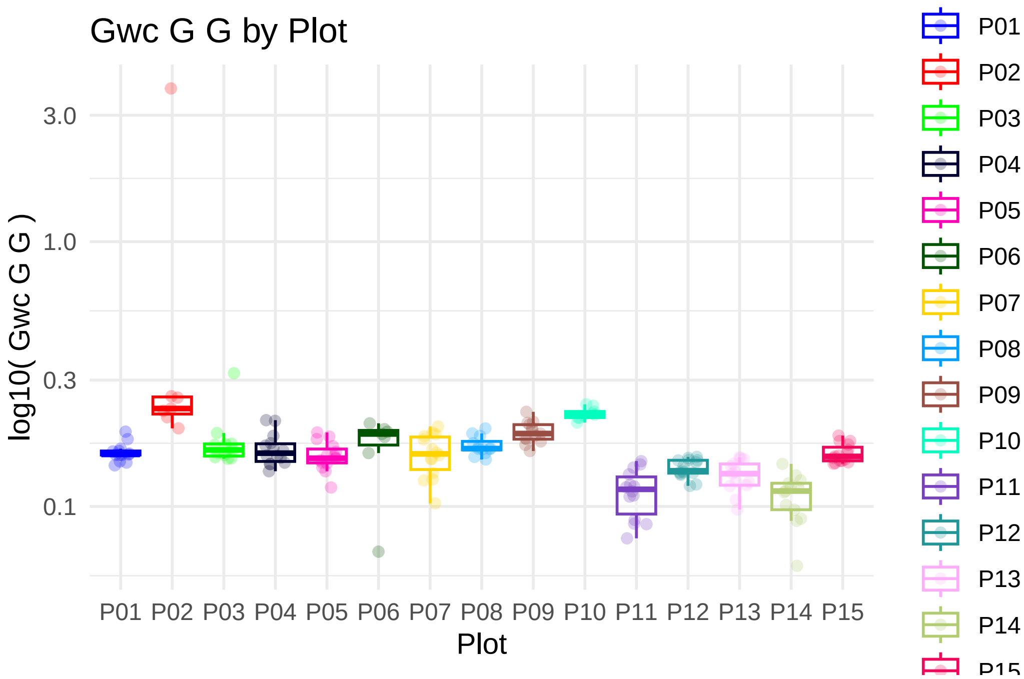
$plant


Cnor B by Plant
10 000 000 000
3 000 000 000
1 000 000 000
log10(Cnor B)
Corn
SorghumSorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

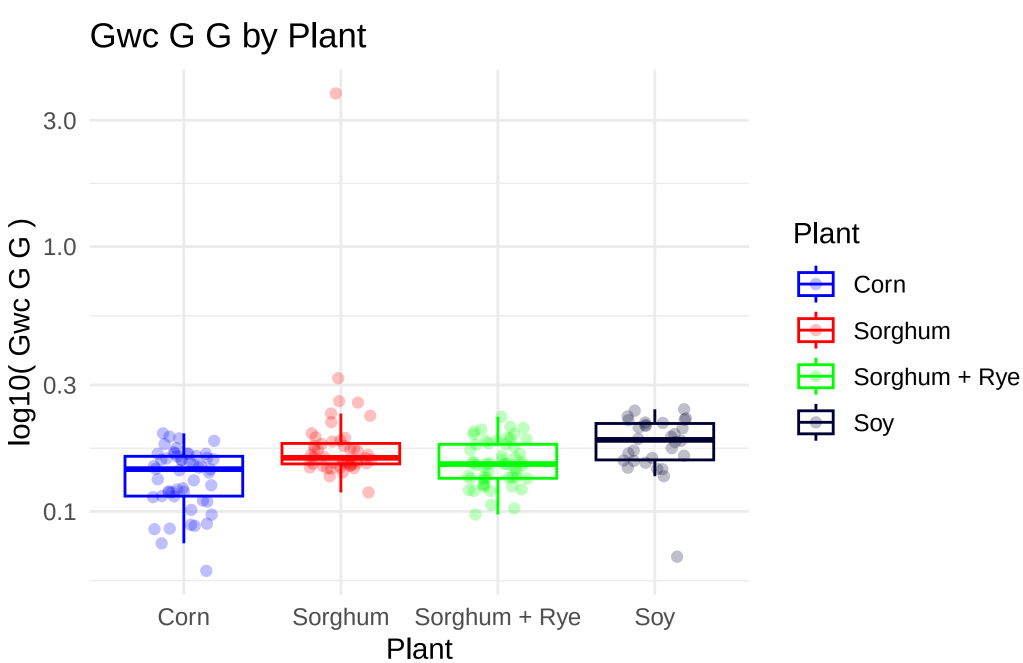
```
$sampling_location
```



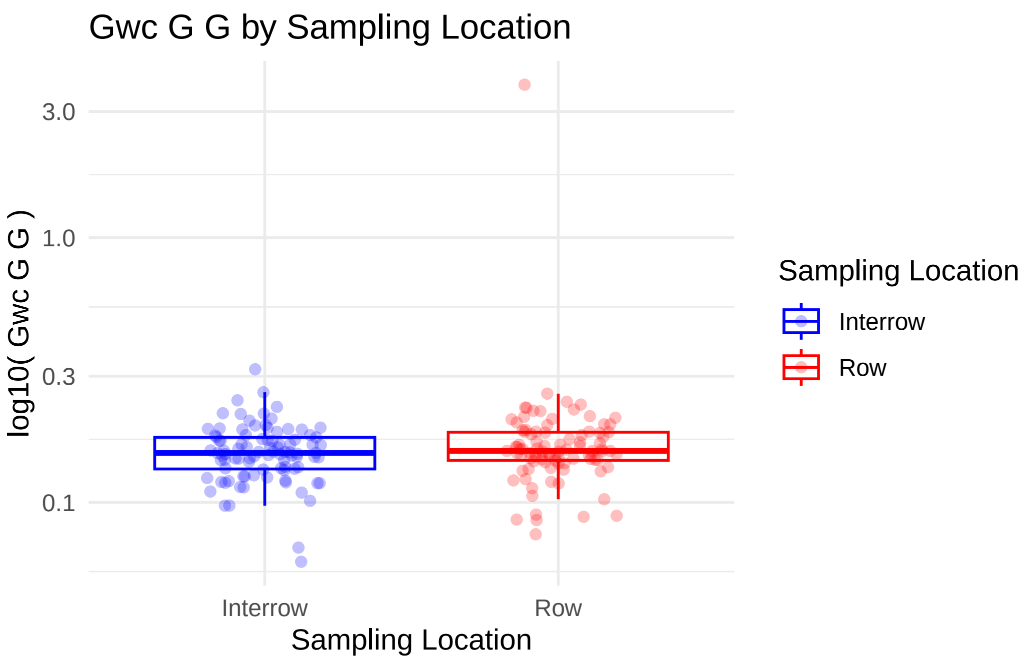

```
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
FEATURE: GWC_G_G
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$plot
```


Gwc G G by Plot
log10(Gwc G G)
3.0
1.0
0.3
0.1
P01 P02 P03 P04 P05 P06 P07 P08 P09 P10 P11 P12 P13 P14 P15
Plot
P01
P02
P03
P04
P05
P06
P07
P08
P09
P10
P11
P12
P13
P14
P15

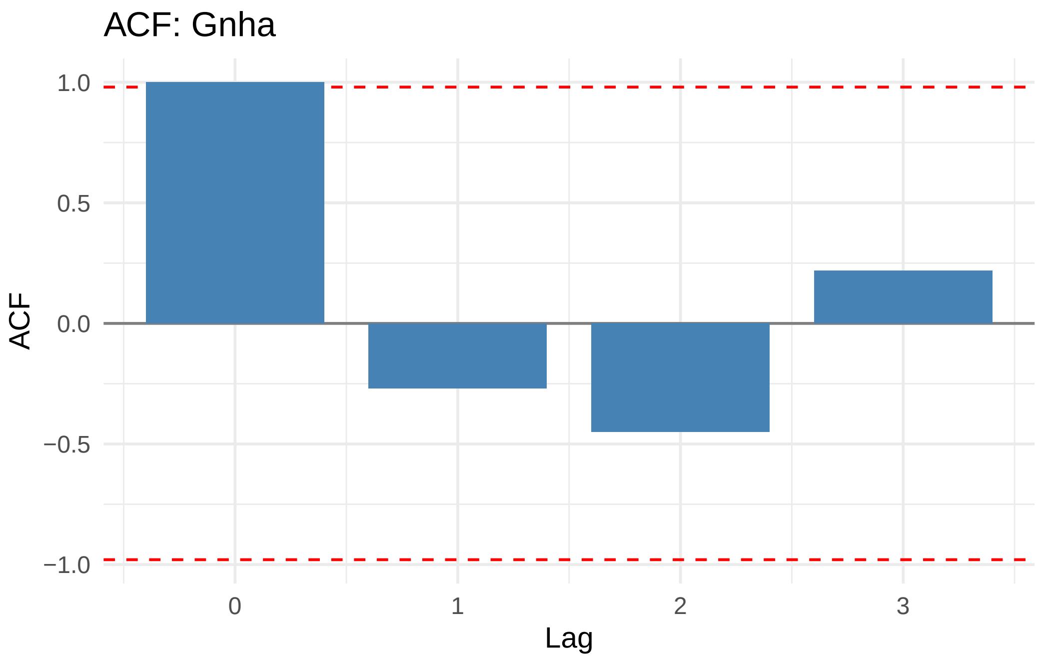
$plant


Gwc G G by Plant
log10(Gwc G G)
3.0
1.0
0.3
0.1
Corn
Sorghum
Sorghum + Rye
Soy
Plant
Plant
Corn
Sorghum
Sorghum + Rye
Soy

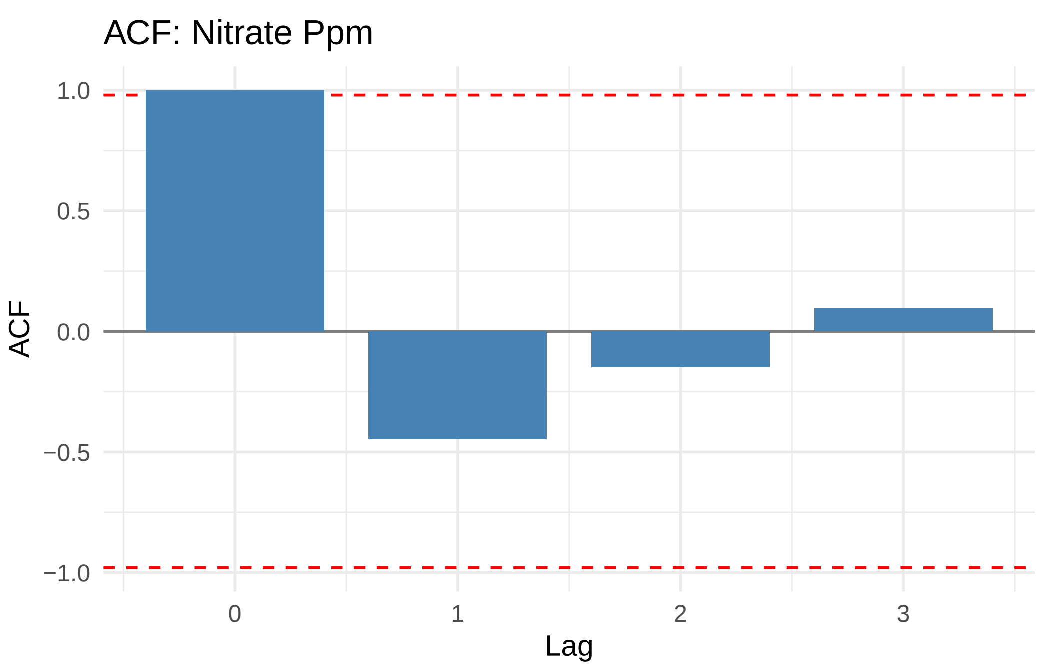
```
$sampling_location
```



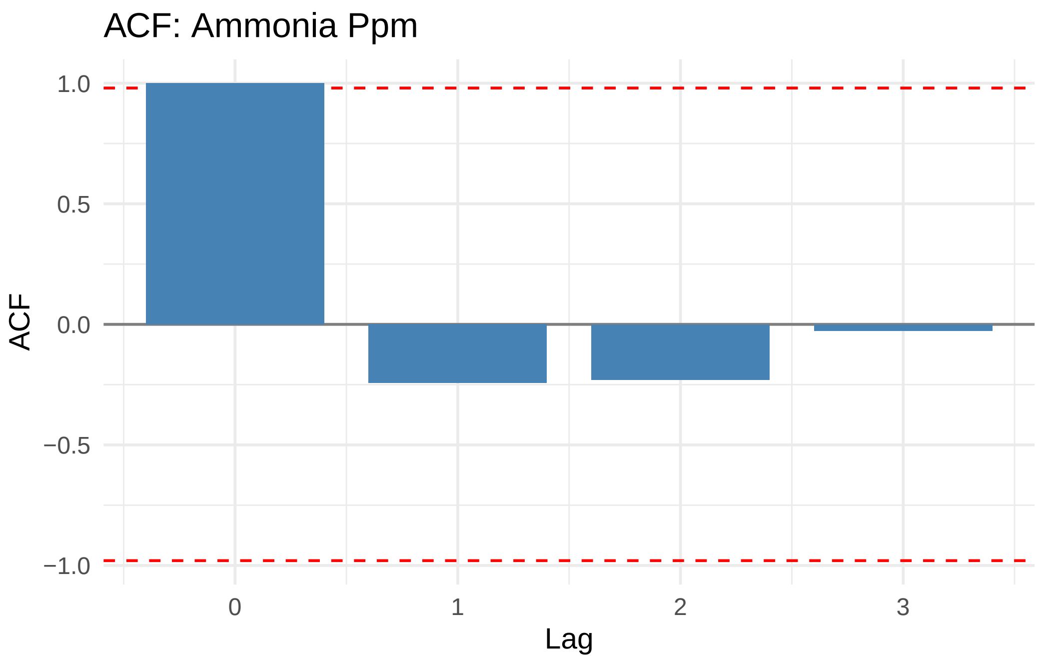

```
> acf_global <- map_dfr(features, function(feat) {
+     series <- acf_preprocess(main_rarefied_df, feature = feat,
+         time_col = "sampling_da ..." ... [TRUNCATED]

> acf_global <- acf_global %>% distinct(feature) %>%
+     mutate(plots = map(feature, ~acf_plots(df = main_rarefied_df,
+         feature = .x, tim .... [TRUNCATED]

> acf_global$plots
[[1]]
```


ACF: Gnha
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

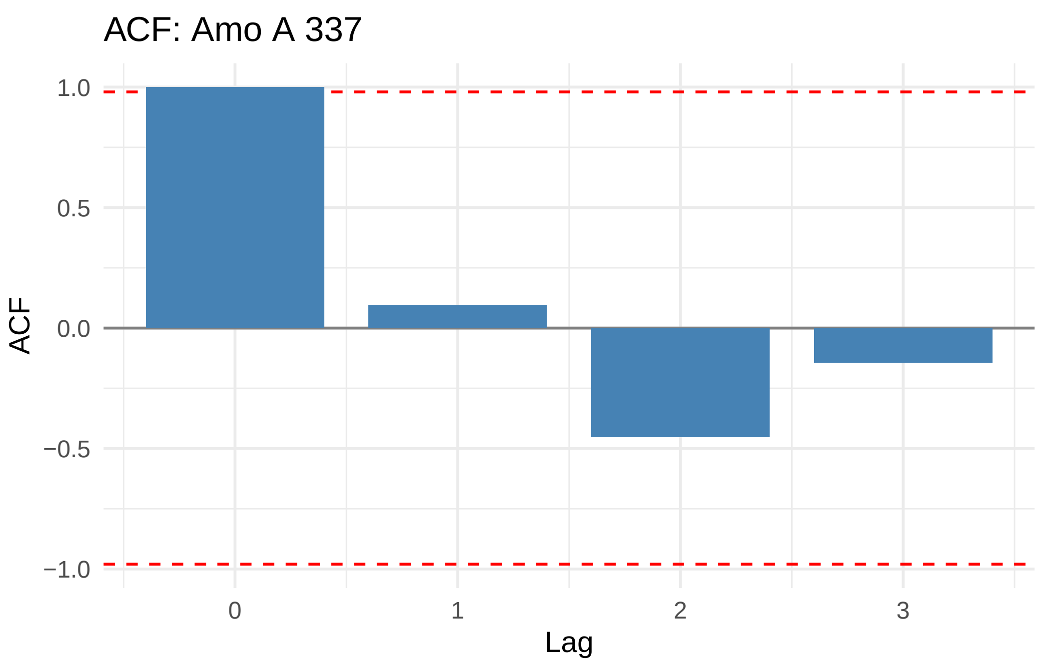
[[2]]


ACF: Nitrate Ppm
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

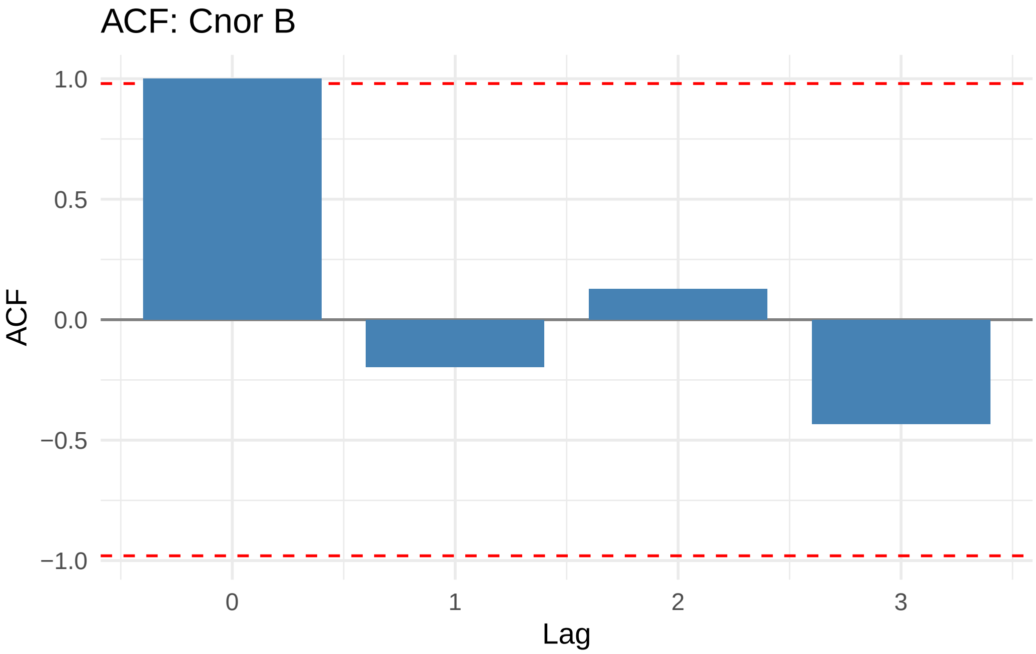
[[3]]


ACF: Ammonia Ppm
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

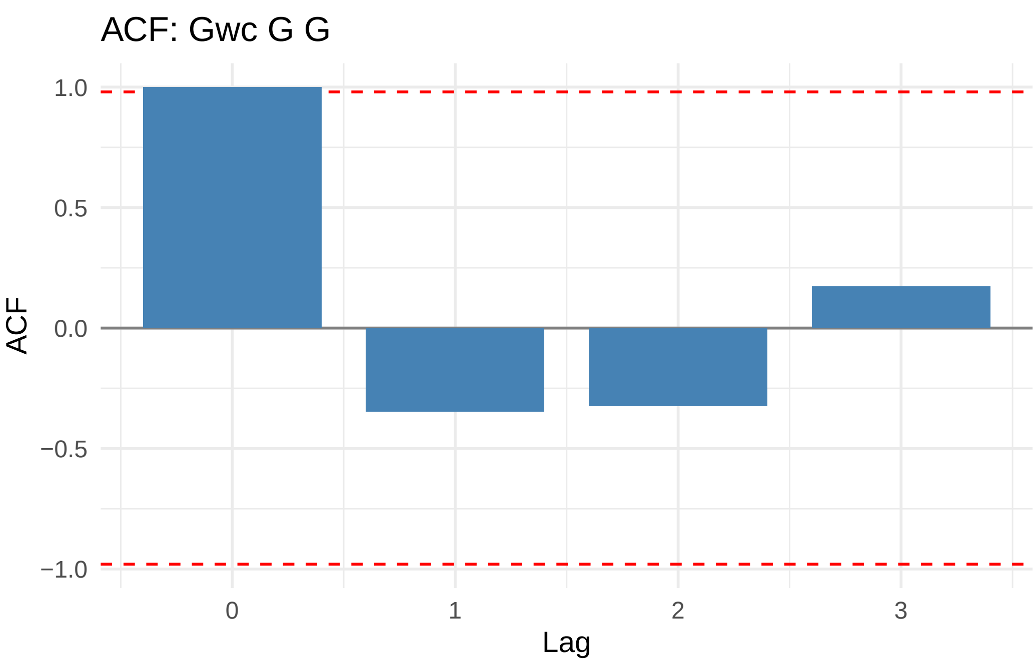
[[4]]


ACF: Amo A 337
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

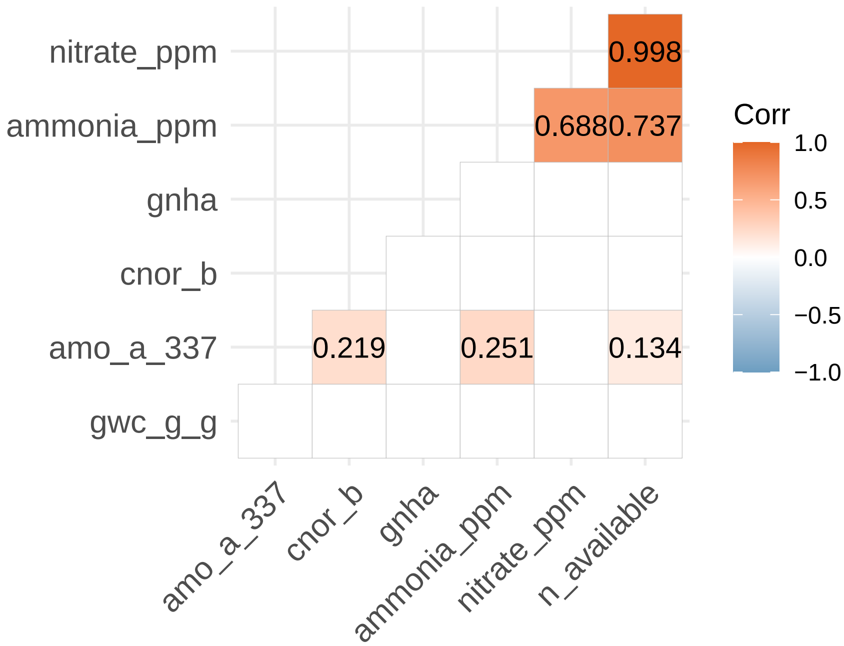
[[5]]


ACF: Cnor B
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

[[6]]


ACF: Gwc G G
1.0
0.5
0.0
−0.5
−1.0
ACF
0
1
2
3
Lag

```
> acf_by_group <- map_dfr(parameters, function(par) {
+     if (!par %in% names(main_rarefied_df)) {
+         return(tibble())
+     }
+     main_rar .... [TRUNCATED]

> metadata_matrix <- select(sabr_2023_metadata_clean,
+     c(gnha, amo_a_337:gwc_g_g))

> cor(metadata_matrix, use = "complete.obs", method = "pearson")
                  gnha  amo_a_337      cnor_b nitrate_ppm ammonia_ppm
gnha         1.00000000 0.01097846 -0.07158056  0.09104686 -0.01516781
amo_a_337    0.01097846 1.00000000  0.21915506  0.11832293  0.25119533
cnor_b      -0.07158056 0.21915506  1.00000000 -0.07982804 -0.06843006
nitrate_ppm  0.09104686 0.11832293 -0.07982804  1.00000000  0.68829633
ammonia_ppm -0.01516781 0.25119533 -0.06843006  0.68829633  1.00000000
n_available  0.08333207 0.13437491 -0.08093123  0.99755883  0.73727365
gwc_g_g     -0.01101017 0.05034588  0.03745435 -0.01970619 -0.02053414
            n_available     gwc_g_g
gnha         0.08333207 -0.01101017
amo_a_337    0.13437491  0.05034588
cnor_b      -0.08093123  0.03745435
nitrate_ppm  0.99755883 -0.01970619
ammonia_ppm  0.73727365 -0.02053414
n_available  1.00000000 -0.02032907
gwc_g_g     -0.02032907  1.00000000

> ggcorrplot::ggcorrplot(cor(metadata_matrix, use = "complete.obs",
+     method = "pearson"), method = "square", hc.order = TRUE,
+     lab = TRUE, .... [TRUNCATED]
```

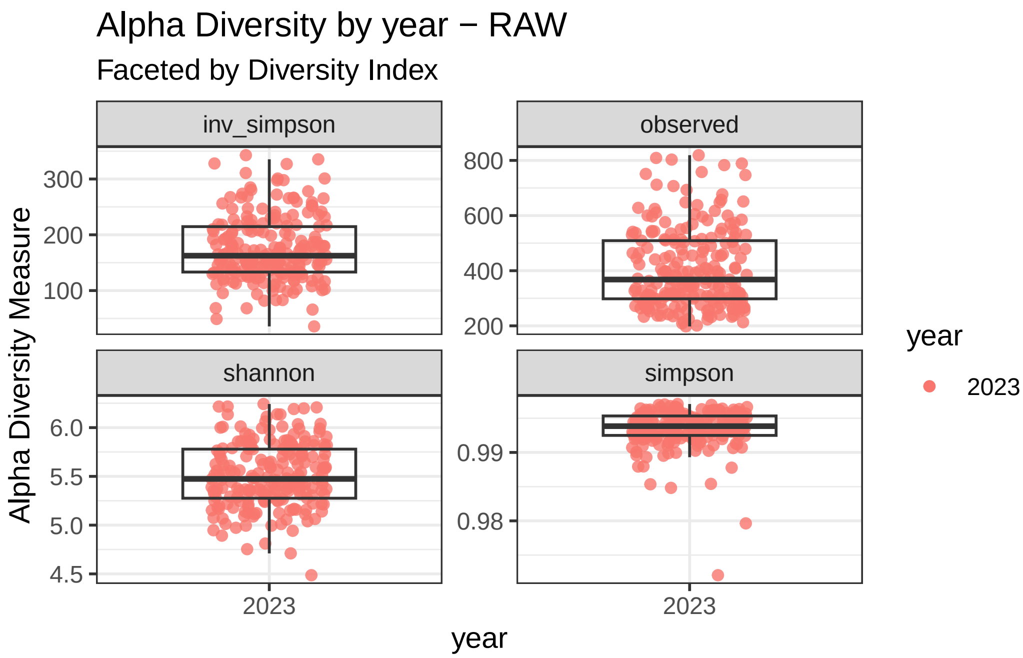


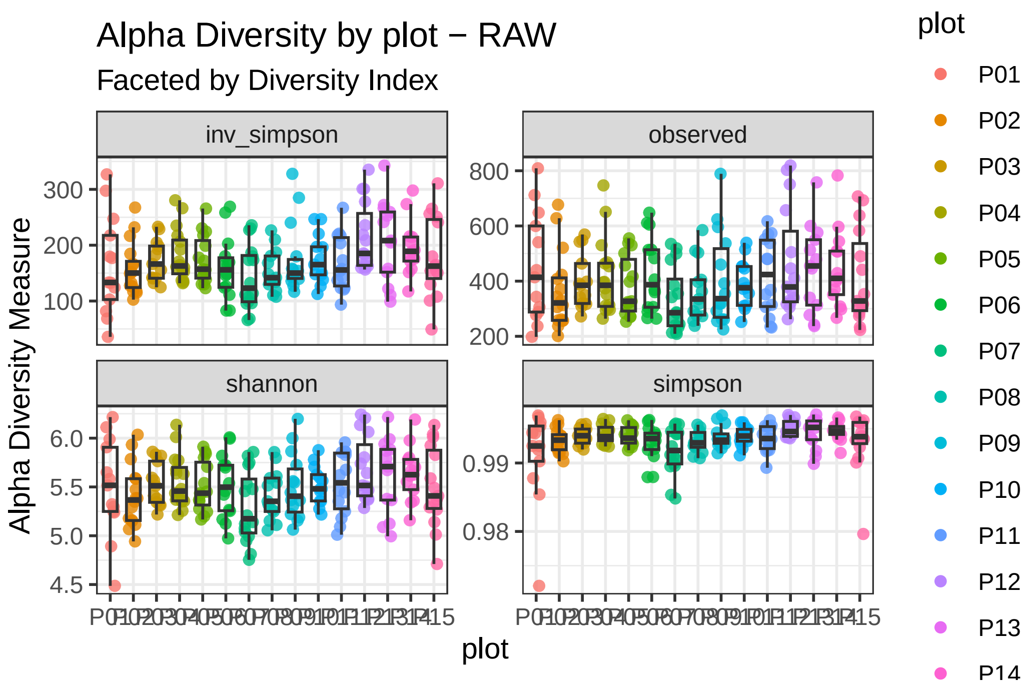
Section 2 — Alpha Diversity Analyses

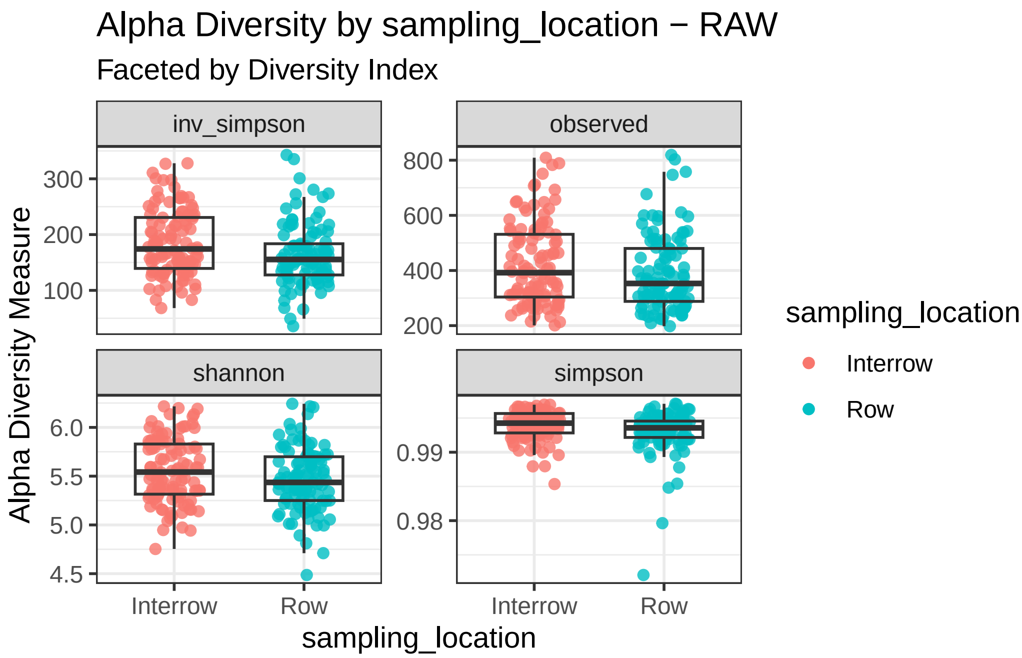
This section runs 004_alpha_diversity.R, including:

- Richness and alpha diversity summaries
- Alpha diversity vs. categorical features
- Simple linear regressions of alpha diversity vs. continuous features

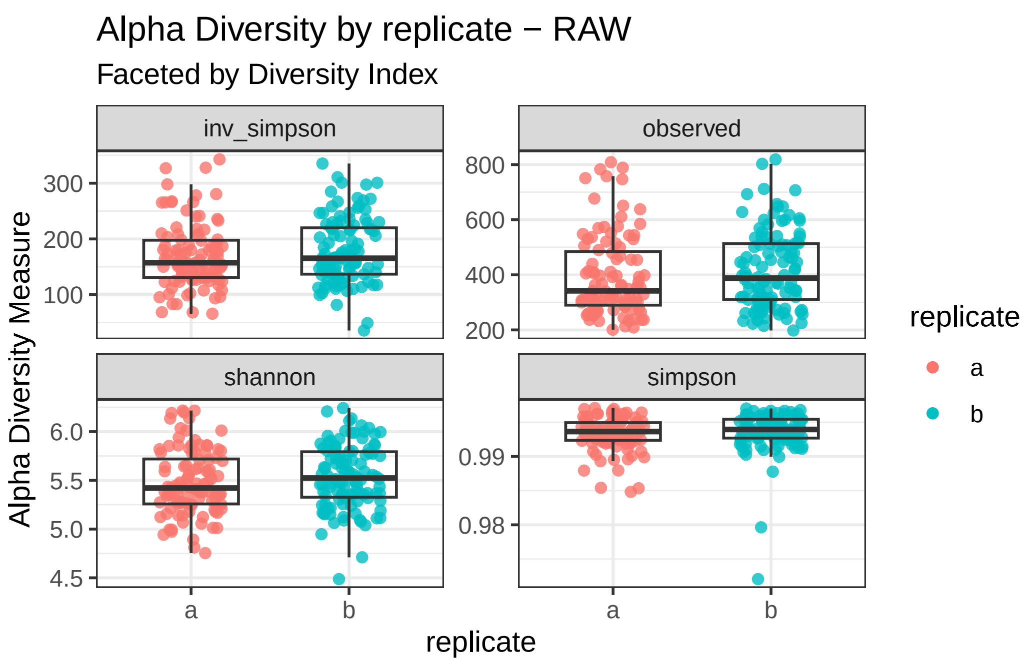
```
source("R/004_alpha_diversity.R", local = knitr::knit_global())
```



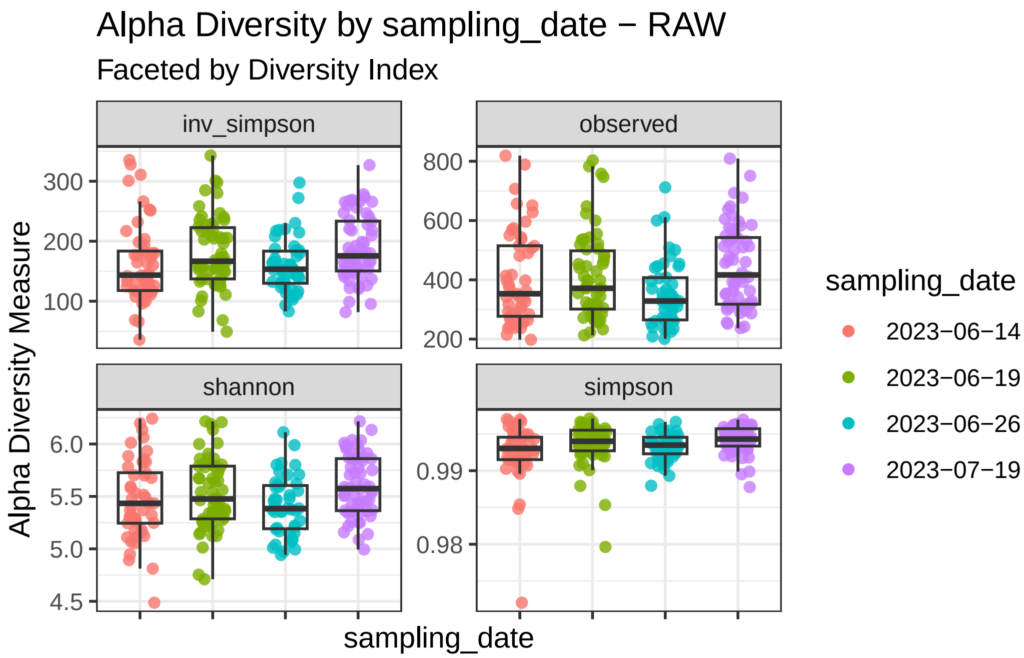


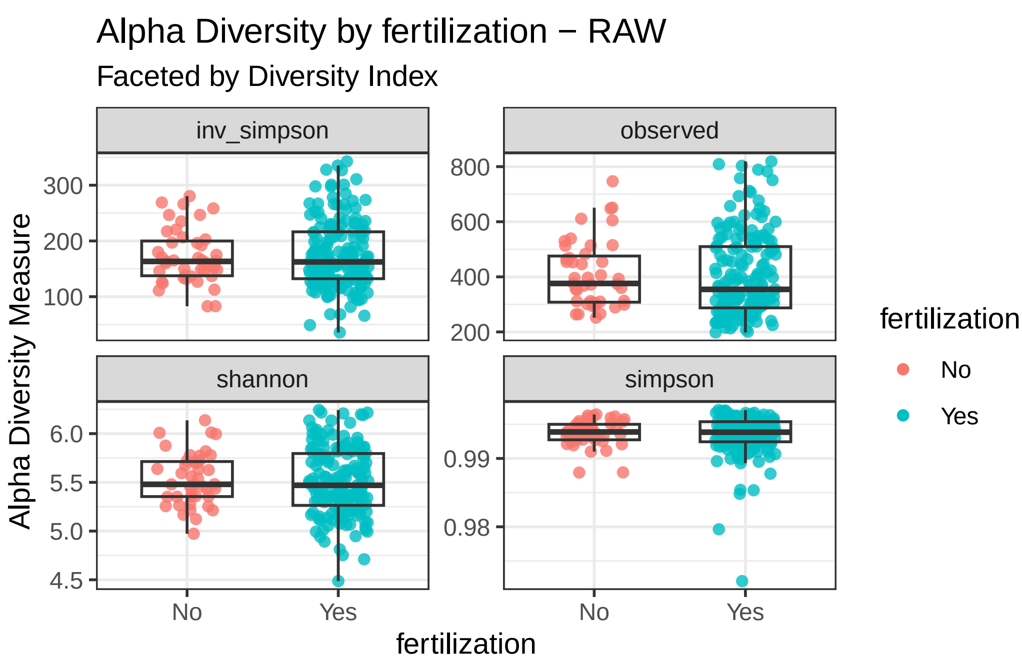


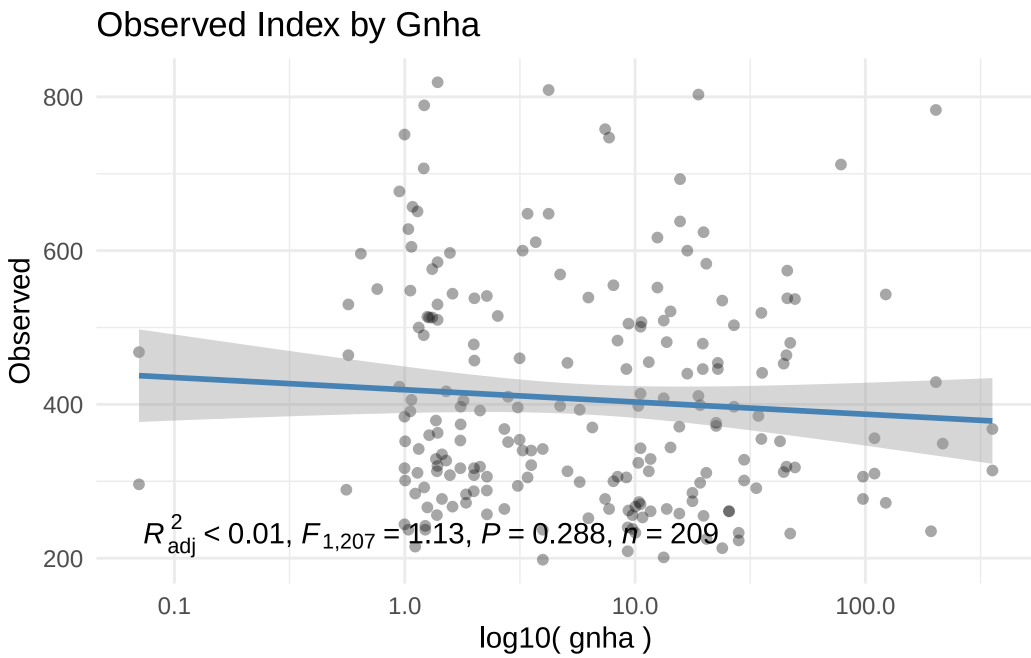
Alpha Diversity by sampling_location − RAW
Faceted by Diversity Index
inv_simpson
observed
shannon
simpson
Alpha Diversity Measure
300
200
100
800
600
400
200
6.0
5.5
5.0
4.5
0.99
0.98
Interrow
Row
Interrow
Row
sampling_location
sampling_location
Interrow
Row



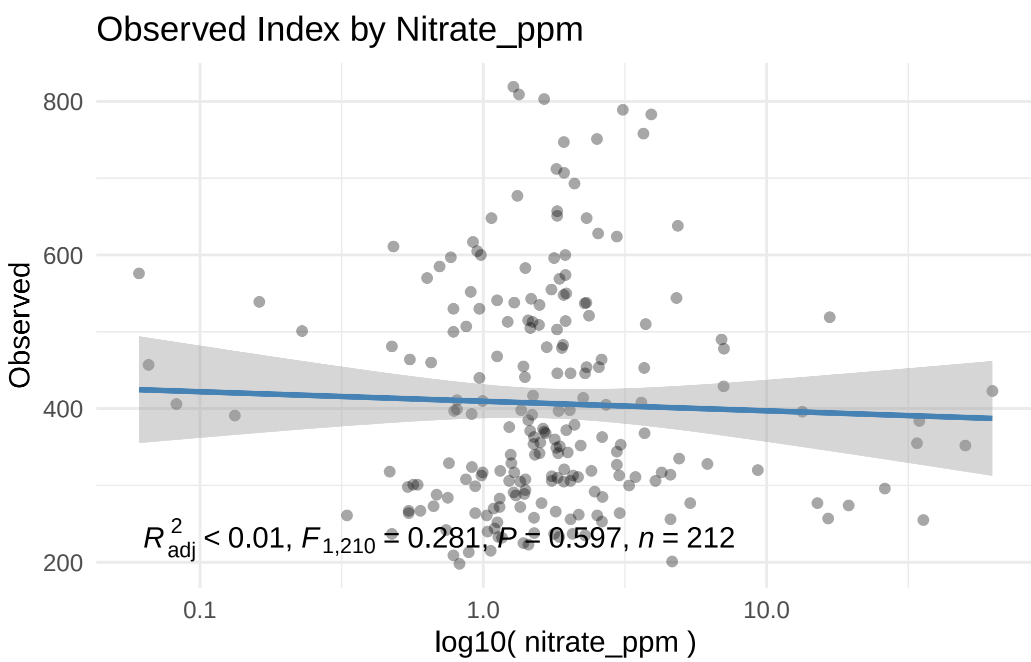
Alpha Diversity by replicate − RAW
Faceted by Diversity Index
inv_simpson
observed
shannon
simpson
Alpha Diversity Measure
300
200
100
800
600
400
200
6.0
5.5
5.0
4.5
0.99
0.98
a
b
a
b
replicate
replicate
a
b


Alpha Diversity by sampling_date − RAW
Faceted by Diversity Index
inv_simpson
observed
shannon
simpson
Alpha Diversity Measure
sampling_date
2023−06−14
2023−06−19
2023−06−26
2023−07−19

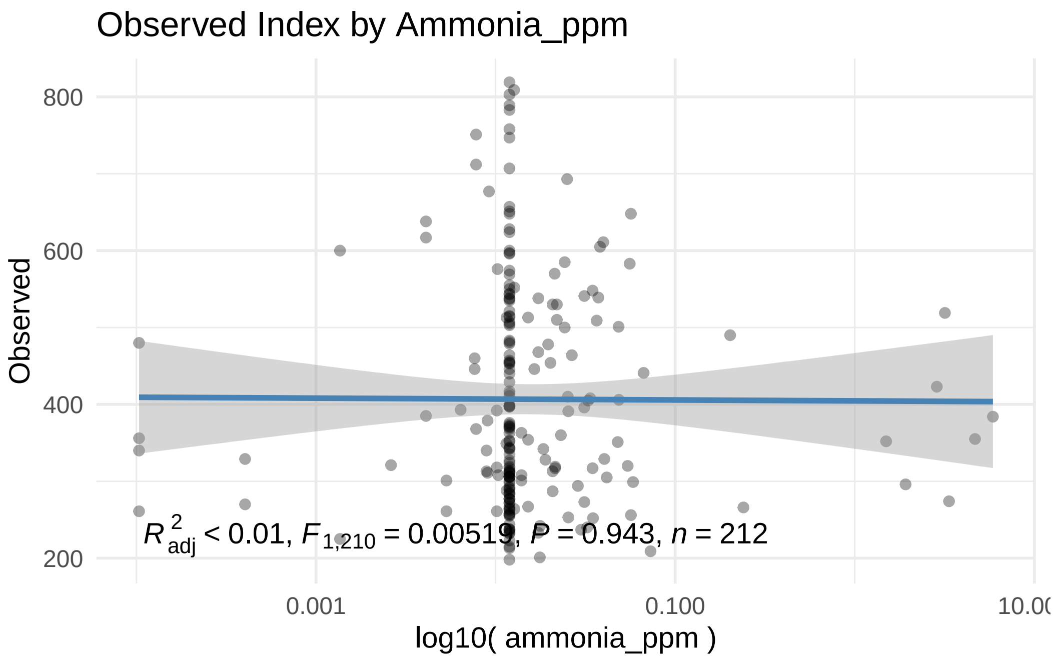

Alpha Diversity by fertilization − RAW
Faceted by Diversity Index
inv_simpson
observed
shannon
simpson
Alpha Diversity Measure
fertilization
No
Yes

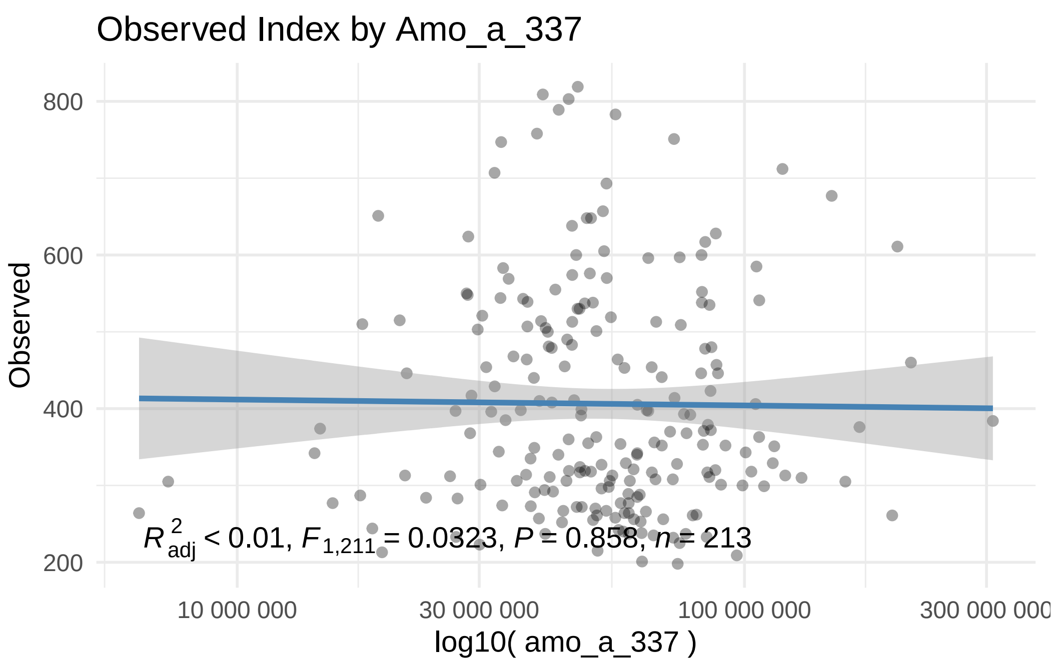


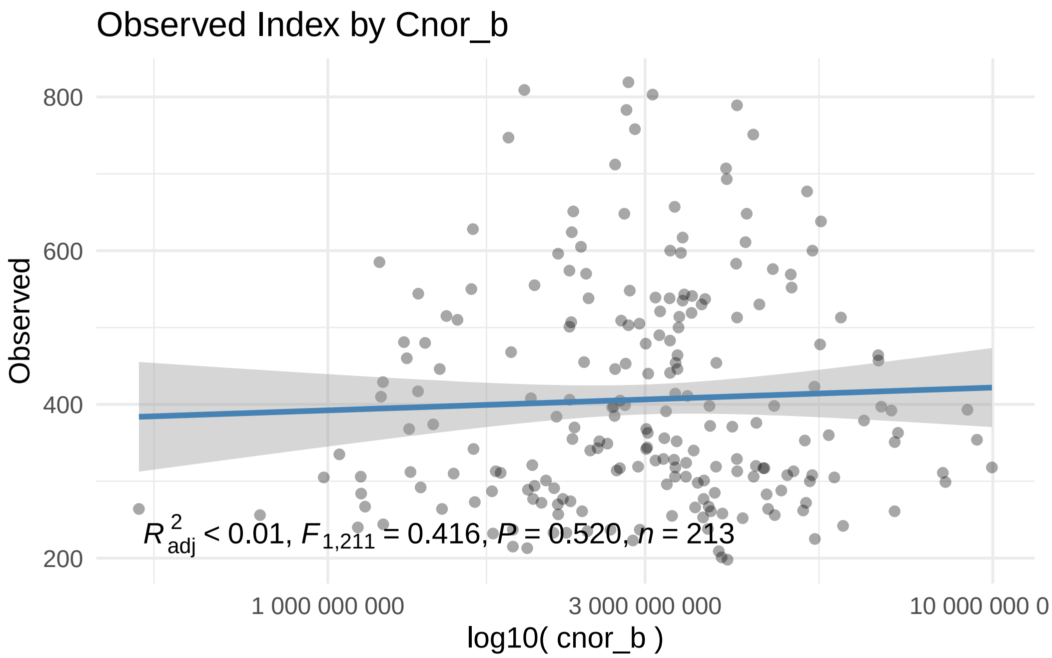
Observed Index by Gnha
Observed
800
600
400
200
0.1
1.0
10.0
100.0
log10(gnha)
$R^2_{adj} < 0.01$, $F_{1,207} = 1.13$, $P = 0.288$, $n = 209$

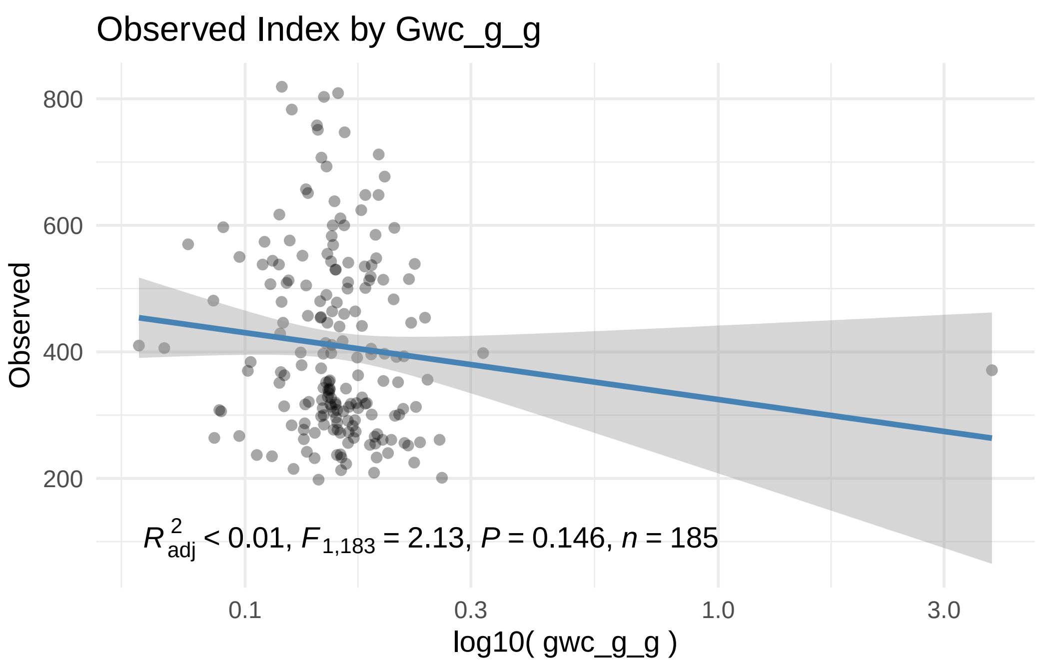


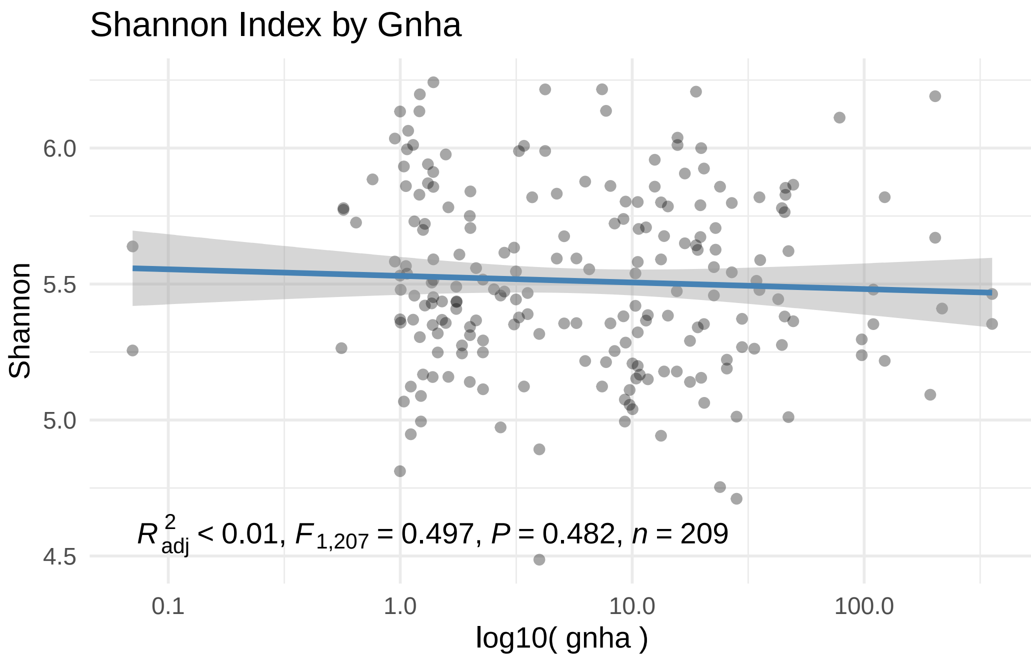
Observed Index by Nitrate_ppm
Observed
800
600
400
200
0.1
1.0
10.0
log10(nitrate_ppm)
$R^2_{adj} < 0.01$, $F_{1,210} = 0.281$, $P = 0.597$, $n = 212$

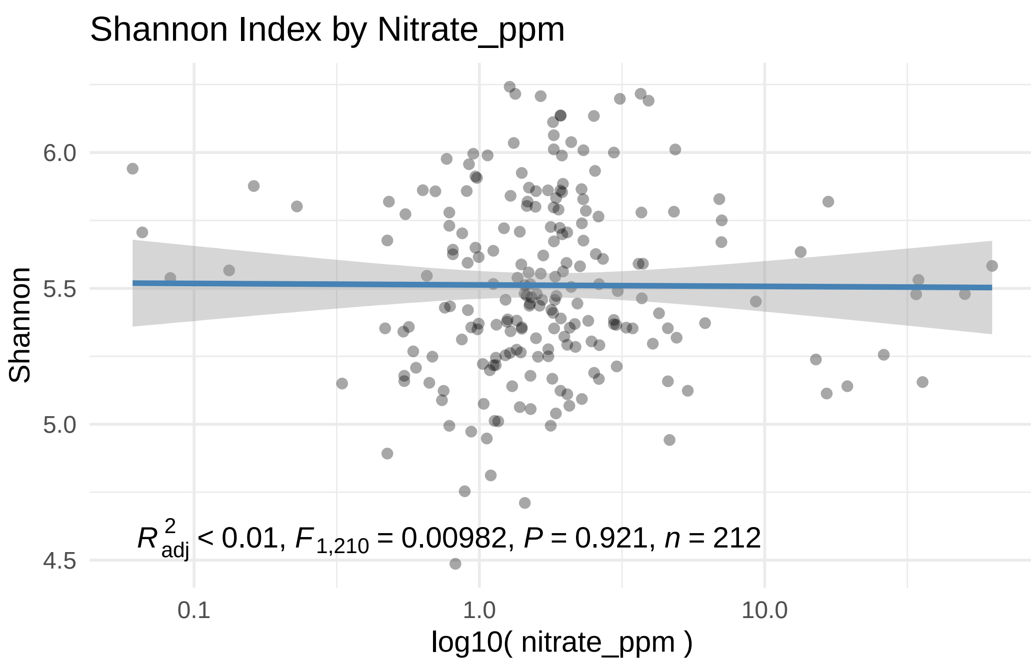

Observed Index by Ammonia_ppm
Observed
800
600
400
200
R^2_{adj} < 0.01, $F_{1,210}$ = 0.00519, P = 0.943, n = 212
0.001
0.100
10.00
log10(ammonia_ppm)

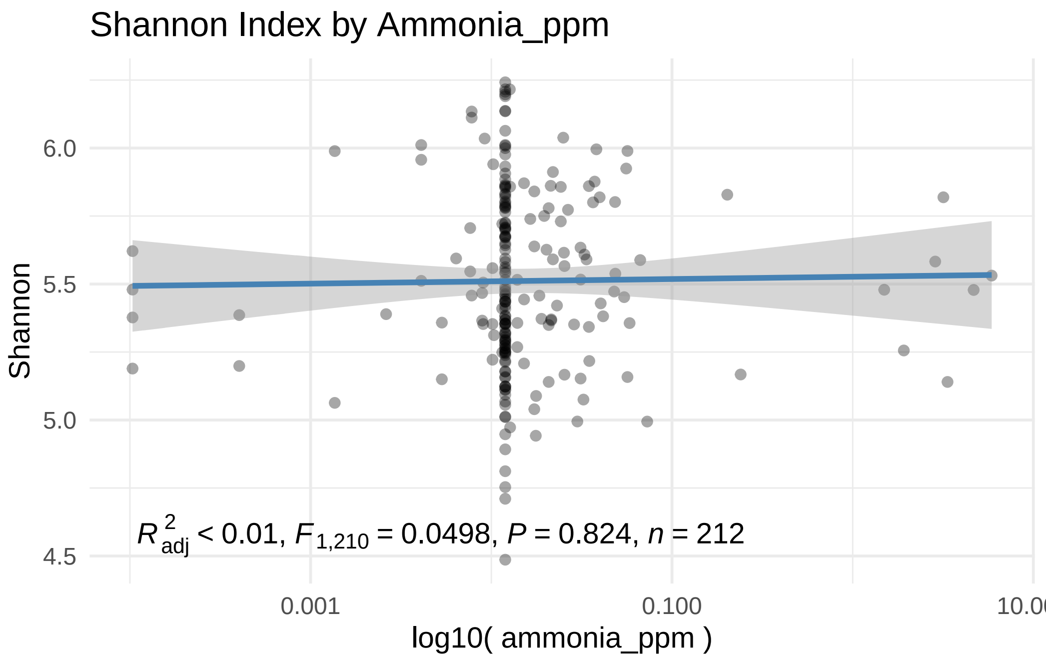

Observed Index by Amo_a_337
Observed
800
600
400
200
R^2_{adj} < 0.01, $F_{1,211}$ = 0.0323, P = 0.858, n = 213
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)


Observed Index by Cnor_b
Observed
800
600
400
200
$R^2_{adj} < 0.01$, $F_{1,211} = 0.416$, $P = 0.520$, $n = 213$
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)

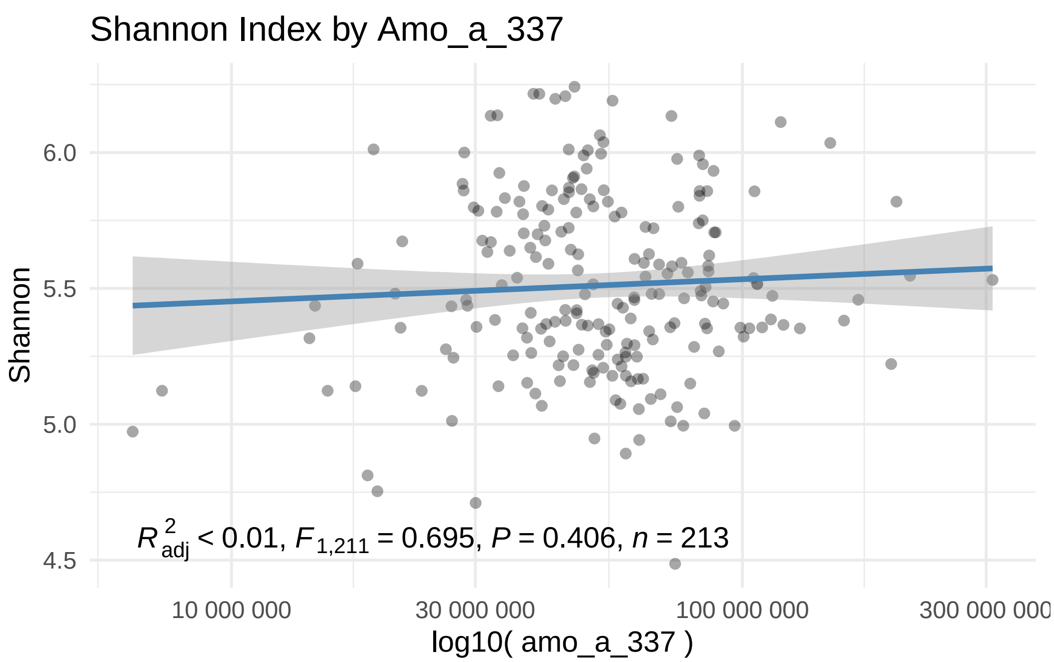

Observed Index by Gwc_g_g
Observed
800
600
400
200
$R^2_{adj} < 0.01$, $F_{1,183} = 2.13$, $P = 0.146$, $n = 185$
0.1
0.3
1.0
3.0
log10(gwc_g_g)


Shannon Index by Gnha
Shannon
6.0
5.5
5.0
4.5
R 2 adj < 0.01, F 1,207 = 0.497, P = 0.482, n = 209
0.1
1.0
10.0
100.0
log10(gnha)


Shannon Index by Nitrate_ppm
Shannon
6.0
5.5
5.0
4.5
R 2 adj < 0.01, F 1,210 = 0.00982, P = 0.921, n = 212
0.1
1.0
10.0
log10(nitrate_ppm)

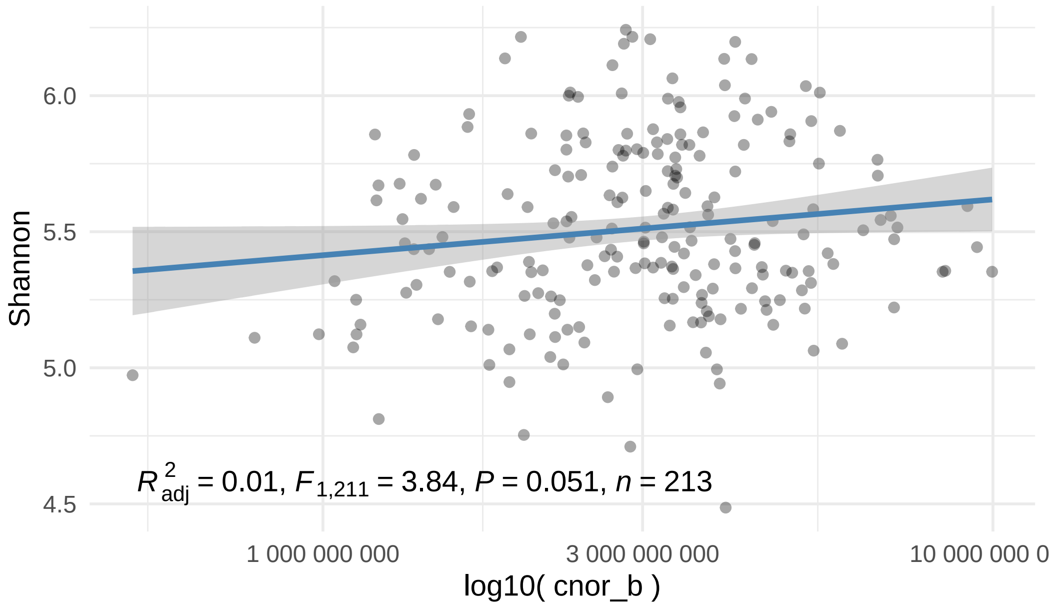


Shannon Index by Ammonia_ppm
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0498$, $P = 0.824$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)

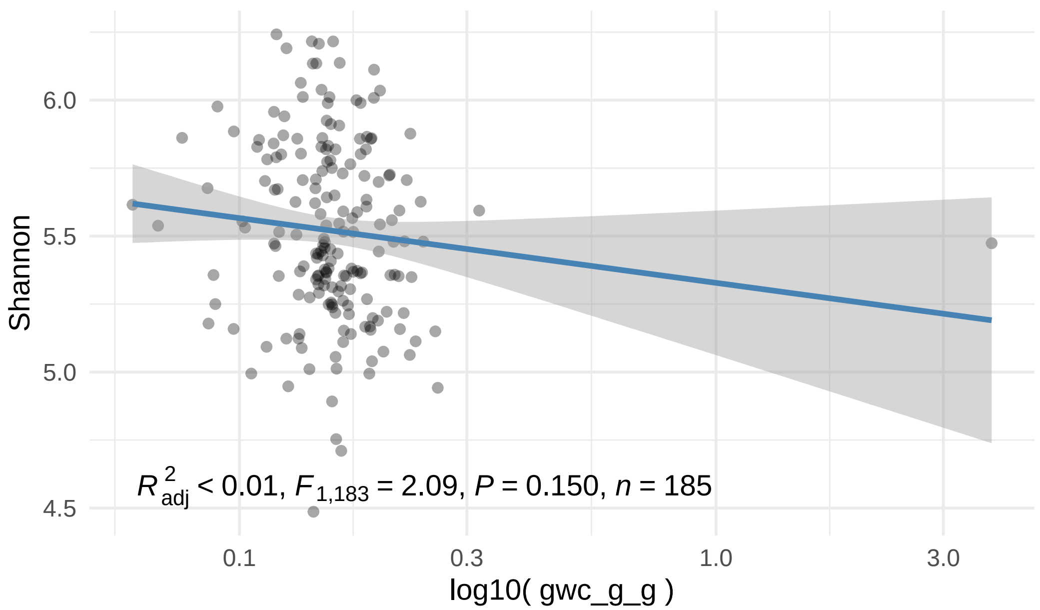


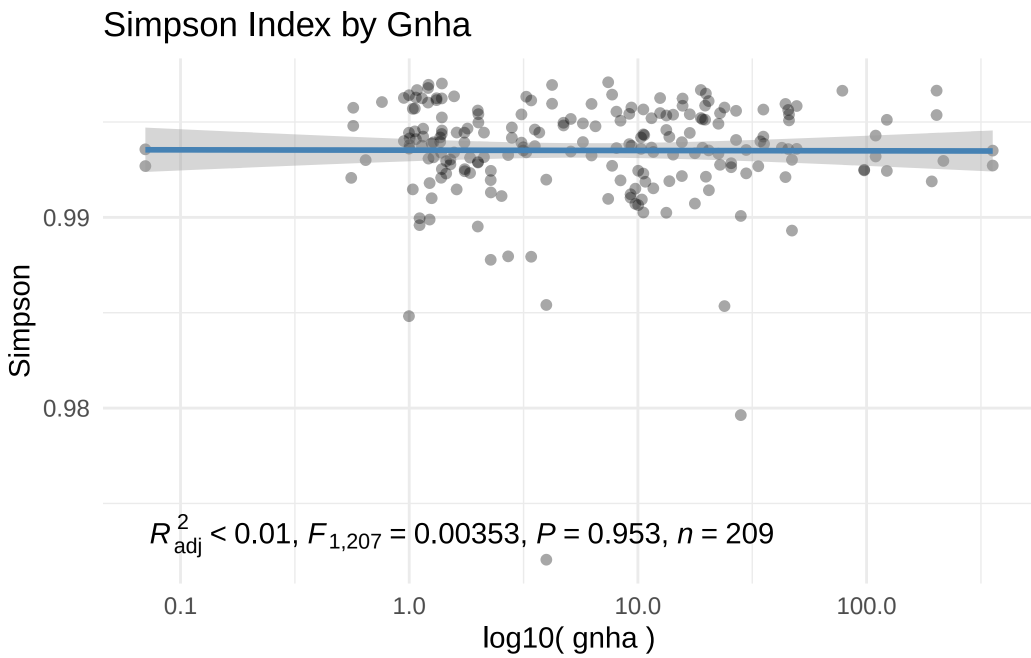
Shannon Index by Amo_a_337
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,211} = 0.695$, $P = 0.406$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)

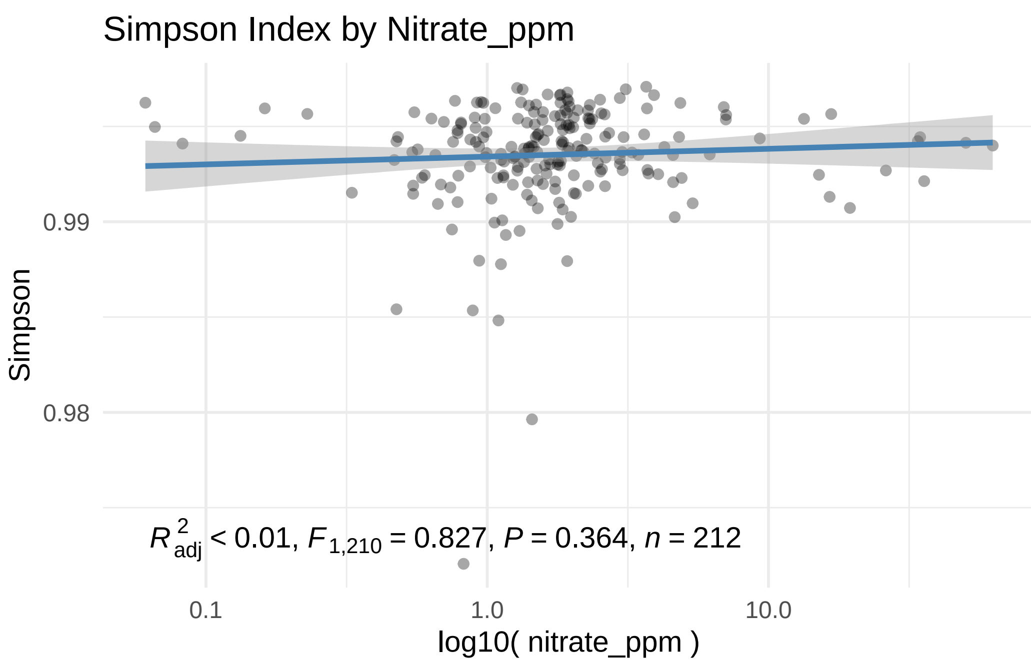
Shannon Index by Cnor_b

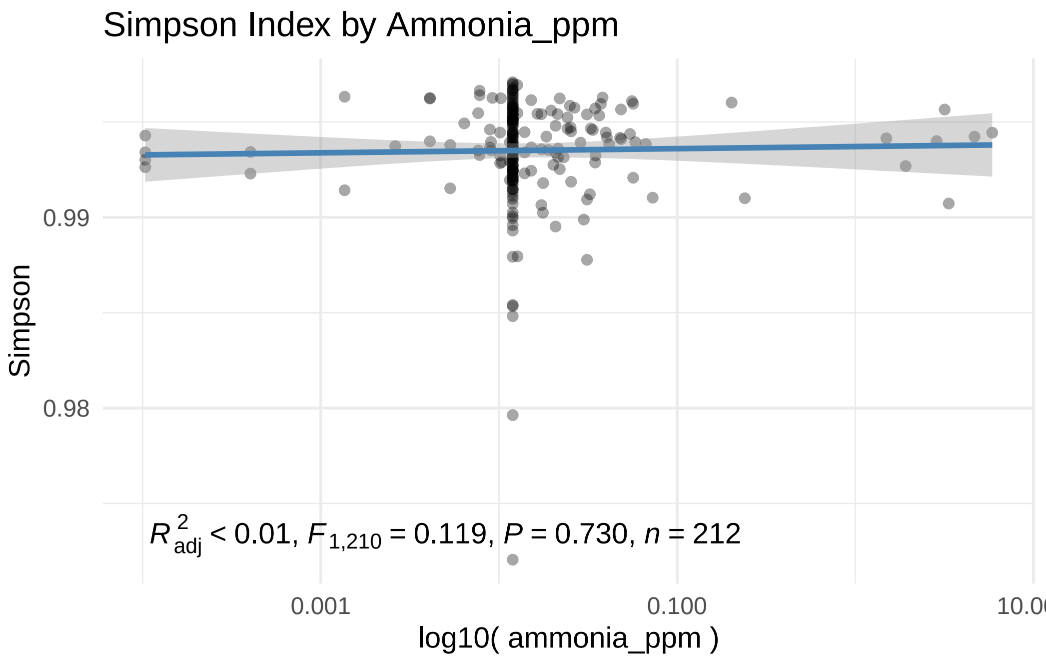


Shannon Index by Gwc_g_g

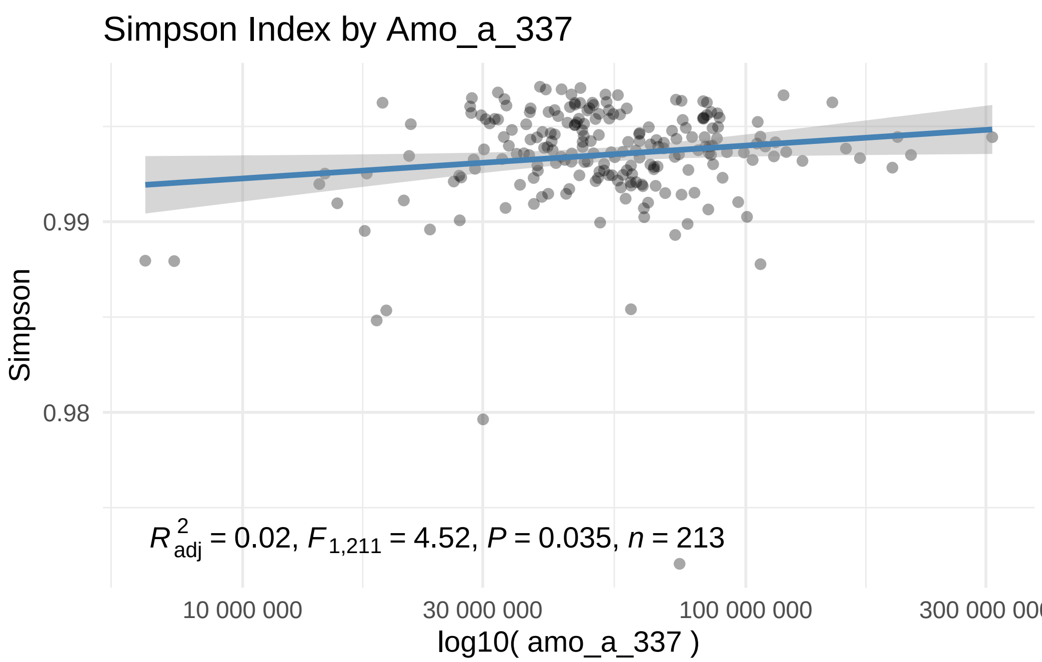



Simpson Index by Gnha
Simpson
0.99
0.98
$R^2_{adj} < 0.01$, $F_{1,207} = 0.00353$, $P = 0.953$, $n = 209$
0.1
1.0
10.0
100.0
log10(gnha)

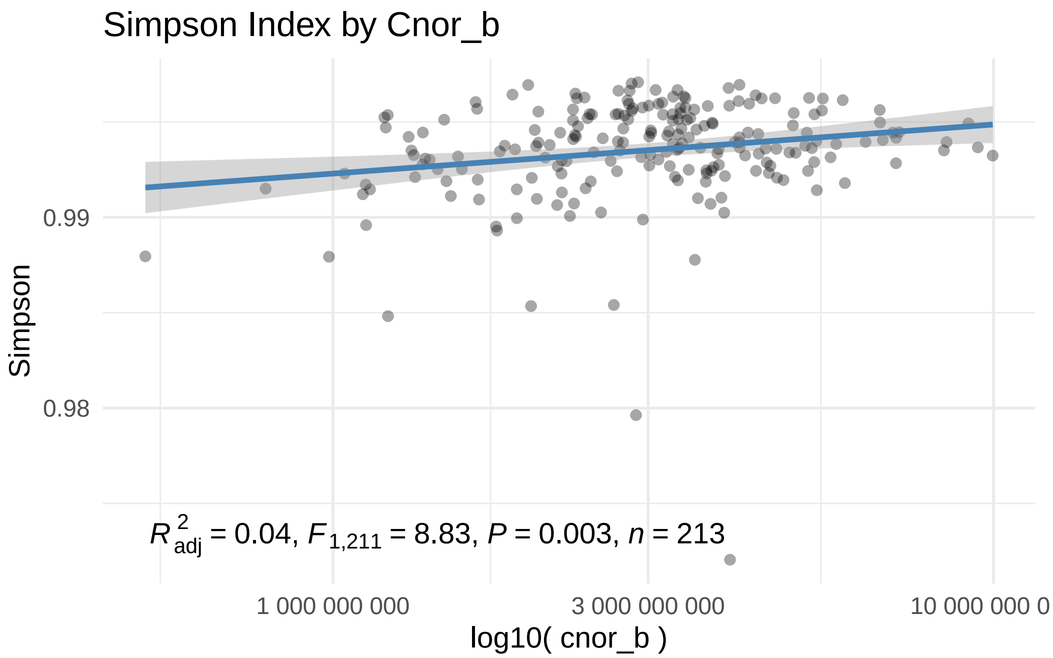

Simpson Index by Nitrate_ppm
Simpson
0.99
0.98
$R^2_{adj} < 0.01$, $F_{1,210} = 0.827$, $P = 0.364$, $n = 212$
0.1
1.0
10.0
log10(nitrate_ppm)

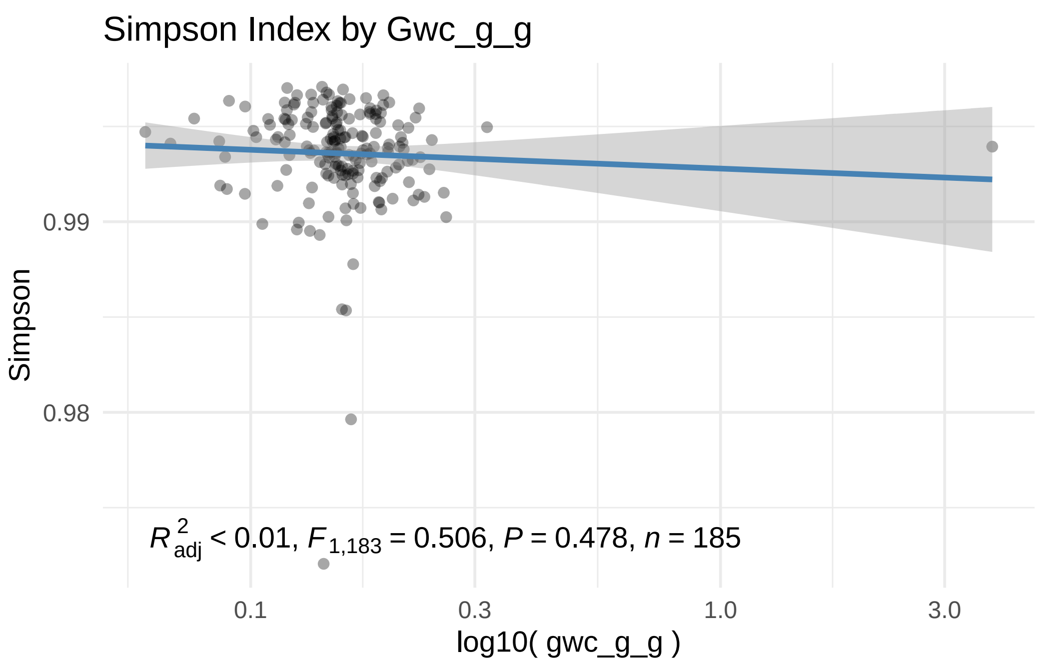


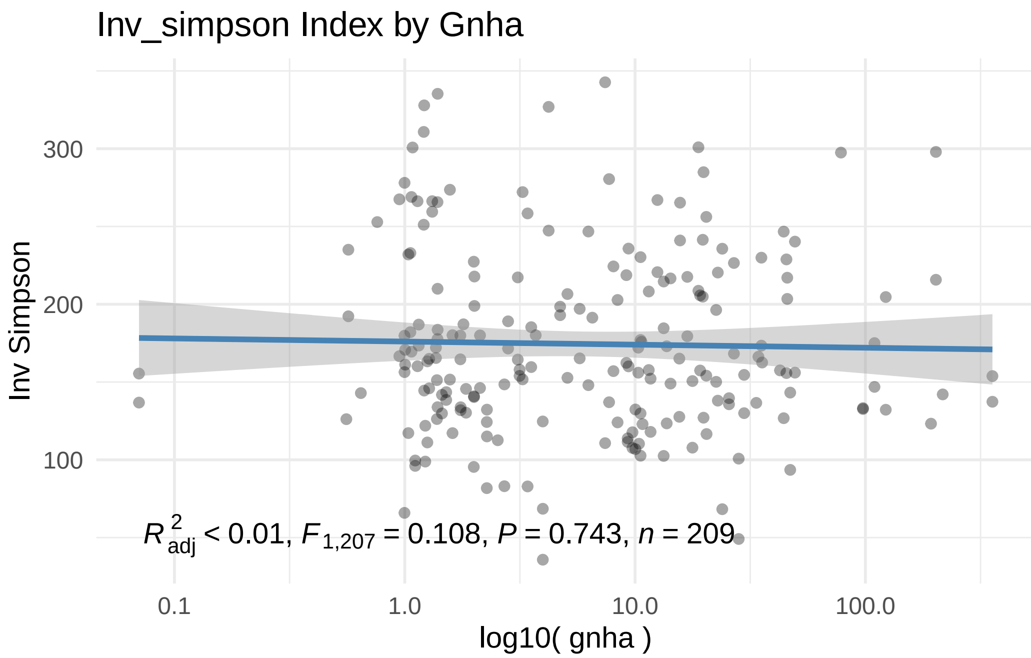
Simpson Index by Ammonia_ppm
Simpson
0.99
0.98
$R^2_{adj} < 0.01$, $F_{1,210} = 0.119$, $P = 0.730$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)

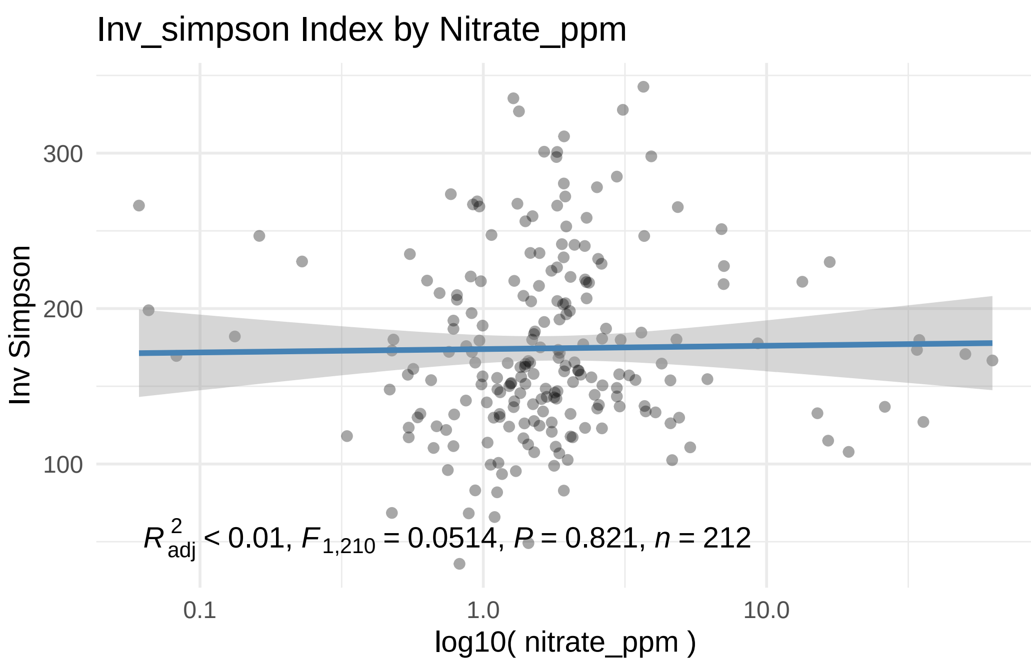


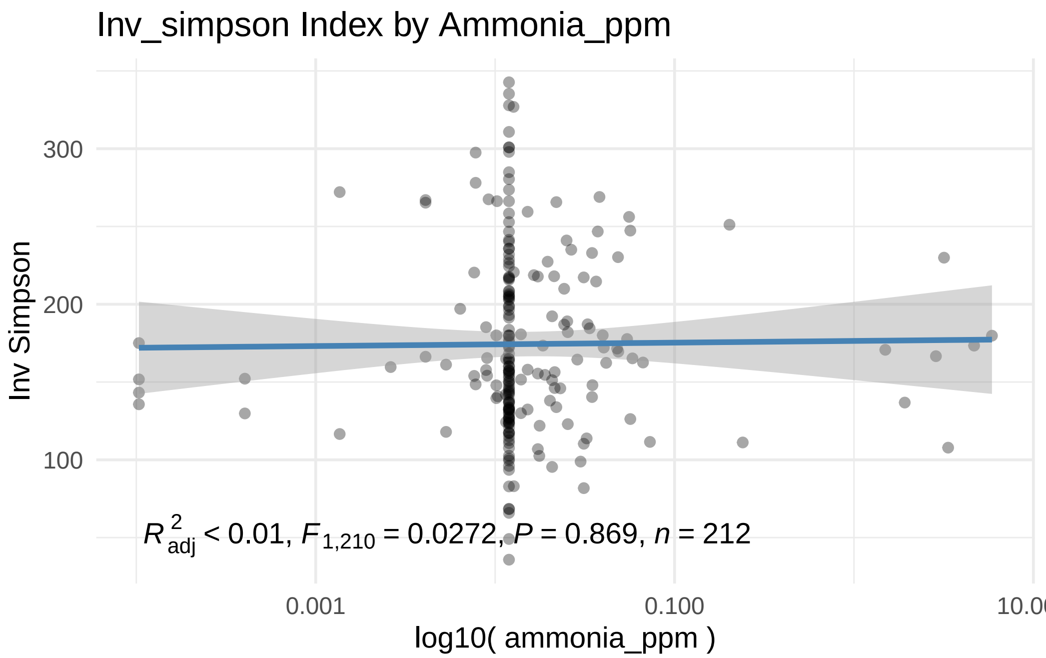
Simpson Index by Amo_a_337
Simpson
0.99
0.98
$R^2_{adj} = 0.02$, $F_{1,211} = 4.52$, $P = 0.035$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)


Simpson Index by Cnor_b
Simpson
0.99
0.98
R^2_{adj} = 0.04, $F_{1,211}$ = 8.83, P = 0.003, n = 213
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)

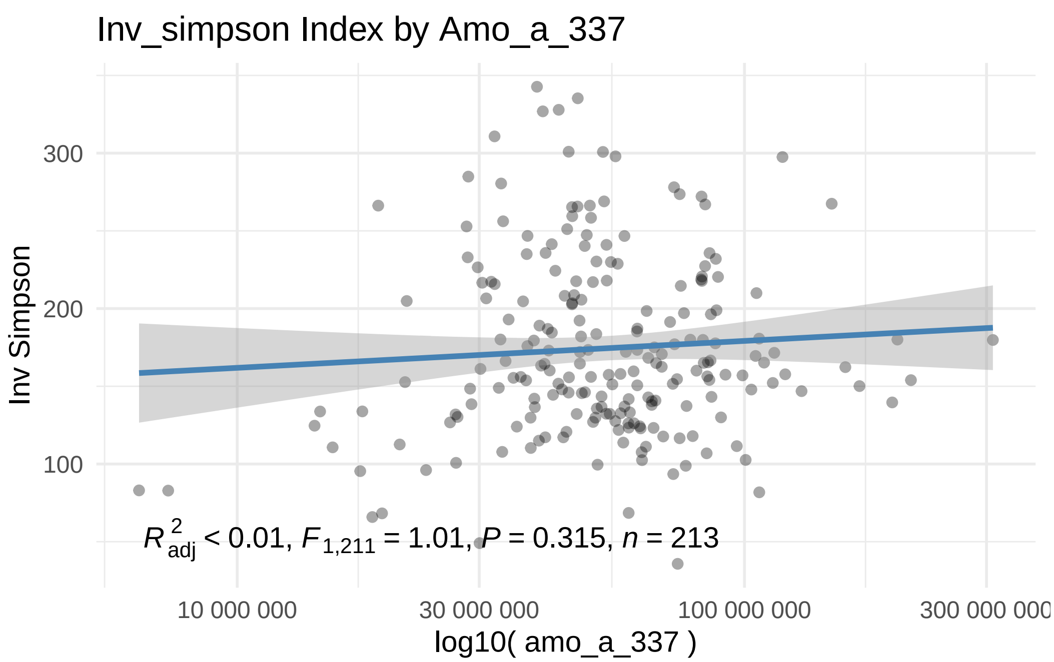

Simpson Index by Gwc_g_g
Simpson
0.99
0.98
R^2_{adj} < 0.01, $F_{1,183}$ = 0.506, P = 0.478, n = 185
0.1
0.3
1.0
3.0
log10(gwc_g_g)


Inv_simpson Index by Gnha
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,207} = 0.108$, $P = 0.743$, $n = 209$
0.1
1.0
10.0
100.0
log10(gnha)

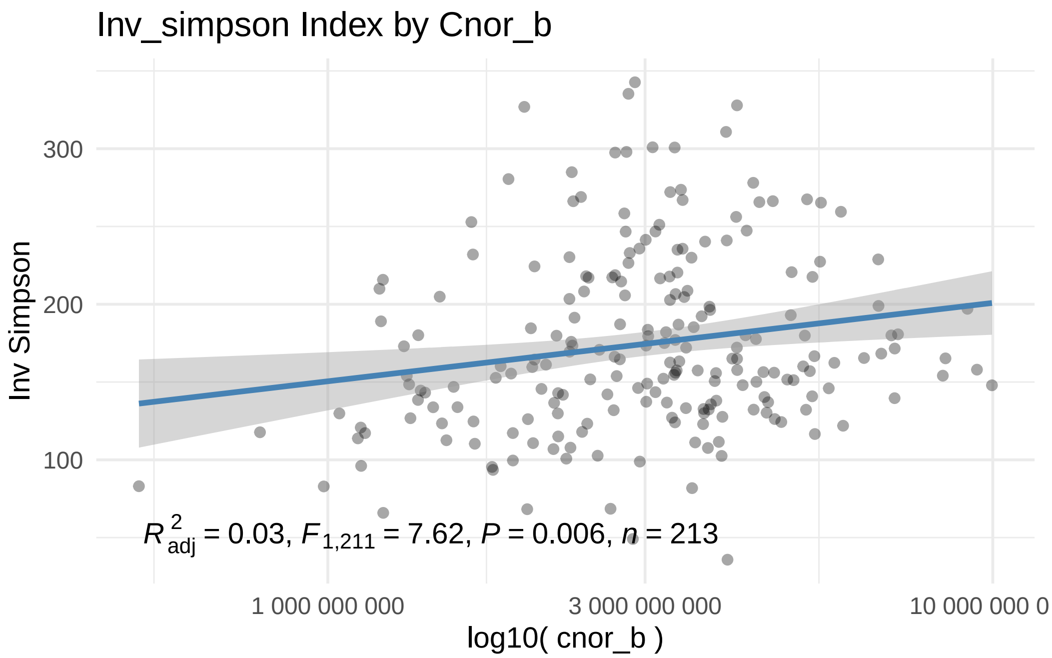

Inv_simpson Index by Nitrate_ppm
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0514$, $P = 0.821$, $n = 212$
0.1
1.0
10.0
log10(nitrate_ppm)

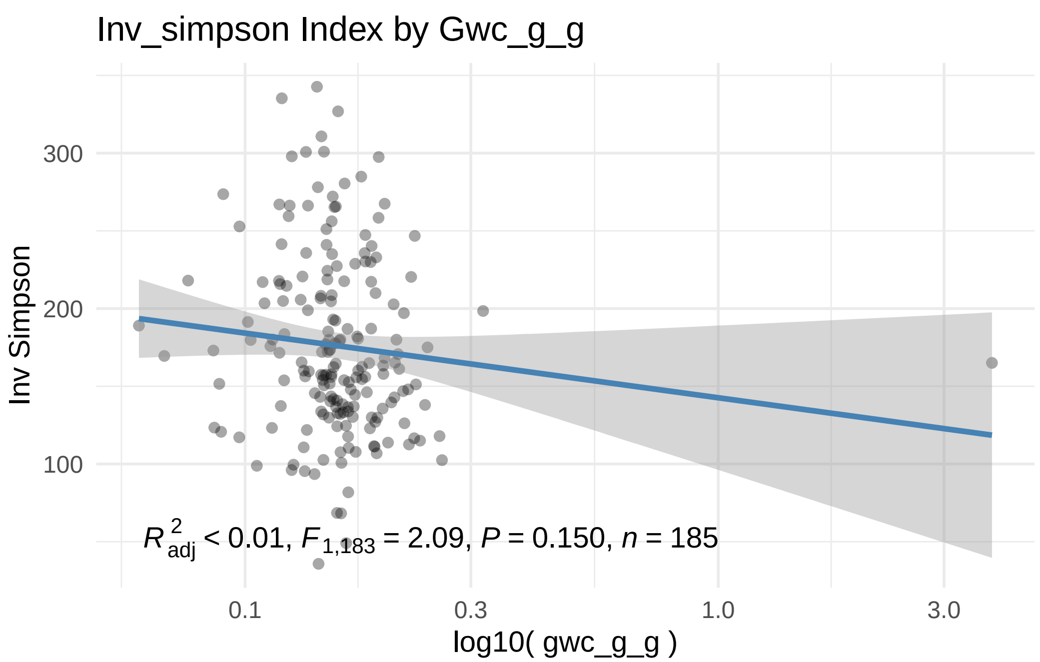


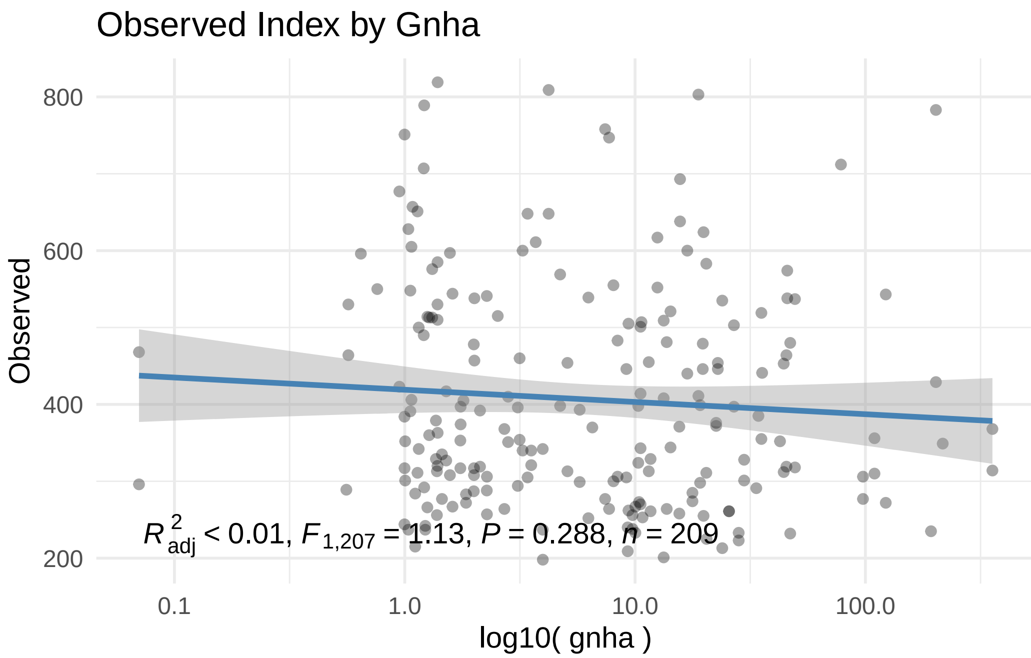
Inv_simpson Index by Ammonia_ppm
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0272$, $P = 0.869$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)

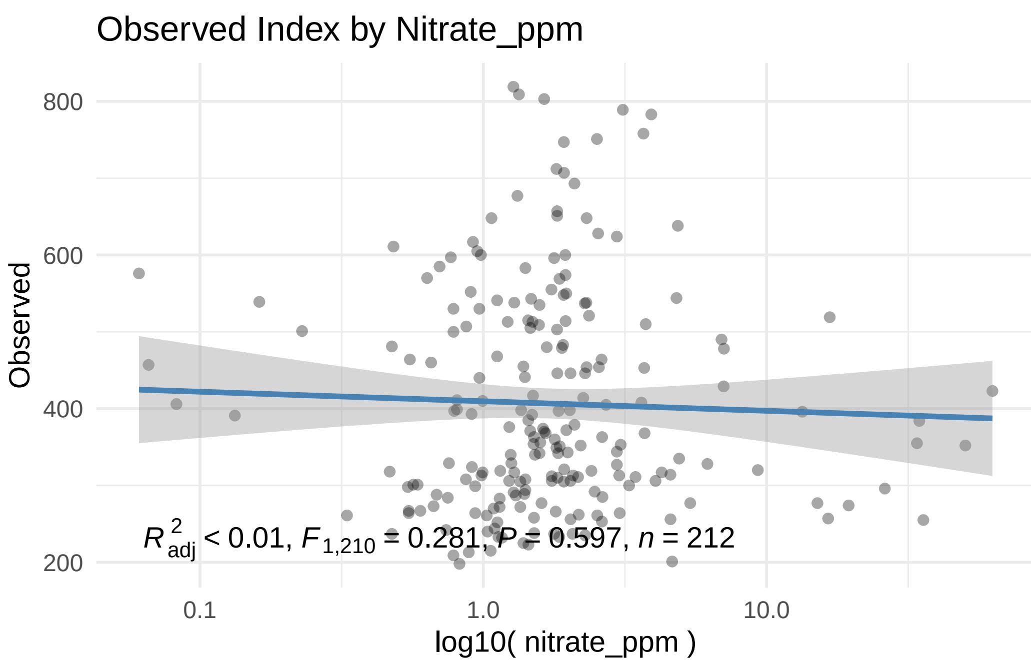


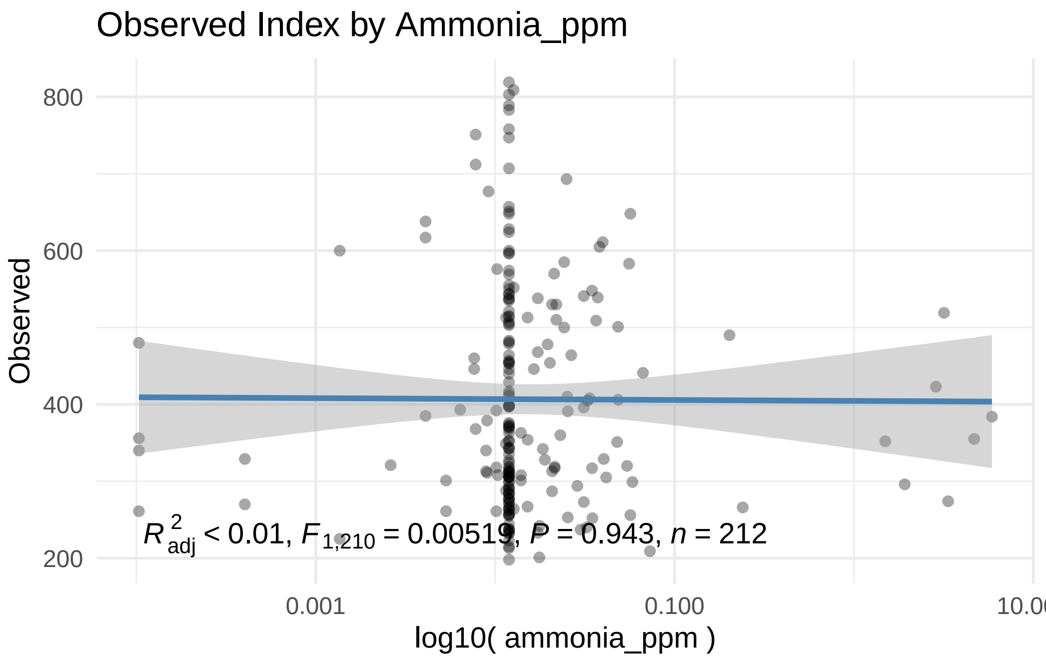
Inv_simpson Index by Amo_a_337
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,211} = 1.01$, $P = 0.315$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)

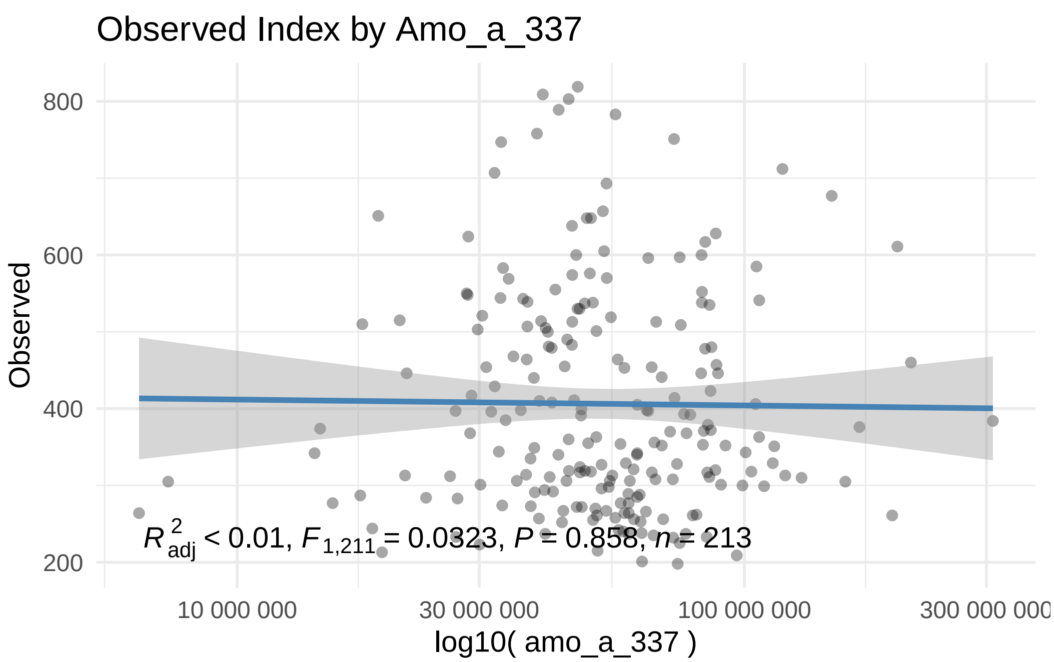

Inv_simpson Index by Cnor_b
Inv Simpson
300
200
100
R^2_{adj} = 0.03, $F_{1,211}$ = 7.62, P = 0.006, n = 213
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)

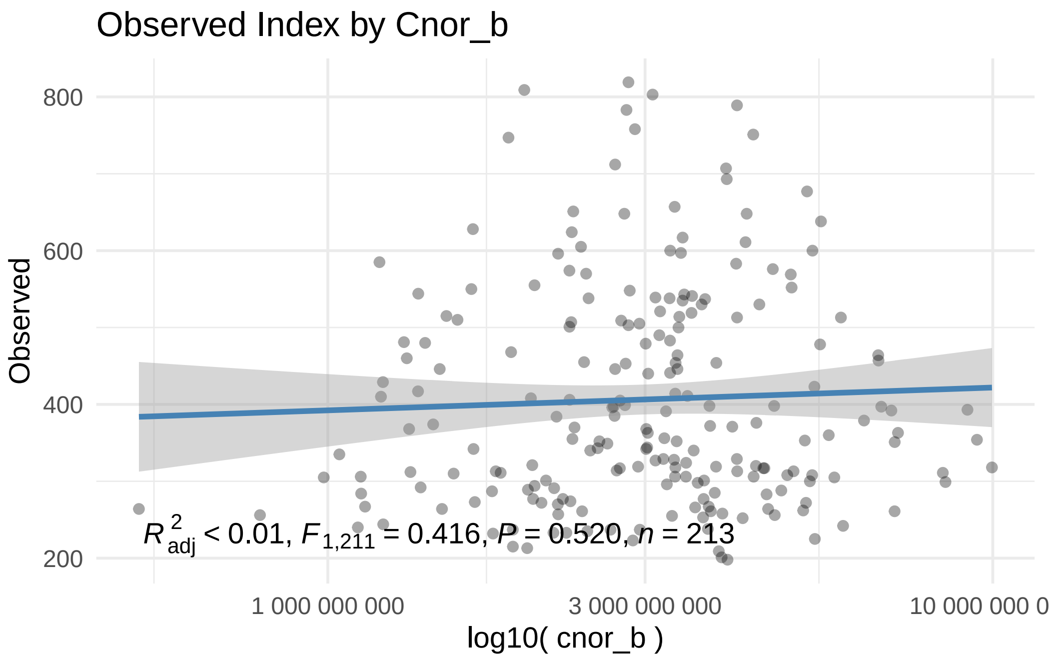

Inv_simpson Index by Gwc_g_g
Inv Simpson
300
200
100
R^2_{adj} < 0.01, $F_{1,183}$ = 2.09, P = 0.150, n = 185
0.1
0.3
1.0
3.0
log10(gwc_g_g)

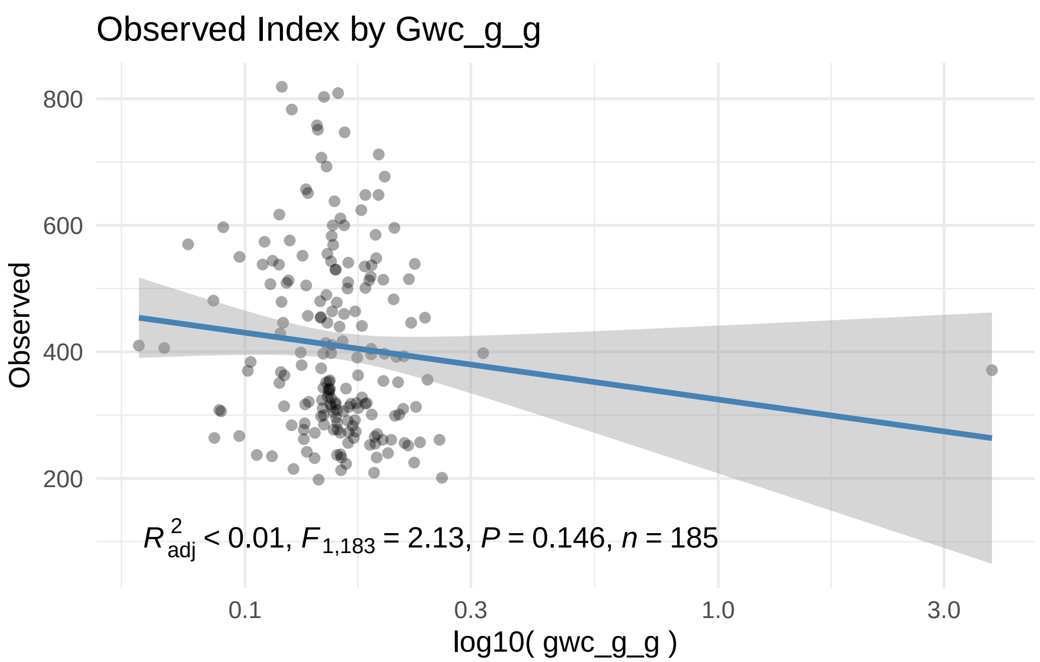

Observed Index by Gnha
Observed
800
600
400
200
0.1
1.0
10.0
100.0
log10(gnha)
R 2 adj < 0.01, F 1,207 = 1.13, P = 0.288, n = 209

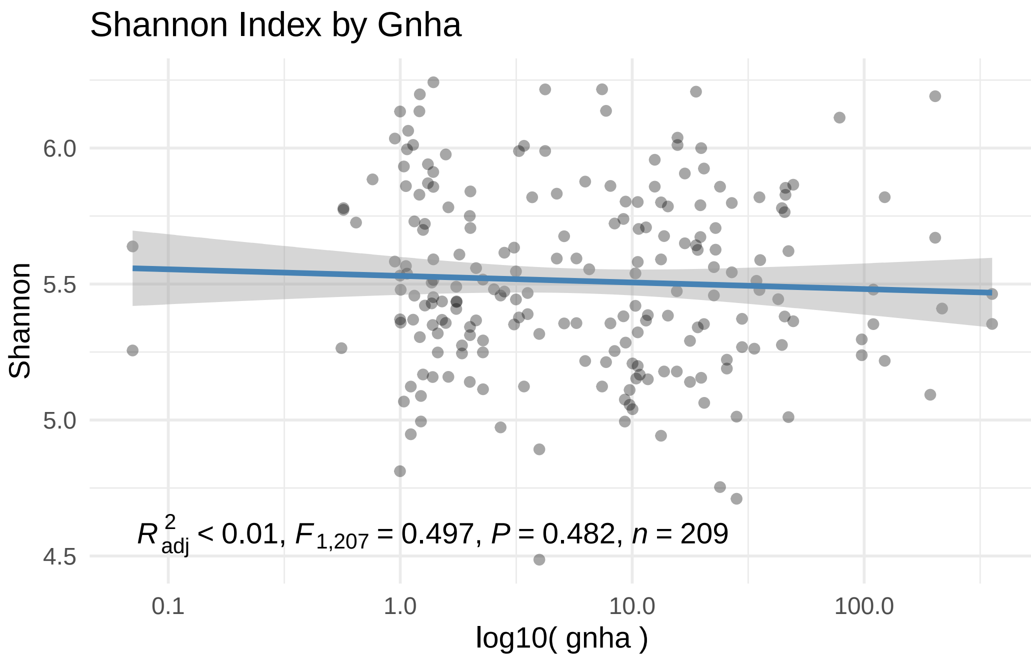

Observed Index by Nitrate_ppm
Observed
800
600
400
200
0.1
1.0
10.0
log10(nitrate_ppm)
R 2 adj < 0.01, F 1,210 = 0.281, P = 0.597, n = 212


Observed Index by Ammonia_ppm
Observed
800
600
400
200
$R^2_{adj} < 0.01$, $F_{1,210} = 0.00519$, $P = 0.943$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)

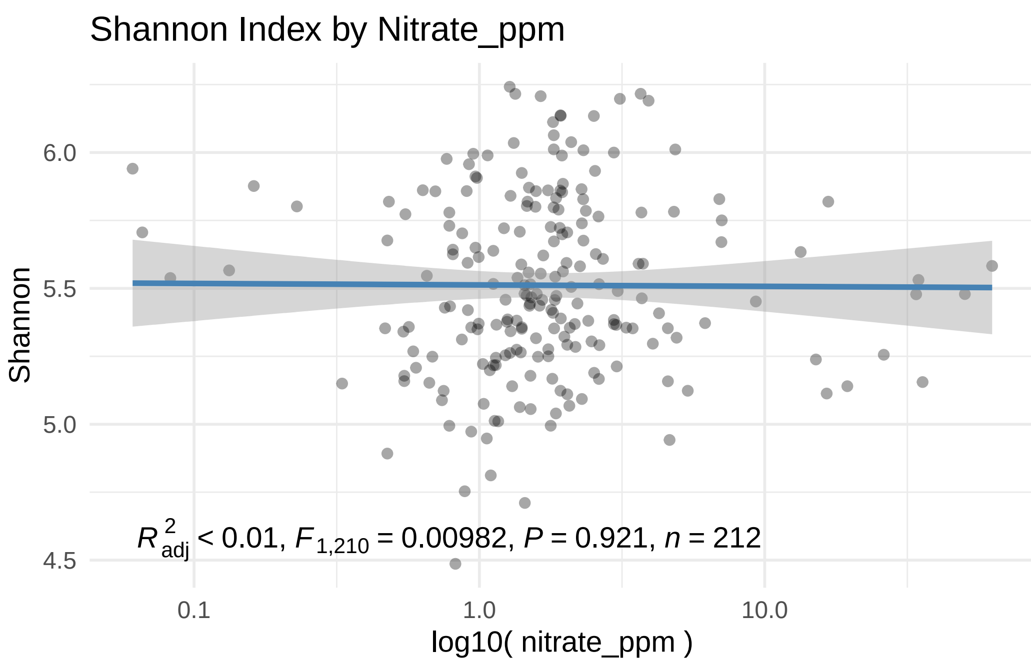

Observed Index by Amo_a_337
Observed
800
600
400
200
$R^2_{adj} < 0.01$, $F_{1,211} = 0.0323$, $P = 0.858$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)


Observed Index by Cnor_b
Observed
800
600
400
200
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)
$R^2_{adj} < 0.01$, $F_{1,211} = 0.416$, $P = 0.520$, $n = 213$

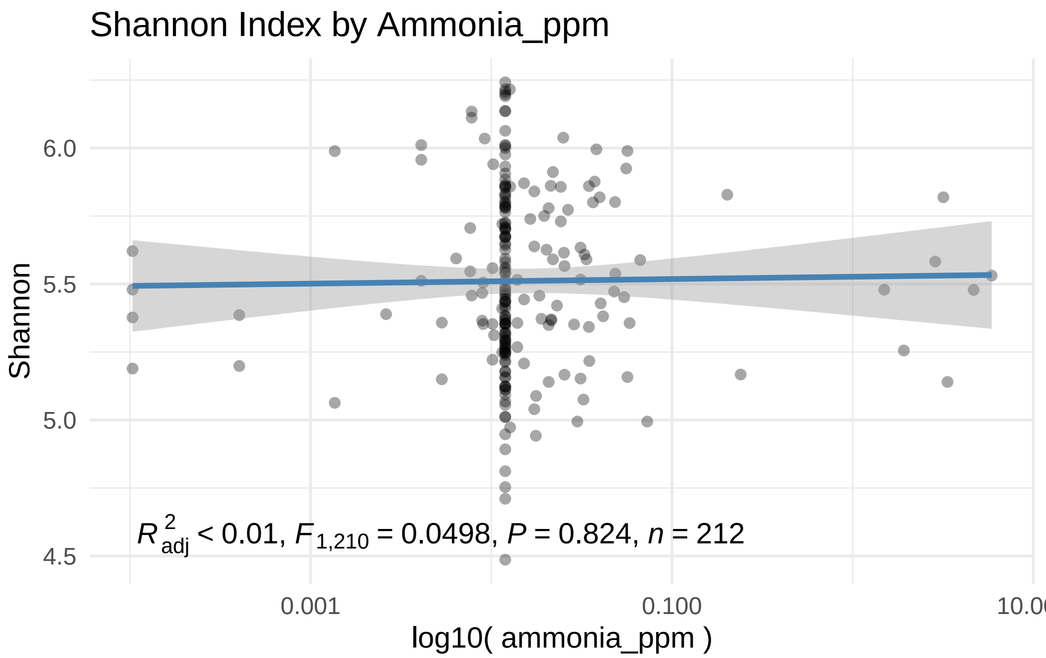

Observed Index by Gwc_g_g
Observed
800
600
400
200
0.1
0.3
1.0
3.0
log10(gwc_g_g)
$R^2_{adj} < 0.01$, $F_{1,183} = 2.13$, $P = 0.146$, $n = 185$



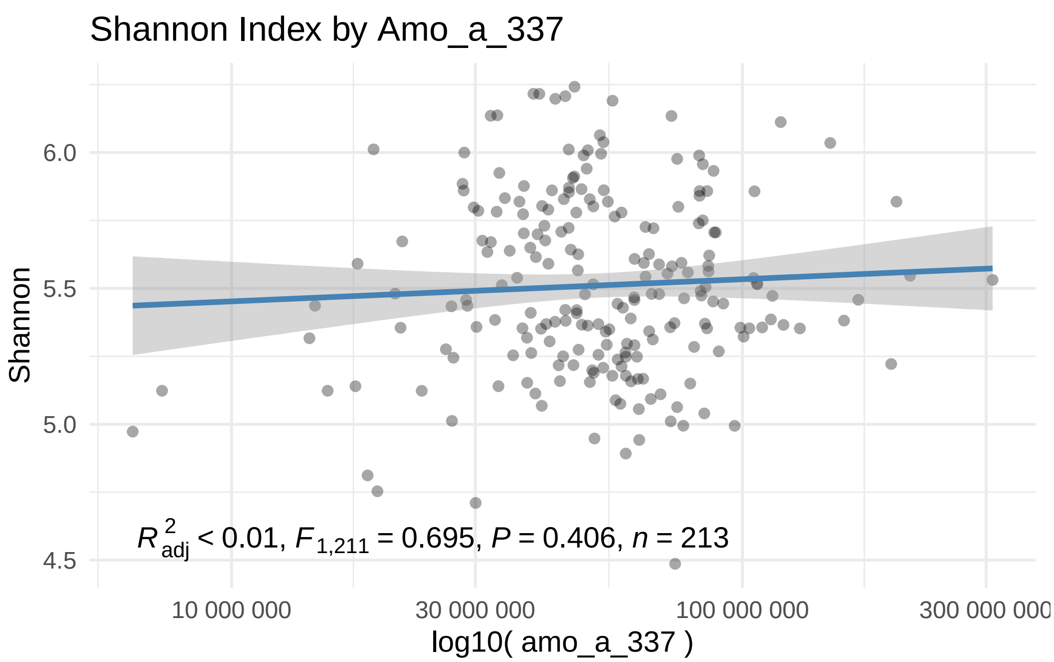
Shannon Index by Gnha
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,207} = 0.497$, $P = 0.482$, $n = 209$
0.1
1.0
10.0
100.0
log10(gnha)



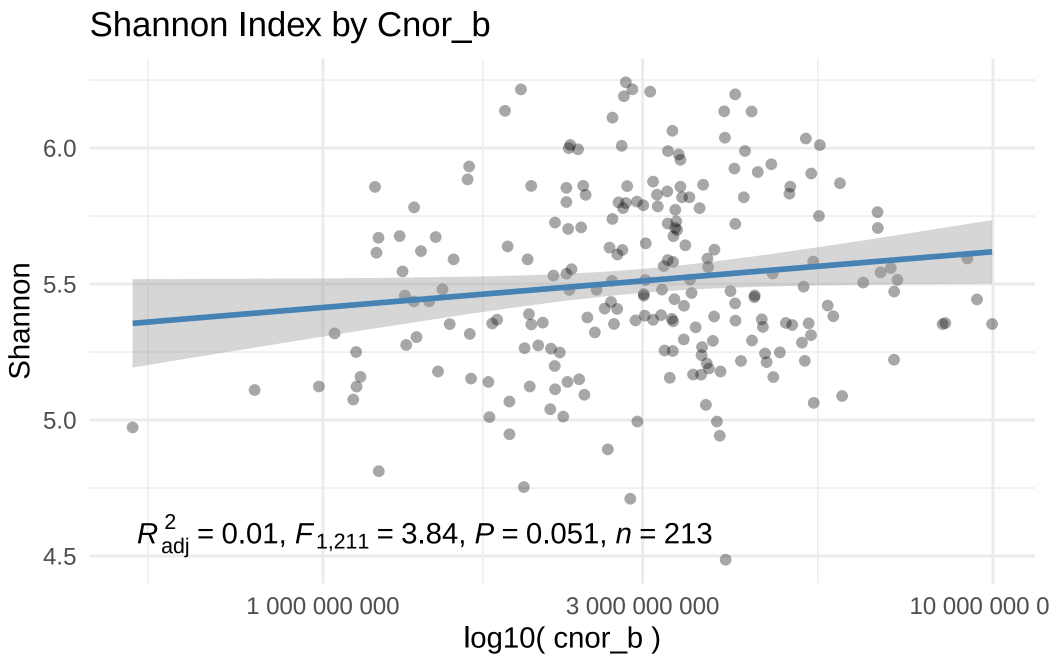
Shannon Index by Nitrate_ppm
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,210} = 0.00982$, $P = 0.921$, $n = 212$
0.1
1.0
10.0
log10(nitrate_ppm)



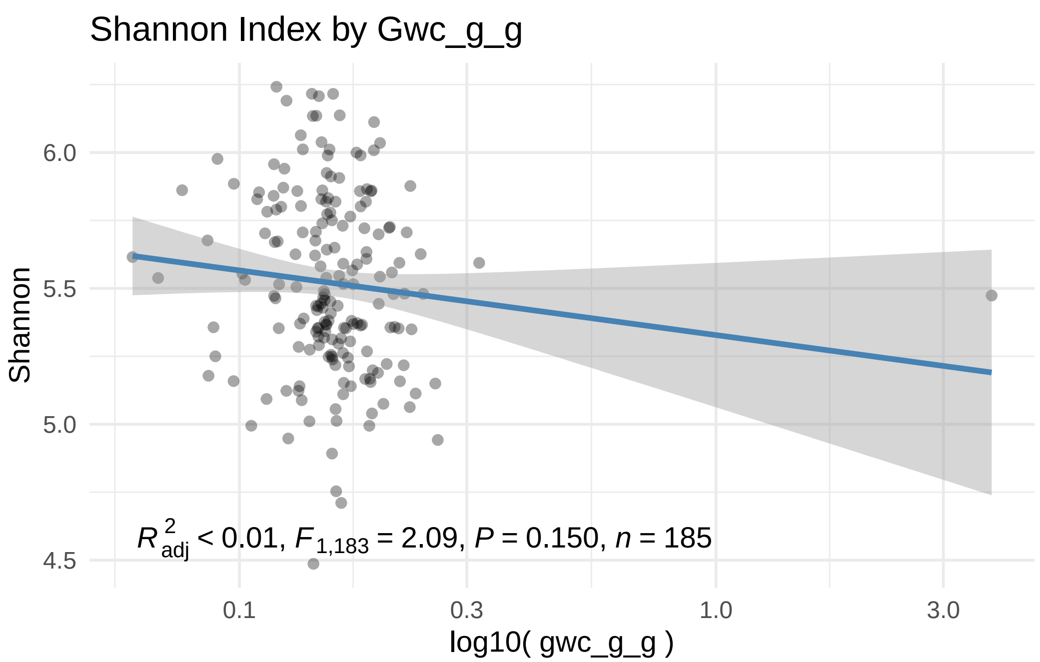
Shannon Index by Ammonia_ppm
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0498$, $P = 0.824$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)



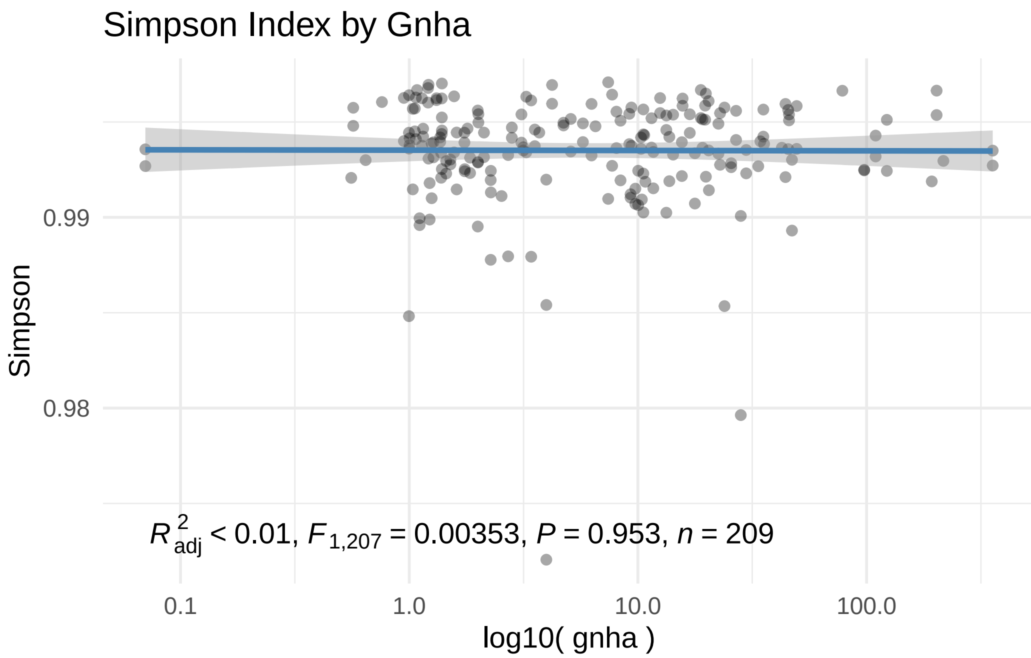
Shannon Index by Amo_a_337
Shannon
6.0
5.5
5.0
4.5
$R^2_{adj} < 0.01$, $F_{1,211} = 0.695$, $P = 0.406$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)

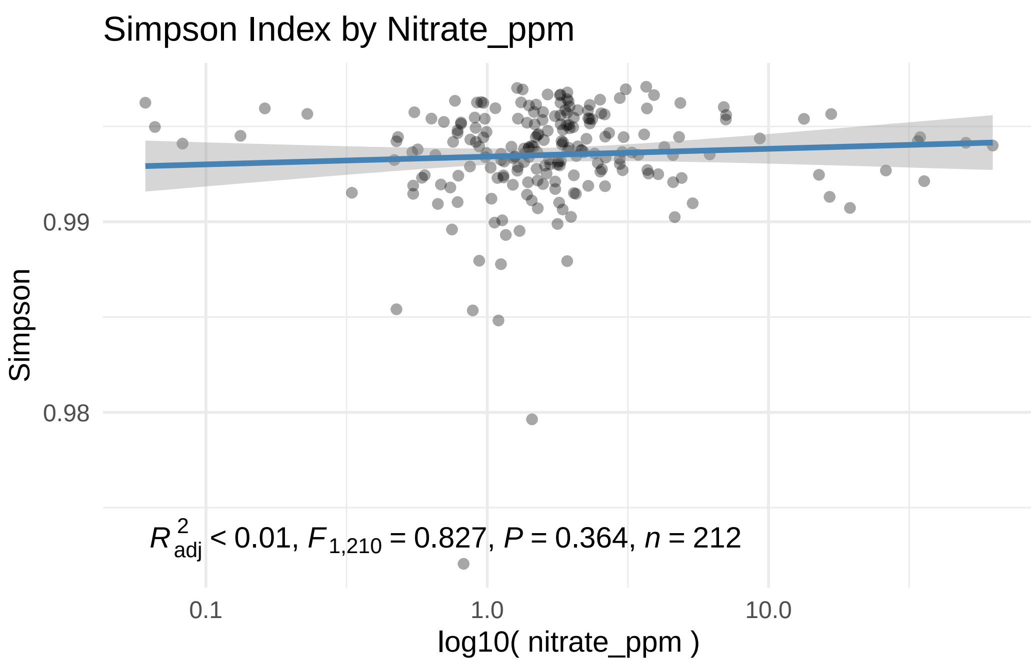


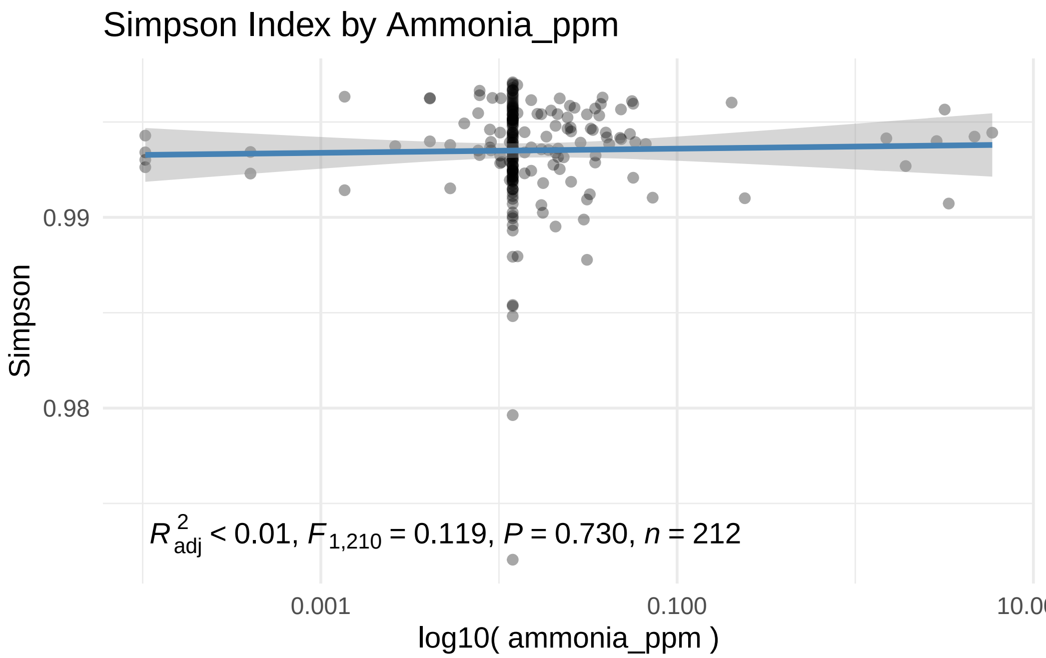
Shannon Index by Cnor_b
Shannon
6.0
5.5
5.0
4.5
R^2_{adj} = 0.01, $F_{1,211}$ = 3.84, P = 0.051, n = 213
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)

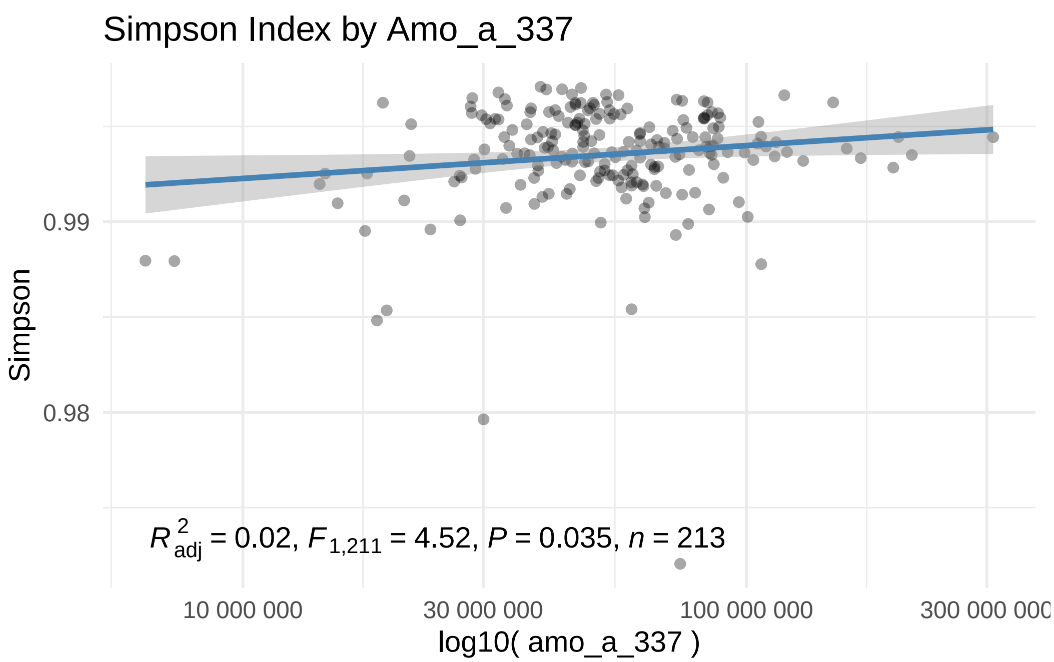


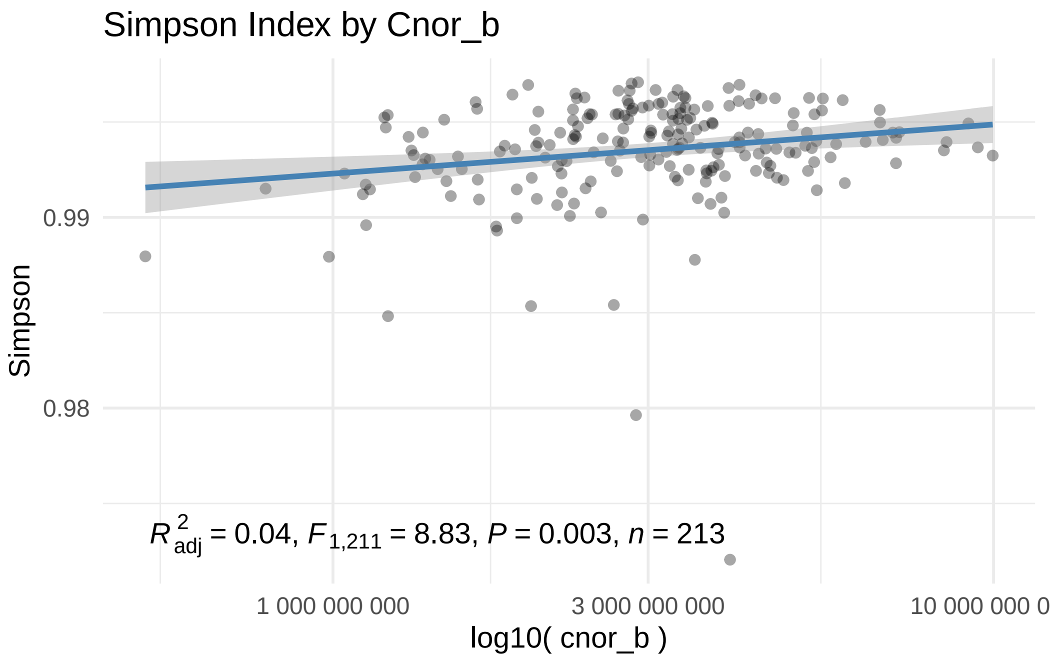
Shannon Index by Gwc_g_g
Shannon
6.0
5.5
5.0
4.5
R^2_{adj} < 0.01, $F_{1,183}$ = 2.09, P = 0.150, n = 185
0.1
0.3
1.0
3.0
log10(gwc_g_g)

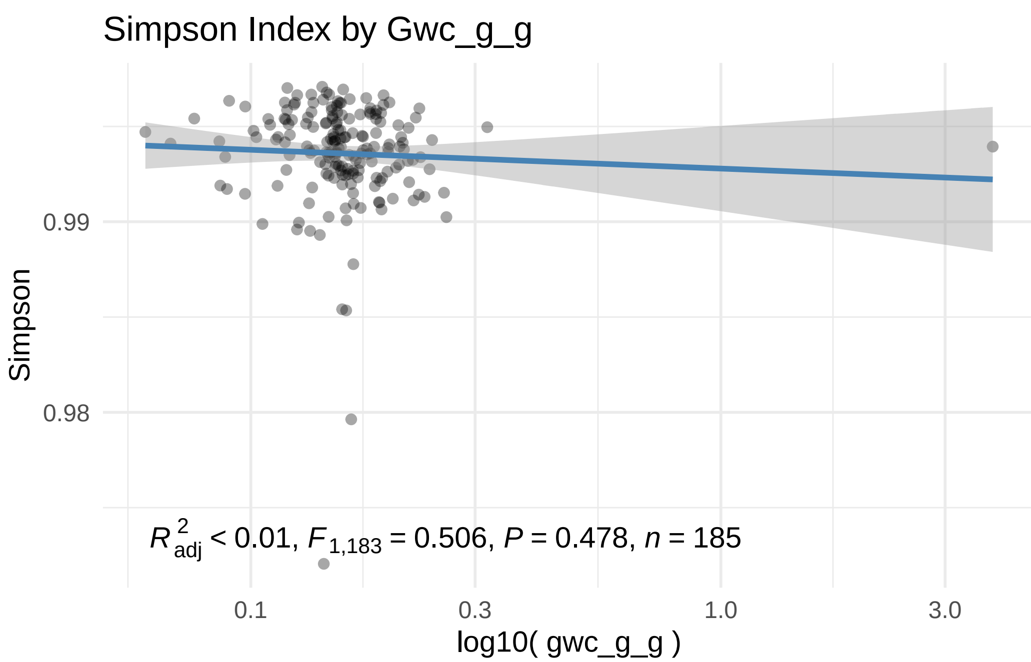

Simpson Index by Gnha
Simpson
0.99
0.98
R 2 adj < 0.01, F 1,207 = 0.00353, P = 0.953, n = 209
0.1
1.0
10.0
100.0
log10(gnha)

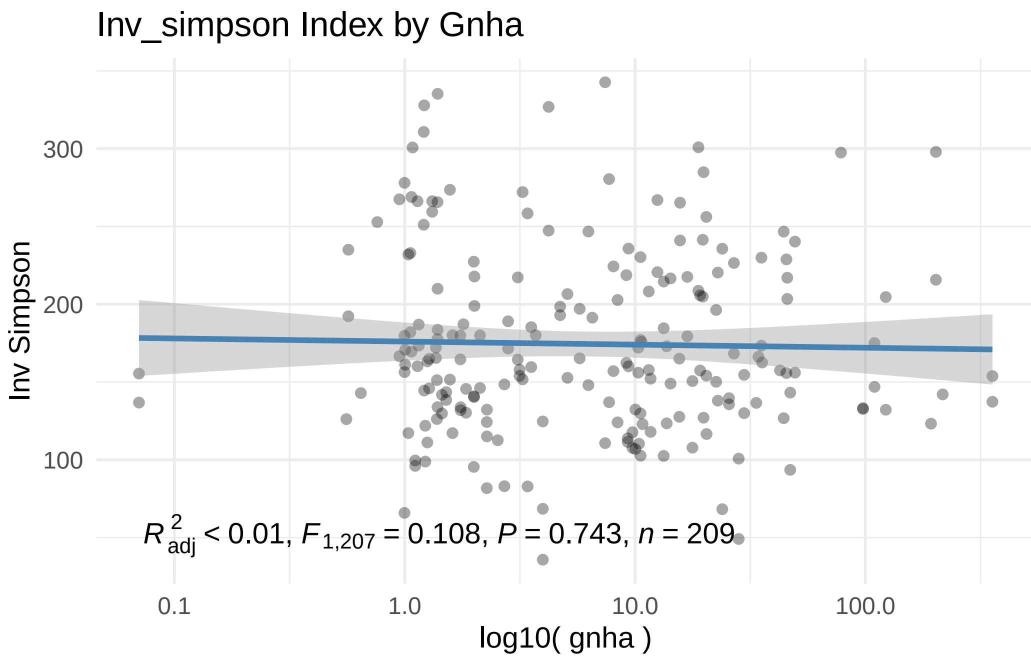

Simpson Index by Nitrate_ppm
Simpson
0.99
0.98
R 2 adj < 0.01, F 1,210 = 0.827, P = 0.364, n = 212
0.1
1.0
10.0
log10(nitrate_ppm)

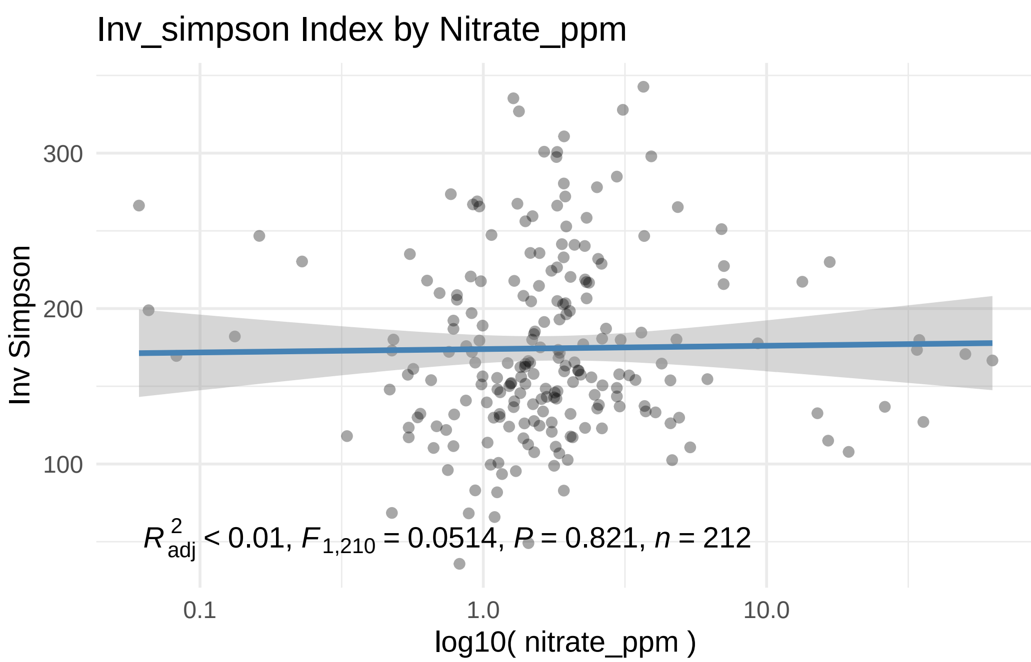

Simpson Index by Ammonia_ppm
Simpson
0.99
0.98
$R^2_{adj} < 0.01$, $F_{1,210} = 0.119$, $P = 0.730$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)

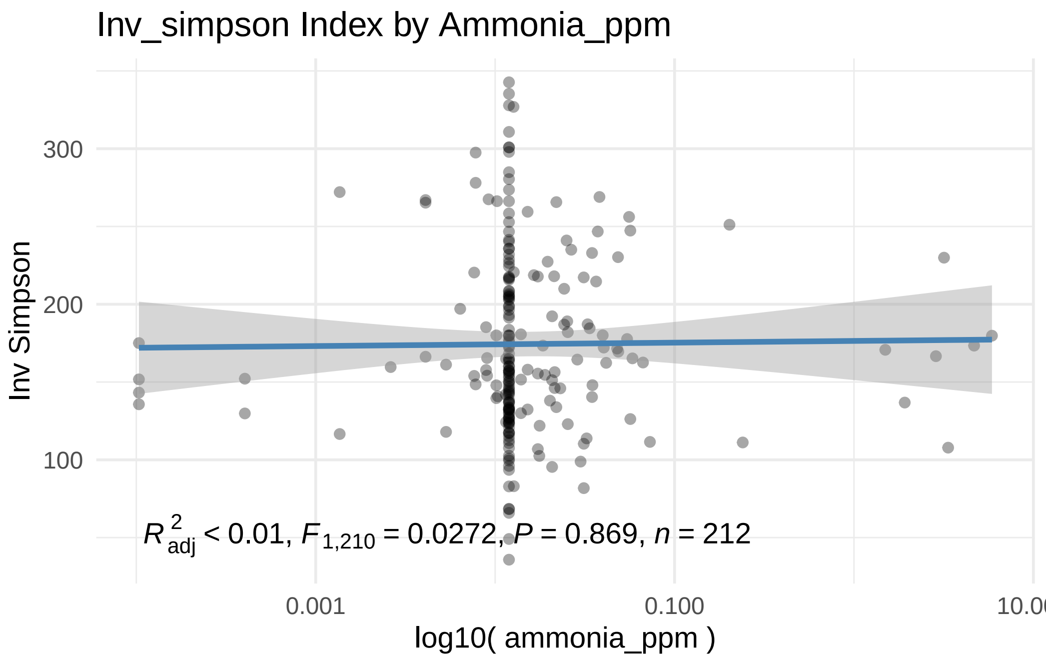

Simpson Index by Amo_a_337
Simpson
0.99
0.98
$R^2_{adj} = 0.02$, $F_{1,211} = 4.52$, $P = 0.035$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)

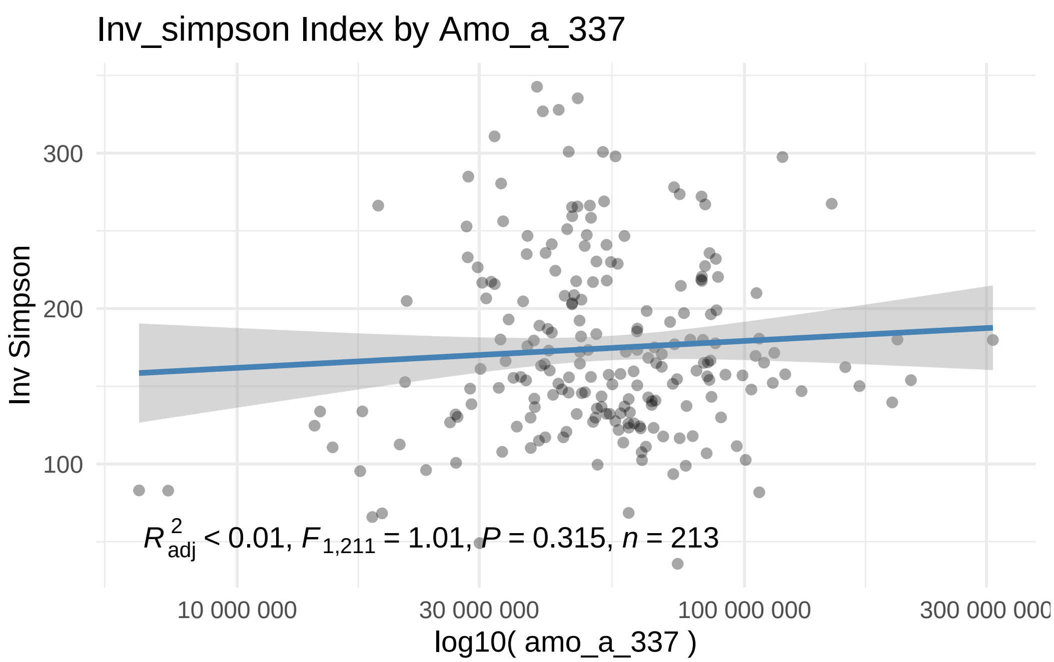

Simpson Index by Cnor_b
Simpson
0.99
0.98
R^2_{adj} = 0.04, $F_{1,211}$ = 8.83, P = 0.003, n = 213
1 000 000 000
3 000 000 000
10 000 000 0
log10(cnor_b)


Simpson Index by Gwc_g_g
Simpson
0.99
0.98
R^2_{adj} < 0.01, $F_{1,183}$ = 0.506, P = 0.478, n = 185
0.1
0.3
1.0
3.0
log10(gwc_g_g)

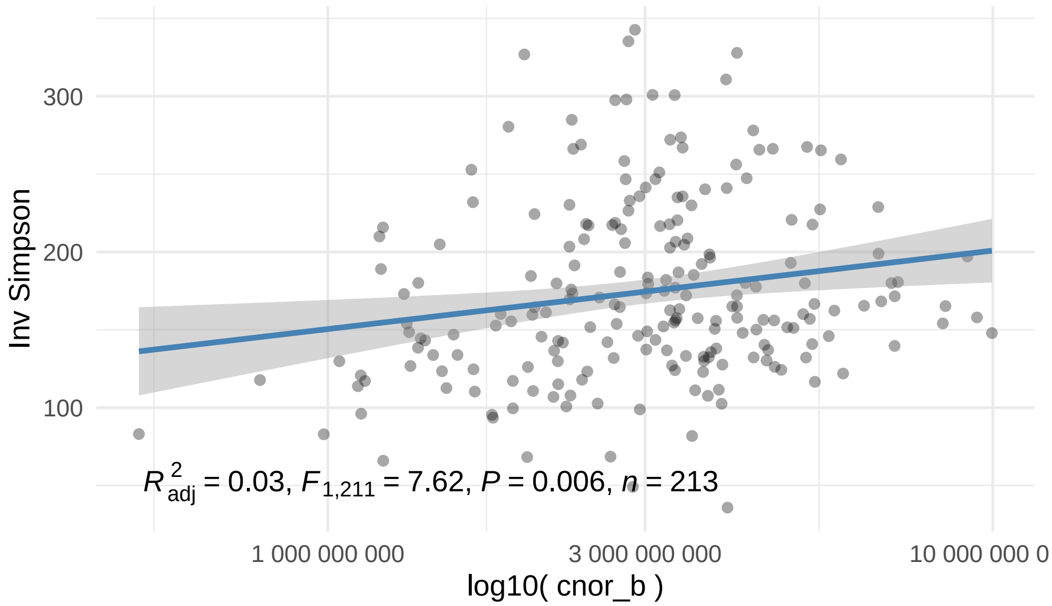

Inv_simpson Index by Gnha
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,207} = 0.108$, $P = 0.743$, $n = 209$
0.1
1.0
10.0
100.0
log10(gnha)


Inv_simpson Index by Nitrate_ppm
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0514$, $P = 0.821$, $n = 212$
0.1
1.0
10.0
log10(nitrate_ppm)

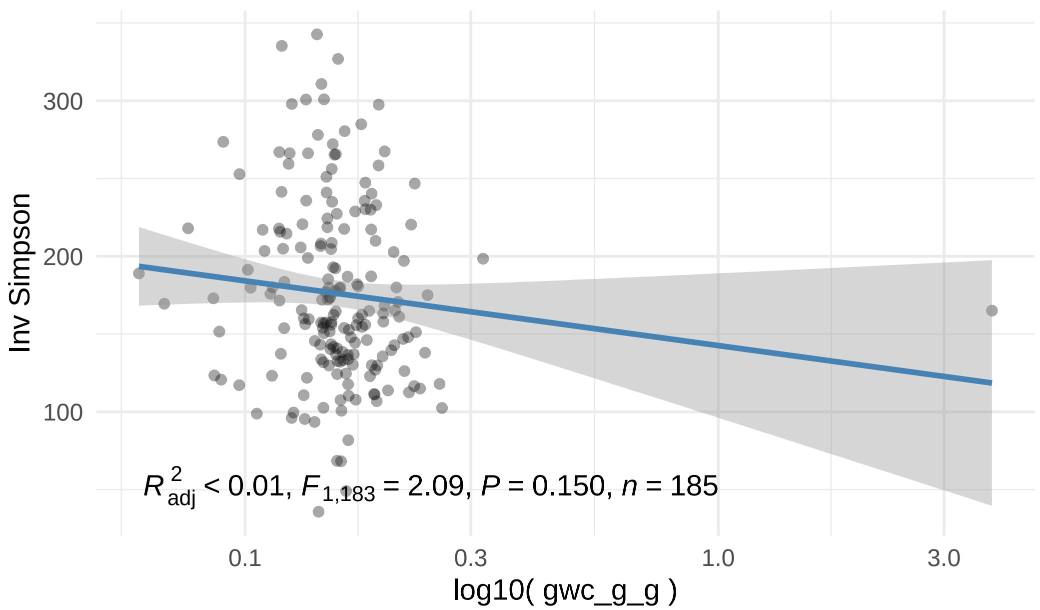

Inv_simpson Index by Ammonia_ppm
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,210} = 0.0272$, $P = 0.869$, $n = 212$
0.001
0.100
10.00
log10(ammonia_ppm)


Inv_simpson Index by Amo_a_337
Inv Simpson
300
200
100
$R^2_{adj} < 0.01$, $F_{1,211} = 1.01$, $P = 0.315$, $n = 213$
10 000 000
30 000 000
100 000 000
300 000 000
log10(amo_a_337)

Inv_simpson Index by Cnor_b



Inv_simpson Index by Gwc_g_g




```
$raw
$raw$year

$raw$plot
```

```
$raw$sampling_location

$raw$replicate

$raw$sampling_date

$raw$fertilization


$observed
$observed$gnha

$observed$nitrate_ppm

$observed$ammonia_ppm

$observed$amo_a_337

$observed$cnor_b

$observed$gwc_g_g


$shannon
$shannon$gnha

$shannon$nitrate_ppm

$shannon$ammonia_ppm

$shannon$amo_a_337

$shannon$cnor_b

$shannon$gwc_g_g


$simpson
$simpson$gnha

$simpson$nitrate_ppm

$simpson$ammonia_ppm
```

```
$simpson$amo_a_337

$simpson$cnor_b

$simpson$gwc_g_g


$inv_simpson
$inv_simpson$gnha

$inv_simpson$nitrate_ppm

$inv_simpson$ammonia_ppm

$inv_simpson$amo_a_337

$inv_simpson$cnor_b

$inv_simpson$gwc_g_g


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DIVERSITY INDEX: OBSERVED
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$gnha

$nitrate_ppm

$ammonia_ppm

$amo_a_337

$cnor_b

$gwc_g_g



- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
DIVERSITY INDEX: SHANNON
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$gnha
```

```
$nitrate_ppm

$ammonia_ppm

$amo_a_337

$cnor_b

$gwc_g_g


- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
DIVERSITY INDEX: SIMPSON
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$gnha

$nitrate_ppm

$ammonia_ppm

$amo_a_337

$cnor_b

$gwc_g_g


- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
DIVERSITY INDEX: INV_SIMPSON
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
$gnha

$nitrate_ppm

$ammonia_ppm

$amo_a_337

$cnor_b
```

$gwc_g_g

Session information

```r
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Arch Linux

Matrix products: default
BLAS/LAPACK: /usr/lib/libopenblas.so.0.3;  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
 [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
 [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
[10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C

time zone: America/Chicago
tzcode source: system (glibc)

attached base packages:
[1] stats4    stats     graphics  grDevices datasets  utils     methods
[8] base

other attached packages:
 [1] BRCore_0.0.2                    metagMisc_0.5.0
 [3] microbiome_1.28.0               mia_1.14.0
 [5] TreeSummarizedExperiment_2.14.0 Biostrings_2.74.1
 [7] XVector_0.46.0                  SingleCellExperiment_1.28.1
 [9] MultiAssayExperiment_1.32.0     SummarizedExperiment_1.36.0
[11] Biobase_2.66.0                  GenomicRanges_1.58.0
[13] GenomeInfoDb_1.42.3             IRanges_2.40.1
[15] S4Vectors_0.44.0                BiocGenerics_0.52.0
[17] MatrixGenerics_1.18.1           matrixStats_1.5.0
[19] janitor_2.2.1                   lubridate_1.9.4
[21] forcats_1.0.0                   stringr_1.5.1
[23] dplyr_1.1.4                     purrr_1.0.4
[25] readr_2.1.5                     tidyr_1.3.1
```

```
[27] tibble_3.2.1                    ggplot2_3.5.2
[29] tidyverse_2.0.0                 vegan_2.6-10
[31] lattice_0.22-6                  permute_0.9-7
[33] phyloseq_1.50.0                 conflicted_1.2.0

loaded via a namespace (and not attached):
  [1] splines_4.4.2               ggpp_0.5.9
  [3] cellranger_1.1.0            minpack.lm_1.2-4
  [5] confintr_1.0.2              rpart_4.1.23
  [7] DirichletMultinomial_1.48.0 lifecycle_1.0.4
  [9] Rdpack_2.6.4                doParallel_1.0.17
 [11] rprojroot_2.0.4             pals_1.10
 [13] MASS_7.3-61                 backports_1.5.0
 [15] magrittr_2.0.3              Hmisc_5.2-3
 [17] rmarkdown_2.29              yaml_2.3.10
 [19] mapproj_1.2.12              DBI_1.2.3
 [21] minqa_1.2.8                 RColorBrewer_1.1-3
 [23] ade4_1.7-23                 maps_3.4.3
 [25] multcomp_1.4-28             abind_1.4-8
 [27] zlibbioc_1.52.0             Rtsne_0.17
 [29] fillpattern_1.0.2           yulab.utils_0.2.0
 [31] nnet_7.3-19                 TH.data_1.1-4
 [33] sandwich_3.1-1              GenomeInfoDbData_1.2.13
 [35] ggrepel_0.9.6               irlba_2.3.5.1
 [37] ggpmisc_0.6.2               tidytree_0.4.6
 [39] iNEXT_3.0.1                 MatrixModels_0.5-4
 [41] rbiom_2.2.0                 parallelly_1.44.0
 [43] DelayedMatrixStats_1.28.1   codetools_0.2-20
 [45] DelayedArray_0.32.0         xml2_1.3.8
 [47] scuttle_1.16.0              ggtext_0.1.2
 [49] tidyselect_1.2.1            UCSC.utils_1.2.0
 [51] farver_2.1.2                lme4_1.1-37
 [53] ScaledMatrix_1.14.0         viridis_0.6.5
 [55] base64enc_0.1-3             jsonlite_2.0.0
 [57] BiocNeighbors_2.0.1         multtest_2.62.0
 [59] decontam_1.26.0             Formula_1.2-5
 [61] survival_3.7-0              scater_1.34.1
 [63] iterators_1.0.14            emmeans_1.11.1
 [65] systemfonts_1.2.3           foreach_1.5.2
 [67] tools_4.4.2                 ggnewscale_0.5.1
 [69] ragg_1.4.0                  treeio_1.30.0
 [71] Rcpp_1.0.14                 glue_1.8.0
 [73] gridExtra_2.3               SparseArray_1.6.2
```

```
 [75] xfun_0.52                   here_1.0.1
 [77] mgcv_1.9-1                  withr_3.0.2
 [79] BiocManager_1.30.25         fastmap_1.2.0
 [81] boot_1.3-31                 rhdf5filters_1.18.1
 [83] bluster_1.16.0              SparseM_1.84-2
 [85] digest_0.6.37               rsvd_1.0.5
 [87] timechange_0.3.0            R6_2.6.1
 [89] estimability_1.5.1          textshaping_1.0.1
 [91] colorspace_2.1-1            lpSolve_5.6.23
 [93] dichromat_2.0-0.1           utf8_1.2.5
 [95] generics_0.1.3              renv_1.1.1
 [97] data.table_1.17.0           DECIPHER_3.2.0
 [99] httr_1.4.7                  htmlwidgets_1.6.4
[101] S4Arrays_1.6.0              pkgconfig_2.0.3
[103] gtable_0.3.6                htmltools_0.5.8.1
[105] biomformat_1.34.0           scales_1.4.0
[107] reformulas_0.4.1            snakecase_0.11.1
[109] knitr_1.50                  rstudioapi_0.17.1
[111] tzdb_0.5.0                  reshape2_1.4.4
[113] checkmate_2.3.2             nlme_3.1-166
[115] nloptr_2.2.1                ggcorrplot_0.1.4.1
[117] cachem_1.1.0                zoo_1.8-14
[119] rhdf5_2.50.2                parallel_4.4.2
[121] vipor_0.4.7                 foreign_0.8-87
[123] pillar_1.10.2               grid_4.4.2
[125] vctrs_0.6.5                 slam_0.1-55
[127] BiocSingular_1.22.0         beachmat_2.22.0
[129] xtable_1.8-4                cluster_2.1.6
[131] beeswarm_0.4.0              htmlTable_2.4.3
[133] evaluate_1.0.3              tinytex_0.57
[135] mvtnorm_1.3-3               cli_3.6.5
[137] compiler_4.4.2              rlang_1.1.6
[139] crayon_1.5.3                labeling_0.4.3
[141] mediation_4.5.0             plyr_1.8.9
[143] fs_1.6.6                    ggbeeswarm_0.7.2
[145] stringi_1.8.7               viridisLite_0.4.2
[147] BiocParallel_1.40.2         lazyeval_0.2.2
[149] quantreg_6.1                Matrix_1.7-1
[151] patchwork_1.3.0             hms_1.1.3
[153] sparseMatrixStats_1.18.0    Rhdf5lib_1.28.0
[155] rbibutils_2.3               gridtext_0.1.5
[157] igraph_2.1.4                memoise_2.0.1
[159] readxl_1.4.5                polynom_1.4-1
```

```
[161] ape_5.8-1
```